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Epistemologia, Lógica e Linguagem

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Ana Pato
e Juan Redmond

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Prefácio

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De 29 a 31 de Outubro de 2012 realizou-se na Faculdade de Ciências da Universidade de Lisboa o Simpósio Internacional de Epistemologia, Lógica e Linguagem (ISELL). O evento foi organizado pelo Centro de Filosofia das Ciências da Universidade de Lisboa (CFCUL) e pelo Grupo de Lógica, Lenguaje e Información (GILLIUS) da Universidade de Sevilha no contexto do projecto internacional "Knowledge Dynamics in the Field of Social Sciences: Abduction, Intuition and Invention" - Acções Integradas Luso-Espanholas. Este Simpósio encontra-se assim na continuidade da fecunda e já longa parceria entre o CFCUL e o GILLIUS que se tem materializado na organização, iniciada em 2007, de uma serie de "Jornadas Ibéricas" de Filosofia da Ciência à qual corresponde a publicação de 5 volumes¹. A publicação do presente volume vem agora suprir uma lacuna no trabalho editorial que se vem realizando no âmbito destas Jornadas, actualmente na sua VIII edição.

¹ Olga Pombo, Ángel Nepomuceno (eds), 2009, *Lógica e Filosofia da Ciência*, Lisboa: CFCUL. David Fernández Duque, Emilio F. Gómez-Camínere Parejo, Ignacio Hernández Antón (eds), 2010, *Estudios de Lógica, Lenguaje y Epistemología - IV Jornadas Ibéricas*, Sevilla: Universidad de Sevilla. Olga Pombo (ed.), 2011, *Lógica Universal e Unidade da Ciência*, Lisboa: CFCUL. Luis Fernández Moreno, Francisco J. Salguero Lamillar, Cristina Barés Gómez (eds.), 2012, *Ensayos sobre Lógica, Lenguaje, Mente y Ciencia*, Sevilla: Ediciones Alfar. Olga Pombo, Paulo Castro (eds.), 2017, *Debates da Filosofia da Ciência Contemporânea - VIII Jornadas Ibéricas de Filosofia da Ciência*, Lisboa: CFCUL.

Cumprindo o dever de divulgação e promoção da investigação que se realiza no CFCUL, o volume que agora se publica reúne 12 artigos de entre as 39 comunicações apresentadas no Simpósio². Esses estudos ocupam-se de temas diversos no âmbito geral da filosofia da ciência. É, contudo, possível assinalar alguns cruzamentos temáticos, nomeadamente, em torno da filosofia da física e da matemática, da teoria dos modelos, das questões da crença e da lógica sobretudo abductiva.

J. R. Croca ocupa-se dos desenvolvimentos da Física Eurítmica – uma nova física complexa e não-linear – apresentando a lei da atracção universal como caso particular do princípio da Euritmia e o postulado da invariância da luz como expressão da natureza complexa não-linear dos fotões.

No artigo de Gil C. Santos, é a partir do campo fértil da Teoria da Relatividade Restrita que são pensadas as questões do tempo, da mudança e do devir e questionadas as implicações de tais questões no próprio estatuto da Lógica e do seu objecto na sua relação com a ontologia e as ciências naturais.

Geometria, Caos, Simetria, Princípios, Convencionalismo são os cinco conceitos escolhidos por Isabel Serra para por em evidência os aspectos matemáticos, físicos e epistemológicos do pensamento de Poincaré.

Já Matías A. Saracho e José Gustavo Morales, no campo da filosofia da matemática, ocupam-se do papel desempenhado pelas definições procurando mostrar, através de um estudo de caso – a definição proposta por Leibniz de “infinitamente pequeno” –, o valor heurístico da definição no contexto da resolução de problemas em matemática.

No artigo da dupla Diego L. Valladares e Juan Redmond é a noção de átomo que é abordada, a partir da proposta de Bohr de 1913, mas aqui com o objectivo de pensar o papel dos modelos em ciência enquanto ferramentas ou mediadores epistémicos.

O papel epistemológico desempenhado pela crença é abordado em mais do que um texto. Fernando R. Velázquez-Quesada propõe uma generalização do “quadro da plausibilidade”, uma das diferentes formas de representação da noção de crença. Reinhard Kahle, por sua vez, analisa a abordagem AGM à revisão das crenças, hoje tida como a abordagem standard a este problema, e propõe, apesar dos seus reconhecidos méritos,

² Não tendo sido feita qualquer revisão científica e crítica dos textos, os mesmos são da inteira responsabilidade dos seus autores.

uma avaliação crítica das suas assumpções fundamentais. Na área da lógica formal, o volume inclui um estudo de Marc Meléndez Schofield sobre um tipo peculiar de falácia lógica, a falácia modal. Munido deste instrumento, o autor procura perceber em que medida a lógica quântica suplanta, ou não, a lógica clássica. Por seu lado, Hans van Ditmarsch e Ángel Nepomuceno-Fernández procuram pensar a prática científica, e, associada a ela, a importante noção de abdução para, em seguida, as colocarem em relação com o problema da expansão das crenças.

O importante campo da relação entre ciência e sociedade é abordado pelo texto de João 'Cão' Duarte no qual o movimento de participação pública da investigação científica é posto em relação com os conceitos de abdução e pensamento diagramático de Peirce.

Por último, Sergi Oms aborda o problema da verdade analisando a teoria minimalista da verdade e os desafios que lhe são colocados pelo paradoxo do Mentiroso.

A todos os autores que contribuíram com os seus textos para o presente volume, assim como a todos os participantes no ISELL, queremos expressar os nossos agradecimentos.

Olga Pombo
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Universal attraction law and the postulate of invariance of the velocity of light in Eurhythmic Physics

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Abstract

The universal attraction law and the invariance of the speed of light have been the cornerstones of classical mechanics and relativity. These important traditional postulates, very adequate at a given scale of description of Reality, may now be understood as a simple consequence of deeper Eurhythmic nonlinear complex inter-relational physical processes.

Keywords: Classical Mechanics, relativity, universal attraction law, invariance of the speed of light, eurhythmic physics, nonlinearity, complexity.

1 | Introduction

The concept of Eurhythm, in its current form, was, for the first time, presented in a talk, *The Principle of Eurhythm a Key to The Unity of Physics* at The First Lisbon Colloquium for the Philosophy of Science - Unity of Science, Non-traditional Approaches, Lisbon, October, 25-28, 2006. Now, more than a decade after its initial formulation, this foundational concept was extended, giving origin to the development of a New Physics of Nonlinear and complex processes, Eurhythmic Physics¹.

In the framework of this New Physics, the traditional fundamental postulates such as: the universal attraction law and the invariance of the speed of the light may be understood in an easy and intuitive way.

¹ Croca, 2003 and 2015.

2 | Eurhythmic Physics

Eurhythmic Physics, an inter-relational nonlinear physics of the complex, rejects both the absolute determinism, also called Laplacian determinism and the indeterminism, by assuming a kind of middle way. This complex inter-relational approach, is based on five basic assumptions:

The first is related with a realistic posture. Consisting in the admission that there is an objective Reality. The observer is a part of this very Reality, changing it and being changed at the same time in a greater or lesser degree. This assumption is contrary to certain extreme idealistic interpretations of Orthodox Quantum Mechanics², in which and in last instance, Reality is a making of the observer consciousness.

The second, *according to* Anaximander and following de Broglie³, is the admission that there is a basic physical medium of indefinite nature, the subquantum medium. All physical processes do occur in this basic medium.

The third assumption describes the physical inter-relational physical entities, such as particles, fields and so. Stating that physical processes are no more than relative stable, at a certain given scale of observation and description, predictive organizations of the basic subquantum medium.

The definition of complex inter-relational particle constitutes the fourth assumption. A complex inter-relational particle may not be characterized by a simple point-like position. It shares both properties of relative localization and of extension, being composed of two inter-related parts: The extended wave, the theta wave, and a relatively well localized and highly energetic and complex part named the acron, or the kernel or even the corpuscle.

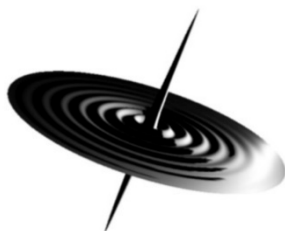


Fig.1 - Complex particle

² Bohr, 1928, 580, 1985; Dirac, 1958; Pauli, 1980; Bohm, 1951, Tomonaga, 1966, Von Newman, 1996

³ De Broglie, 1969.

Finally, the last assumption, the fifth, concerns the genetic organizing principle of eurhythmy⁴. This basic principle of Nature states the acron immersed in the theta wave field with a relatively minute energy follows a nonlinear stochastic process that, on average, leads it to the regions of the field where the energetic density is greater.

3 | Universal attraction law

Newton in his *Principia*⁵ postulated what he called universal gravitational law, having the mathematical form

$$F = G \frac{m_1 m_2}{r^2}. \quad (1)$$

This physical expression tells us that the gravitic force, F , between two bodies is proportional to the product of their masses, m_1 and m_2 , and varies inversely with the square of the distance, r^2 , between them. G is the universal gravitation constant.

The plot of this function, of the gravitational force in function of the distance is shown in next picture, Fig.2.



Fig.2 - Gravitational force in function of the distance

⁴ Croca, 2010; Croca, 2012.

⁵ Newton, 1952.

As may be seen from the picture, this said Universal gravitation force comes to infinity as the distance decreases,

$$F \rightarrow \infty, \text{ as } r \rightarrow 0.$$

According to the universal attraction gravitational law, for very short distances the force approaches infinity.

Many efforts were made to understand the why of this law⁶: Descartes⁷ assumed that motions of planets were due to a kind of vortex (*tourbillon*) of a celestial fluid. While the first trial to explain mathematically the motion of the celestial bodies was due to Huygens. Nicholas Fatio, a friend of Newton, presented in 1660 to the Royal Society of London a corpuscular theory for gravitation.

Nevertheless, it was George-Louis Le Sage (1724-1803) the one that proposed a more developed theory aiming to explain not only the how gravity works but also the why of these forces. Naturally, as expected, Le Sage was full immersed in Descartes linear conceptual paradigm in which the whole is always assumed to be equal to the sum of the constituent parts. Parts that in any possible combination maintain always its own individuality. In such conditions, his explanation of the gravitic interaction was, naturally, based on a linear simplistic and individualistic ontology. After having enjoyed a certain degree of success, the model was put aside mainly due to the work of Lord Kelvin, who pointed out certain inconsistencies of the theory with the energy conservation principle. Le Sage gravitation theory has had a certain revival at the end of last century. Still, all these improvements to the model were made in the Cartesian linear ontology in which the reaction is assumed to be always equal to the action, so that in any interacting physical situation each system maintain its own individuality.

3.1 | Problems with the “universal” gravitation law

The first problem to consider is related with the fact that this expression, to describe how gravity works, was developed by human beings immersed in the linear conceptual and experimental framework of the seventeenth century. Now at the beginning of the XXIth century, the

⁶ Matthew, 2002.

⁷ Descartes, 1979.

experimental observation of Reality has dramatically increased both at probing very short and very large spatial and time scales. So, it is natural to conceive, that this "law" being a human construct, very adequate to describe Reality, at the scale in which it was developed, now, in other more general experimental and conceptual realm, may show its natural, human limitations.

Has we have seen, at very short spatial scales of observation, at the quantum level, this expression gives a practically an infinite force. In such conditions, there must be a distance limit for which the attraction "law" is useful to describe approximately what is going on. This is related with the so-called black holes. Naturally, this does not exclude the possibility of existence of certain particular regions of the universe in which the gravity field is really very intense. Nevertheless, even if much intense the real physical gravitic field is never infinite.

On the other hand, the same is to be expected for very large, that is, at very large distances, at galactic scale. At these this very large spatial scales this so-called universal law may also cease to be adequate. As a consequence, it is possible that the dark matter and concomitant dark energy are no longer necessary to explain the observed anomalies of the attraction law at galactic spatial scales of observation.

Summarizing, if we assume that the gravitational expression, a mere human construct, is absolutely valid, in any physical interaction medium and for every distance, we have then be forced to introduce the concepts of black holes of infinite energy and the hypothetical dark matter and energy constituting the great majority of the universe.

3.2 | Attraction of complex bodies in Eurhythmic Physics

In the nonlinear complex framework of Eurhythmic Physics, in order to understand the attraction of the complex physical entities we have to recur to the principle of eurhythmmy. This principle, a generalization of de Broglie guiding principle, tells us that the corpuscle, the acron moves in the theta wave field following preferentially a stochastic path that leads it, on average, to the regions in which the theta wave field is more intense.

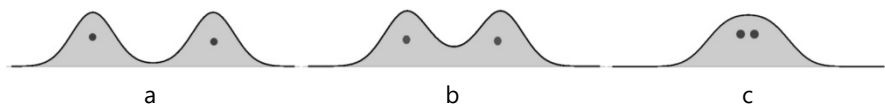


Fig.3 - Average motion of two bodies for different times

Fig.3 tries to illustrate this situation for different time intervals. At time $t = a$, the two theta wave fields have a certain degree of superposition producing an asymmetry toward the centre of the fields. In such case, since each acron sees an asymmetric field in which the intensity is greater in the middle, they, in average, tend to approach. This situation is shown for time $t = b$. As the interactive process goes on, the field tends to coalesce⁸ at time $t = c$. Everything occurs, as if a kind of pushing force lead the corpuscles to approach each other.

In last instance, the gravitic interaction may be described by the coalescence theorem, which by its turn is simple consequence of the Principle of Eurhythmy. Indeed, it is possible to show [see Eurhythmic Physics] that the traditional attraction law is quite adequate to describe gravitic interaction between two extreme values.

$$F \propto \frac{1}{r^2}, \quad \text{for } r_m \leq r \leq r_M$$

For distances, very short, $r \leq r_m$ and very large, $r \geq r_M$ the expression no longer, in general, is adequate to describe what is going on, as indicated in Fig.4.

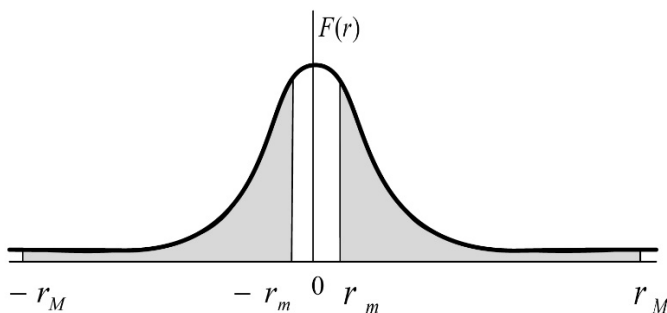


Fig.4 - Limits of adequacy of the attraction "law"

⁸ Gatta, 2010.

Definitely, between these two extreme limits, the eurhythmic tendency for the complex gravitic particle kernel, to approach, that is, the pushing force assumes the mathematical traditional form varying with the inverse of the square of the distance.

4 | The invariance of the speed of light

At the end of the XIXth experimental devices to probe time intervals of femtosecond, $\Delta t \sim 10^{-15} s$, order, were developed. At these very short times, the traditional law of addition of velocities shows certain difficulties in explaining the observed experimental results. Indeed, the experiments done by Michelson and Morley⁹ show that the two-way velocity of light did not change with the motion of the source relative to a hypothetical aether. To explain the so-called “null results” of Michelson-Morley’s experiments the theory of relativity was developed¹⁰.

One of the most fundamental assumptions of relativity is that the speed of light, c , is independent of the relative motion of the emitting source and even more, its speed, in the so-called vacuum, $c = 300\,000 km/s$, is the maximum possible speed for the transmission of any signal. This assumption, is a natural consequence of the fact that people at that time were full formatted into the Cartesian linear framework. In such conditions, since the traditional linear addition rule didn’t work because we are dealing with very complex nonlinear interacting situations the so-called anomaly was simply postulated. That is, the complex intrinsic nonlinear nature of the complex interaction of the photon with the real physical medium was buried by means of a postulate.

This state of affairs, imposed an alleged absolute theoretical and practical boundary for any possible way, existent and to be devised, for sending information. Accordingly, nothing can go faster than the speed c of light. Now, that we are more than a century apart from those heroic times, we may ask if it is possible to explain the why of the postulate of the invariance of the speed of light. Furthermore, if we understand why Galilean’s linear “law” of addition of velocities does not work in such cases, we may eventually conceive other physical conditions, in which we may

⁹ Michelson and Morley, 1887, 333-345.

¹⁰ Einstein, 1905; Einstein, 1923.

eventually overcome the supposed unsurpassable barrier of $v \leq c = 300\,000\text{km/s}$.

Let us now see how, the inadequacy of the linear additive rule for light, in certain physical situations, has an easy and intuitive explanation in Eurhythmic Physics.

4.1 | Velocity of the particle in different interacting physical media

A physical complex particle, when moving in an extended homogeneous medium, as shown in Fig.5, naturally interacts with it, in a greater or lesser degree.

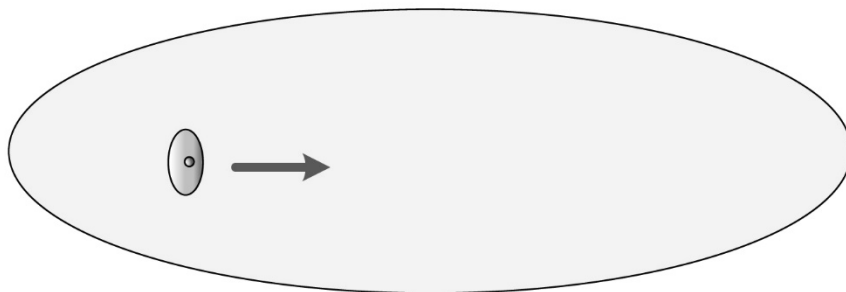


Fig.5 - A complex particle is moving in an extended homogeneous field

The relative velocity depends on degree of reciprocal interaction between the particle own field and the extended field. Two extreme situations may occur:

1 - The intensity of the own field of the complex particle, I_P , is relatively much more intense than the extended field, I_M , the medium in which is moving,

$$I_P \gg I_M.$$

In this particular interacting conditions, the particle practically ignores the extended field in which is moving and consequently the traditional addition law, traducing the relative independence of the physical entities, particle-medium, is quite adequate to describe the situation. For this case, and at this level of description of Reality, the Cartesian linear

independence, may be assumed and consequently particle and medium may be treated as practically independent. Furthermore, it allows for photonic speeds greater than c , as observed in some very particular experimental situations¹¹. Still, since these observed results are against relativity, some conservative authors¹², by the addition of the necessary *ad hoc* assumptions, are able to "show" post-factum, that there is no discrepancy with the postulate of invariance of the speed of light.

2 - The intensity of the extended field in which the complex particle moves is relatively much more intense,

$$I_P \ll I_M.$$

For these concrete interacting conditions, the kernel of the particle, the acron, in practice, "sees" only the extended field. In such circumstances, the high energetic part of the particle, the acron, nearly "ignores" the origin and consequently, its velocity, relative to the extended medium, is independent on the relative velocity of the source. In such case, since there is no independence but interdependence, the traditional law of addition of velocities is not good to describe what is going on. This happens, because, the supposed absolute Cartesian independence, between particle and the medium, is no longer an adequate assumption. In this case, the speed of the light is constant and independent of the relative motion of the source. Then, in these experimental situations, we are in the realm of adequacy of relativity.

¹¹ Sagnac, 1913, 708-710; Sagnac, 1913, 1410-1413; Wang et al. 2013; Wang et al. 2014; Nimtz et al. 1993; Nimtz et al. 1993, 1993, 9605.

¹² Langevin, 1921, 831.

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A Física e a Ontologia nas Origens da Lógica Temporal

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Introdução

A problemática filosófica sobre o devir (*genesis*), como processo de mudanças (*metabolê*, *kinêsis*), o tempo (*chrônos*), a substância permanente ou persistente (*ousia*), e as entidades (*ta onta*, *pragmata*) susceptíveis de persistir e de mudar remonta às próprias origens da filosofia na Grécia antiga.

Fosse ela entendida como um princípio (*archê*) universal, activo, permanente e subjacente à génese, à corrupção e às mudanças de todas as coisas – tal como se entendeu no quadro da concepção milésia da *physis* –, fosse ela entendida como uma causa formal ou organizacional (*eidos*), imanente e ontologicamente individuada, da consistência ôntica, da unidade e da identidade de cada coisa singular – de acordo com a concepção aristotélica¹ –, através da noção de substância (*ousia*) procurou-se fornecer um primeiro tipo de resposta ao problema da conciliação da permanência com o devir. De Parménides a Heraclito e aos Atomistas, as diferentes concepções de ser ou substância propostas corresponderam a diferentes teorias e concepções do devir e da permanência².

O desenvolvimento da filosofia grega ilustra bem que entre as diferentes categorias de *entidades* fundamentais propostas pelas mais variadas ontologias, e a concepção das próprias noções de *mudança*, de *persistência* e de *identidade diacrónica*, bem como até a discussão sobre a

¹ Cf. Aristóteles, *Metaphysics*, VII, 17, 1041 a-b 11-14.

² “The beginnings of an awareness of the problem of change can be traced back to Milesian speculation about the primary substance” (Lloyd, 1970, *Early Greek Science*, p. 36). Para uma análise breve das diferentes noções de substância, dos seus significados e das questões que elas suscitam, vide Lowe, 1988, *Substance*, 255-278; e Wiggins, 2016, *Continuants: Their Activity, Their Being and Their Identity*, 41-70.

própria (ir)realidade objectiva destas noções, há uma relação de codependência e determinação recíproca.

Foi esta mesma estrutura problemática que reemergiu, com particular vigor e de uma forma original, a partir dos inícios do séc. XX, no contexto de uma interpretação sobre o significado e o alcance *ontológicos* outorgáveis à Teoria da Relatividade Restrita (TRR) de Einstein, bem como no contexto de uma controvérsia sobre as suas possíveis consequências para o domínio da *Lógica*.

Mais concretamente, o problema centrou-se no debate em torno das razões filosóficas e científicas que justificariam, ou não, a construção de uma lógica temporal, e sobre que tipo de lógica temporal mereceria ser construído, tendo em conta as concepções de tempo, de devir e dos objectos que eram atribuídas à TRR. Ora, esta controvérsia viria a ter um papel relevante na história contemporânea da Lógica, na medida em que ela suscitou o debate acerca do próprio estatuto disciplinar da Lógica, quer quanto às suas relações de dependência e/ou de autonomia face à ontologia e às ciências naturais, quer quanto à definição do seu objecto de estudo próprio.

1 | A Ontologia Quadri-extensionalista

No contexto da filosofia contemporânea da lógica, a avaliação da possibilidade de construção de uma lógica temporal reflectiu, em boa medida, a controvérsia que se estabeleceu em torno do significado e do alcance ontológicos atribuíveis à TRR. Este debate foi protagonizado por Arthur Prior e Willard Quine.

No que diz respeito ao debate propriamente ontológico, duas posições opostas emergiram entre os filósofos (impropriamente chamados quadri-dimensionalistas) que – como Quine – defenderam que a TRR exigia uma redefinição fundamental das noções pré-relativistas de tempo e devir, assim como da natureza, da identidade e da persistência dos objectos³,

³ Esta perspectiva geral remonta a Bertrand Russell, desde a publicação de *Our Knowledge of the External World* (1914), de *The ABC of Relativity* (1925) e de *The Analysis of Matter* (1927), tendo sido posteriormente desenvolvida por Rudolf Carnap, em *Der logische Aufbau der Welt* (1928), por Hans Reichenbach, em *Philosophie der Raum-Zeit-Lehre* (1928) e em *The Direction of Time* (postumamente publicada em 1956) e, mais recentemente, por autores como J. J. C. Smart, W. Quine, David Lewis, Mark Heller ou Theodore Sider.

e aqueles filósofos (impropriamente chamados tri-dimensionalistas) que – como Prior – defenderam a conservação, em termos gerais, de tais noções – seja, porque criticavam a própria interpretação da TRR promovida pelos primeiros, seja porque recusavam à TRR a única ou a definitiva palavra sobre o que é o tempo, o devir, a identidade e a persistência dos objectos.

De acordo com a primeira perspectiva, uma metafísica que pretenda ser minimamente consistente com os desenvolvimentos da ciência contemporânea deve assumir, em acordo com a TRR,

(1) a defesa de uma ontologia fundamental de *acontecimentos*, à luz da qual as noções clássicas de objecto, identidade e mudança deverão ser reformuladas; e

(2) a defesa de uma ontologia fundada numa representação geocronométrica⁴ da realidade física que conduzirá a uma identificação sumária entre as noções de *dimensão* e *extensão* e, por essa via, a uma concepção extensional (por analogia com o espaço) da dimensão temporal e da persistência dos objectos.

Eis o argumento de base: num universo onde os acontecimentos são tidos como as entidades fundamentais, e as dimensões do espaço e do tempo se encontram interligadas numa estrutura físico-geométrica única – o ‘espaço-tempo’ –, os objectos persistentes devem ser concebidos como séries de acontecimentos, i.e., como estruturas, não apenas espacialmente, mas também temporalmente *extensas*. São entidades quadri-extensionais.

Pese embora esta perspectiva seja comumente conhecida como quadridimensionalista, e a sua oposta, tridimensionalista, tais designações são equívocas e enganadoras. As noções de ‘dimensão’ e ‘extensão’ só se equivalerão à luz de uma assimilação sumária da física à geometria. Porém, a noção de dimensão diz apenas respeito ao número de coordenadas necessárias para a *localização* de um qualquer fenómeno, e neste sentido *qualquer* tipo de entidade é quadridimensional – sejam acontecimentos (não persistentes), sejam objectos duradouros, processos, relações, etc.

⁴ Como sublinhou M.-A. Tonnelat, ao passo que a geometria cartesiana deixava o tempo de fora, reduzindo a sua geometria à pura extensão, a geometria relativista inclui e integra o tempo como uma dimensão própria no seu *continuum* métrico. Cf. Tonnelat, 1971, *Histoire du Principe de Relativité*, 455; Merleau-Ponty e Morando, 1971, *Les Trois Étapes de la Cosmologie*, 201, n. 1; Lévy-Leblond, 1996, *Aux Contraires – L'exercice de la pensée et la pratique de la science*, 135; e Lochak, 1994, *La Géométrisation de la Physique*.

Nada aqui há que justifique – como ironizava Einstein – a irrupção de “um sentimento não diferente daquele que é despertado pelos pensamentos do oculto”, quando se ouve falar em coisas quadridimensionais. Para perceber isto bastará tomar em consideração os esclarecimentos reiteradamente feitos pelo próprio Einstein. Por exemplo:

“É um erro largamente difundido julgar que a teoria da relatividade terá, num certo sentido, descoberto pela primeira vez ou, em qualquer caso, introduzido de uma nova forma, a quadridimensionalidade do *continuum* físico. Isto não é obviamente o caso. A mecânica clássica também se baseia no *continuum* quadridimensional do espaço e do tempo”⁵. Com efeito, “[m]esmo na física clássica, um acontecimento é localizado por quatro números, três coordenadas espaciais e uma coordenada temporal; a totalidade dos ‘acontecimentos’ físicos era então concebida como imersa numa diversidade contínua quadridimensional”⁶.

A noção de quadridimensionalidade *não* visa, pois, uma nova ontologia da estrutura material do nosso mundo e dos corpos físicos. É uma simples questão de *métrica*. Acresce que o facto de o tempo-coordenada ser representado como um eixo cartesiano num espaço a quatro dimensões não faz do tempo uma dimensão física cujas propriedades sejam assimiláveis às do espaço, tornando-se assim também uma dimensão extensional⁷. Daí, que a oposição ontológica real seja

⁵ Einstein, 1949, *Autobiographical Notes*, 57.

⁶ Einstein, 1954, *Relativity – The Special and the General Theory*, 151. Einstein reforça esta ideia: “The non-mathematician is seized by a mysterious shuddering when he hears of ‘fourdimensional’ things, by a feeling not unlike that awakened by thoughts of the occult. And yet there is no more common-place statement than that the world in which we live is a fourdimensional space-time continuum. Space is a three-dimensional continuum. By this we mean that it is possible to describe the position of a point (at rest) by means of three numbers (co-ordinates) x , y , z , and that there is an indefinite number of points in the neighbourhood of this one, the position of which can be described by co-ordinates such as x_1 , y_1 , z_1 , which may be as near as we choose to the respective values of the co-ordinates x , y , z , of the first point. In virtue of the latter property we speak of a ‘continuum’, and owing to the fact that there are three coordinates we speak of it as being ‘three-dimensional’. Similarly, the world of physical phenomena which was briefly called ‘world’ by Minkowski is naturally four dimensional in the space-time sense. For it is composed of individual events, each of which is described by four numbers, namely, three space coordinates x , y , z , and a time co-ordinate, the time value t .” (Einstein, 1954, *Relativity – The Special and the General Theory*, 56-57).

⁷ Cf. Norton, 2015, *What Can We Learn about the Ontology of Space and Time from the Theory of Relativity?*, 194-195.

entre quadri-extensionalistas e tri-extensionalistas. A controvérsia ontológica reside na questão de se saber se a dimensão física de tempo constitui-se, também ela, como um tipo de *extensão*.

Porém, a confusão que se instalou entre estas noções disseminou-se de uma forma tão inesperada e surpreendente que se chegaria ao cúmulo de poder defender-se que os acontecimentos são entidades *tridimensionais*, porque não persistem...⁸. Faltar-lhes-ia a dimensão temporal. É fácil prever a conclusão a que inevitavelmente chegaríamos se seguissemos esta forma de entendimento: se os acontecimentos são idealmente concebíveis como pontuais, não apenas temporalmente, mas também espacialmente, segue-se que as entidades últimas da realidade física estão privadas de dimensões *físicas*...

Seja como for, é à luz desta ontologia quadri-extensionalista que se elaborará a teoria *perdurabilista* da persistência, de acordo com a qual os objectos persistentes devem ser concebidos como entidades cujas estruturas se prolongam extensionalmente na dimensão temporal, através de uma série de acontecimentos. Neste tipo de representação, qualquer objecto elementar persistente será representado por uma *linha do universo* (*world-line*), ao passo que um corpo composto será representado por um cilindro ou prisma quadrangular (*world-tube*) formado pelas diferentes linhas do universo dos seus múltiplos constituintes. Todos os corpos serão representados como extensões no espaço-tempo, à imagem de um ser cilíndrico e anelídeo (como uma minhoca; *worm-like*), constituído por, e segmentável em, diferentes *partes temporais*.

Neste sentido, um objecto não persiste, porque e enquanto consegue conservar uma mesma e única identidade no curso da sua evolução, apesar das mudanças (quantitativas e qualitativas) a que está naturalmente sujeito. Ao invés, um objecto persiste, porque existe uma relação de *continuidade* espacio-temporal e causal (chamada

⁸ "It is perfectly possible to think of things and processes as four-dimensional space-time entities. The instantaneous state of such a four-dimensional space-time solid will be a three-dimensional 'time-slice' of the four-dimensional solid. Then instead of talking of things and processes changing or not changing we can now talk of one time slice of a four-dimensional entity *being* different or not different from some other time slice. (Note the tenseless participle of the verb 'to be' in the last sentence.)" (Smart, 1963, *The Space-Time World*, 497).

‘genidentidade’⁹) entre diferentes acontecimentos, de tal forma que essa relação de continuidade permitirá encarar essas entidades pontuais, e *numericamente distintas*, enquanto *partes* temporais de um mesmo *todo* temporal chamado objecto¹⁰.

Uma ontologia fundamental de acontecimentos é, assim, defendida¹¹. Como escreveu B. Russell, o mundo concebido pela teoria da relatividade “não é um mundo de ‘coisas’ em ‘movimento’, mas um mundo de ‘acontecimentos’”. É verdade que existem partículas que parecem persistir, mas estas (...) devem ser concebidas como séries de acontecimentos interligados, como as notas sucessivas de uma canção. Os *acontecimentos* é que constituem a matéria-prima da física da

⁹ O termo genidentidade (*Genidentität*) foi um termo originalmente criado pelo psicólogo alemão Kurt Lewin, no seu trabalho *Der Begriff der Genese in Physik, Biologie und Entwicklungsgeschichte*, de 1922. Lewin pertenceu, tal como Reichenbach, à Sociedade de Filosofia Empírica (mais tarde, renomeada Científica) de Berlim – sociedade, essa, que manteria uma cooperação importante com o Círculo de Viena. No contexto da problemática da persistência e da identidade diacrónica dos objectos, o termo ‘genidentidade’ foi pela primeira vez usado por Reichenbach, na sua *Philosophie der Raum-Zeit-Lehre*, interpretando-a como uma conexão *causal* estabelecida pela transmissão de sinais de luz. Mais tarde, Reichenbach viria a defender uma teoria da genidentidade *funcional*, para dar conta da realidade sub-atómica. Cf. Reichenbach, 1956, *The Direction of Time*, 226-236. Rudolf Carnap, Bertrand Russell e muitos outros filósofos quadri-extensionalistas viriam também a adoptar a teoria da genidentidade, se bem que sob modalidades por vezes distintas. A teoria da genidentidade não viria, todavia, a conseguir superar o carácter vago e inconclusivo das condições que ela reclamaria como necessárias (continuidade espacio-temporal e causal), nem a evitar incorrer num círculo vicioso ao ter de pressupor a identidade do todo para poder identificar uma continuidade *qualitativa* entre as suas partes numericamente distintas entre si. O reconhecimento dos limites desta teoria da identidade acabaria, finalmente, por conduzir à grande fuga em frente chamada ‘universalismo mereológico’ ou ‘princípio de composição irrestrita’. Doravante, *qualquer* colecção de porções de matéria conta como *um objecto*, e qualquer par de objectos forma um outro objecto como o todo mereológico desse par. Ou seja, na impossibilidade de se determinar critérios precisos de genidentidade entre partes ou fases que nos permitam *determinar* as relações de continuidade que representam, *distinguindo-as* das séries que não representam, a existência de objectos persistentes, a adesão ao *anything goes* do ‘princípio de composição irrestrita’ pareceu ser a única saída viável para filósofos como W. Quine, D. Lewis, M. Heller, M. Jubien, M. Rea, H. Hudson ou T. Sider.

¹⁰ “Some philosophers believe that everyday objects are four-dimensional space-time worms, that a person (for example) persists through time by having temporal parts, or stages, at each moment of her existence. None of these stages is identical to the person herself; rather, she is the aggregate of all her temporal parts. (...) The metaphysical view (...) called ‘four-dimensionalism’ (...) may be stated roughly as the doctrine that temporally extended things divide into temporal parts.” (Sider, 1996, *All The World’s A Stage*, 433).

¹¹ Cf. Sider, 2001, *Four-Dimensionalism – An Ontology of Persistence and Time*, 53 e 79-80.

relatividade¹². Em vez da permanência de um corpo material, temos uma 'linha do universo' que não é mais do que uma série de acontecimentos causalmente interligados¹³.

Carnap comparará um objecto persistente a um comboio, cujas carruagens seriam as suas partes temporais¹⁴. Reichenbach afirmaria que "uma coisa é uma série de acontecimentos que se sucedem no tempo"¹⁵. Isto significa, como bem observou Quine, que os objectos clássicos passam a ser tratados como coisas processuais (*process-things*): "[o]s objectos físicos, concebidos quadridimensionalmente no espaço-tempo, não se distinguem de acontecimentos ou, no sentido concreto do termo, de processos"¹⁶. Como dirá mais tarde David Lewis, um objecto persistente é um agregado de fases interrelacionadas, ou uma classe de fases, ou uma sequência ordenada de fases ou partes temporais, ou ainda uma "soma mereológica, de tal forma que as fases são literalmente partes do objecto persistente"¹⁷. Quine descreve, desta forma, a representação quadri-extensionalista dos objectos que resulta da espacialização do tempo: "Um corpo é (...) visualizado, de um ponto de vista eterno, como um todo quadridimensional que se estende para cima e para baixo, para norte e para sul, para leste e para oeste, para o antes e para o depois"¹⁸.

Esta é a representação mais literalmente geométrica e extensional da dimensão temporal que é possível oferecer. Lewis preferiu a descrição por analogias: o "perdurar [de um objecto] corresponde ao modo como uma estrada persiste através do espaço; uma das suas partes está aqui, outra das suas partes está ali, e nenhuma das suas partes está integralmente presente em dois locais diferentes"¹⁹. Ou ainda, e de uma forma mais

¹² Russell, 1925, *The ABC of Relativity*, 222-223.

¹³ Russell, 1927, *The Analysis of Matter*, 244-245. Cf. igualmente: Russell, 1948, *Human Knowledge*, 337.

¹⁴ Carnap, 1928, *The Logical Structure of the World*, § 159, 252-253.

¹⁵ Reichenbach, 1956, *The Direction of Time*, 224.

¹⁶ Quine, 1960, *Word and Object*, § 36, 171.

¹⁷ "A continuant person is an aggregate of person-stages, each one I-related to all the rest (and to itself). For short: a person is a an I-interrelated aggregate." (Lewis, 2001, *Survival and Identity*, 399). Numa nota a este passo, acrescenta Lewis: "It does not matter what sort of 'aggregate'. I prefer a mereological sum, so that the stages are literally parts of the continuant. But a class of stages would do as well, or a sequence or ordering of stages, or a suitable function from moments or stretches of time to stages." (Lewis, 2001, *Survival and Identity*, 418, n. 4).

¹⁸ Quine, 1970, *Philosophy of Logic*, 30. Cf. igualmente: Quine, 1987, *Quiddities*, 197.

¹⁹ Lewis, 1986, *On the Plurality of Worlds*, 204.

sugestiva: “um objecto persistente é como um desfile: primeiro aparece uma das suas partes, depois aparece outra” (ressalvando-se, porém, que “a maioria dos objectos persistentes são muito mais contínuos que a maioria dos desfiles”)²⁰.

Quanto à noção de mudança, esta mereceu alguma discórdia no próprio seio dos ontólogos quadri-extensionalistas. Por exemplo, ao passo que J.J.C. Smart defendeu explicitamente uma perspectiva estática da realidade, sem lugar a processos de mudança (na TRR, o tempo não é abolido, mas integrado como uma dimensão estática, não envolvendo, nem mudanças, nem alterações, nem permanência de identidades)²¹, Quine procuraria afirmar a compatibilidade do quadri-extensionalismo, e da espacialização do tempo, com a mudança – pese embora esta devesse ser representada de um ponto de vista estático e intemporal. Com efeito, diz Quine, falar da mudança de um corpo é dizer que as suas partes temporais mais recentes diferem das suas partes temporais mais remotas, tal como uma parte espacial de um corpo pode distinguir-se de outra parte espacial do mesmo corpo²².

2 | O Tempo na Lógica

Que consequências deverão ser extraídas desta interpretação ontológica da TRR para o domínio da Lógica? Justificar-se-á uma lógica temporal dinâmica, quando esta noção de tempo parece ter sido rejeitada pela TRR? E em que termos poder-se-á justificar a representação lógica de objectos duradouros, quando os filósofos quadri-extensionalistas procuraram mostrar que uma tal concepção havia sido igualmente recusada por esta teoria física?

²⁰ Lewis, 2002, *Tensing the Copula*, 1.

²¹ Temos, assim, um universo estático: “quando pensamos quadridimensionalmente, substituímos as noções de ‘mudança’ e de ‘permanência de um mesmo’ pelas noções de ‘semelhança’ e de ‘dissimelhança’ das partes temporais dos sólidos quadridimensionais” (Smart, 1963, 497). Cf., ainda, Smart, 1955, *Spatialising Time*, 239-241.

²² “Change is still there, with all its fresh surprises. It is merely incorporated. To speak of a body as changing is to say that its latter stages differ from its earlier stages, just as its upper parts differ from its lower parts. Its later shape need be no more readily inferred from its earlier shape than its upper shape its lower.” (Quine, 1987, *Quiddities*, 197). Cf. também: Quine, 1981, *Things and Their Place in Theories*, 10.

Para Arthur Prior, a questão deve começar por ser colocada sobre uma outra base: se a Lógica trabalha com proposições ou enunciados (*statements*) e suas relações inferenciais, não há qualquer justificação *lógica* para que ela se limite ao estudo das proposições que expressam verdades imutáveis, por exclusão das proposições sujeitas a flexões verbais temporais (no presente, passado e futuro) e, por conseguinte, susceptíveis de mudar de valor de verdade²³. Acresce que, para Prior, existem 'estruturas' sólidas no mundo da mudança e da sucessão temporal que os lógicos devem saber descobrir. Não há, por isso, qualquer justificação para abandonar o estudo da natureza dinâmica do mundo para os domínios exclusivos das filosofias da existência ou das teorias da argumentação. Mais: segundo Arthur Prior, esse estudo propriamente lógico poderá, inclusivamente, ajudar a uma descrição filosófica mais precisa da realidade do devir e da mudança²⁴.

As razões teóricas que explicam a omissão do tempo na Lógica foram, de resto, distintas ao longo da história²⁵. No caso da Lógica moderna teve, naturalmente, um papel fundamental as motivações teóricas dos seus fundadores, como Frege e Peano: "inventar uma notação simbólica e um conjunto de regras que fossem adequadas para todas as *demonstrações dedutivas* - e, em particular, para a análise das *demonstrações matemáticas*"²⁶.

Quine é disso um bom exemplo. O seu objectivo de intemporalizar o discurso lógico precedeu a reivindicação da (sua interpretação da) TRR.

²³ "Logic deals, at bottom, with statements – it enquires into what statements follow from what – but logicians aren't entirely agreed as to what a statement is. Ancient and medieval logicians thought of a statement as something that can be true at one time and false at another." (Prior, 1996, *Some Free Thinking about Time*, 47). Cf. também: Prior, 1976, *What is Logic?*, 128-129.

²⁴ "Propositions which are subject to tense-inflections, such as 'I am falling out of a boat', are liable to be true at one time and false at another. Certainly there are unchanging truths, but there are changing truths also, and it is a pity if logic ignores these, and leaves it to existentialists and to comparatively informal 'dialecticians' to study the more '*dynamic*' aspects of reality. There are clear, hard structures for formal logicians to discover in the world of change and temporal succession. There are practical gains to be had from this study too, for example in the representation of time-delay in computer circuits, but the greatest gain that logic of tenses brings is the accurate philosophical description of the reality of the passage of time." (Prior, 1996, *A Statement of Temporal Realism*, 46).

²⁵ Por exemplo, vários projectos de eliminar o tempo na Lógica foram desenvolvidos, desde a segunda metade do século XIX, seja pela inserção do tempo como um elemento do termo geral que representa a propriedade, seja por considerá-lo como parte do sujeito. Em qualquer uma destas opções, os valores de verdade das frases tornam-se, assim, imutáveis. Cf. Prior, 1957, *Appendix A: Tenses and Truth in the History of Logic*, 104-122.

²⁶ Putnam, 1988, *Lógica*, 27.

Em *Elementary Logic*, Quine já avançava com o programa de eternizar as frases, tratando-se aí de um mero problema de gerência interna do sistema do cálculo lógico. Sendo as indicações temporais do discurso um obstáculo à determinação dos valores de verdade, Quine recusava-se a concebê-las como elementos do cálculo²⁷. Aliás, o próprio Quine nota a independência deste programa face a uma interpretação da TRR, notando que Russell já havia manifestado semelhante preocupação em eliminar o tempo da linguagem lógica nos *Principles of Mathematics*, de 1903 – i.e., antes da publicação do primeiro trabalho de Einstein sobre a relatividade²⁸.

Sucede, todavia, que a esta razão de índole propriamente (ideo)lógica viria a somar-se, reforçando-a, a invocação da TRR²⁹. Com efeito, quando se tratou de confrontar a possibilidade de construção de uma lógica temporal com a interpretação que Quine acolhia da TRR, escreveu o filósofo: “uma lógica das flexões temporais (*logic of tense*) é uma enorme trivialidade que não se justifica, se a nossa preocupação se limitar ao *uso científico* da linguagem, e não ao estudo científico dela”³⁰.

Porém, quando não se trata de definir o objecto de estudo e os objectivos disciplinares da Lógica em função de pressupostos *extra*-lógicos ou metafísicos, é-se levado a encarar o pensamento e o discurso humano enquanto tais, procurando determinar as estruturas e as condições próprias de formalização das suas diferentes *formas* de representação e expressão da realidade. Daí, surge o problema *propriamente lógico* de estudar o tempo, tal como este é representado e expresso no e pelo *logos*.

Como lucidamente observou J.-L. Gardies, “se a lógica restringir as suas pretensões à fundamentação das estruturas matemáticas já elaboradas antes de si, ela não terá que considerar o tempo. Se, pelo contrário, ela

²⁷ “logical analysis is facilitated by requiring rather that each *statement* be true once for all or false once for all, independently of time. This can be effected by rendering verbs tenseless and then resorting to explicit chronological descriptions when need arises for distinctions of time.” (Quine, 1941, *Elementary Logic*, §2, 6).

²⁸ Cf. Quine, 1960, *Word and Object*, § 36, 172.

²⁹ “Einstein’s discovery and Minkowski’s interpretation of it provided an essential impetus, certainly, to spatiotemporal thinking, which came afterward to dominate philosophical constructions in Whitehead and others. But the idea of paraphrasing tensed sentences into terms of eternal relations of things to times was clear enough before Einstein”. Todavia, acrescenta Quine, a descoberta de Einstein – “that space and time are distinguishable only relative to a velocity” – “leaves no reasonable alternative to treating time as spacelike” (Quine, 1960, *Word and Object*, § 36, 172 e n. 3).

³⁰ Quine, 1970, *Methodological Reflections on Current Linguistic Theory*, 397.

pretender ultrapassar o domínio ao qual as matemáticas tradicionalmente se confinaram, e não aceitar renunciar, *a priori*, qualquer forma de racionalidade, em suma, se nada de racional lhe for estranha, então, ela não se reconhecerá no direito de prescindir de uma consideração do tempo”³¹.

Ora, não apenas toda a existência humana, como todas as ciências não-formais, são marcadas pela dinâmica temporal de processos objectivos de persistência e de mudança. O problema colocava-se, portanto, na possibilidade de resposta por parte da Lógica, como teoria formalizante do *logos* humano. Georg von Wright observava, a este respeito, em 1968, que um aspecto característico do desenvolvimento da filosofia no pós-guerra era o crescente interesse em conceitos e em problemas respeitantes à *praxis* humana, numa nova atenção à dinâmica da mudança e da acção no nosso mundo. Ora, não é surpreendente que esse novo interesse da filosofia na *acção* humana, em geral, viesse também a reflectir-se, mais cedo ou mais tarde, na própria Lógica³².

Como observava, ainda, Georg Henrik von Wright (ao qual devemos os primeiros passos para a construção da lógica deôntica e de uma teoria lógica da acção), tratava-se da necessidade da Lógica superar os limites anquilosantes da sua representação *estática* do mundo, onde as coisas simplesmente possuem, ou não possuem, certas propriedades³³. Foi a esta concepção do devir que Prior se referiu como a intemporal “*tapestry view of time*”³⁴.

Acresce, por fim, que não apenas o pensamento humano põe em jogo *formas* lógico-temporais nas suas operações inferenciais, como é

³¹ Gardies, 1975, *La Logique du Temps*, 12-13.

³² “Traditional logic is, basically, a theory of the conceptual scaffolding of a *static* world. The logical study of the various forms of practical discourse [*scil.* tense-logic, logic of change and action], challenges the need for a theory of the conceptual framework of a *dynamic* world, a world of change and process”. Ora, “[a] characteristic feature of the post-war development in philosophy generally is an increased interest in the concepts and problems relating to human action (*praxis*). (...) It is not surprising that the new interest in the philosophy of human action should have become reflected in logic, too” (Wright, 1968, *The Logic of Practical Discourse*, 159 e 141).

³³ “[w]e could say that formal logic, as we know it today, is essentially the logic of a *static* world. Its basic objects are possible states of affairs and their analysis by means of such categories as thing, property, and relation. There is no room for *change* in this world. Propositions are treated as definitely true or false – not as now true, now false. Things are viewed as having or lacking given properties and not as changing from, say, red to not-red.” (Wright, 1963, *Norm and Action – A Logical Inquiry*, vii).

³⁴ Prior, 1996, *Some Free Thinking about Time*, 47.

mesmo possível descobrir formas ou “esquemas especificamente temporais de raciocínio”³⁵.

Já em 1903, observava Charles Peirce: “[o] tempo foi habitualmente considerado pelos lógicos como um tema ‘extra-lógico’. Nunca partilhei essa opinião. Todavia, pensei que a lógica não tinha ainda alcançado um estado suficiente de desenvolvimento para que a introdução de modificações temporais nas suas formas não conduzissem a uma grande confusão”³⁶.

Este foi justamente o objectivo que Arthur Prior assumiu para o seu trabalho, a partir dos anos 50 do século XX: procurar construir uma ‘*lógica temporal flexionada*’ (*tense logic*), i.e., uma lógica construída a partir de proposições nas quais o tempo é expresso pelas flexões temporais dos verbos no presente, no passado e no futuro (e exceptuando, por conseguinte, o infinitivo, o particípio e o gerúndio). Adoptando, como sua linguagem própria, as noções relacionais da série temporal A (na classificação de McTaggart), esta lógica passará a trabalhar com proposições temporais susceptíveis de mudar de valor de verdade, e de representar, por esta via, a realidade objectiva do devir enquanto processo de mudanças.

Nesta ordem de ideias, será fácil compreender que Prior se tenha oposto ao *artifício conceptual* de uma linguagem lógica na qual os sujeitos se transformam em ‘partes temporais’ de objectos, e em que a noção de mudança desapareça para dar lugar a uma representação de meras ordenações de ocorrências estáticas e numericamente distintas entre si. Assim, diz Prior: “um dos objectivos da lógica temporal flexionada é, precisamente, facilitar o discurso acerca dos objectos persistentes, e outro dos seus objectivos é, precisamente, evitar a introdução de pseudo-entidades como ‘eu-em-*t*’, ‘eu-em-*t*’, etc; por conseguinte, um lógico das flexões temporais não irá encarar a frase ‘eu era mais magro do que sou agora’ como a expressão de uma comparação entre tais entidades. Com efeito, ‘eu era mais magro do que sou agora’, não significará, certamente, que ‘foi o caso que (eu sou mais magro do que sou)’, uma vez que isso *nunca* foi o caso”³⁷.

³⁵ Gardies, 1975, *La Logique du Temps*, 12-13.

³⁶ Peirce, 1967, *Collected Papers*, vol. IV, § 523, 406.

³⁷ Prior, 1967, *Past, Present and Future*, 169-170.

De facto, se bem que os “objectos individuais tenham, em geral, partes [espaciais] – os homens têm braços, pernas, e por aí adiante –, e estas partes sejam, elas mesmas, objectos de um certo tipo, nós *não* dizemos que os objectos individuais têm partes ou fases *temporais*, da mesma forma que os processos e as histórias as têm. A minha infância, por exemplo, não é uma parte minha, se bem que seja uma parte da minha história; e não é igualmente o caso que uma parte minha foi um rapaz na Nova Zelândia, ao passo que outra parte minha é um homem em Inglaterra; sou *eu* que fui aquele rapaz, e sou *eu* – o mesmo eu – que sou o homem”³⁸.

Prior defenderá, assim, a construção de uma linguagem temporalmente dinâmica dos objectos, enquanto ‘substâncias’³⁹, por oposição à linguagem intemporal ou temporalmente estática das partes temporais ou dos acontecimentos pontuais no espaço-tempo⁴⁰. Se as partes temporais ou os acontecimentos não persistem, nem mudam, mas simplesmente ocorrem, é claro que uma lógica fundada em tais entidades só poderá ser estática. Mas os acontecimentos são apenas as *acções* exercidas e sofridas pelos objectos, e não coisas suplementares a ser adicionadas às coisas⁴¹. Por isso é que “aquilo que parece ser um discurso sobre mudanças em acontecimentos é, na verdade, uma forma apenas mais retorcida de falar acerca de mudanças em coisas”⁴².

Como já havia observado J.J.C. Smart, “as coisas mudam, os *acontecimentos* ocorrem”⁴³. De facto, se acontecimentos apenas ocorrem, eles não persistem, e se não persistem tão-pouco poderão mudar. Este foi,

³⁸ Prior, 1968, *Papers on Time and Tense*, 78-79.

³⁹ “In a logic with tenses, it is natural to let them stand for the ‘things’ of ordinary speech, that is, ‘substances’ in the old sense, or what W. E. Johnson calls ‘continuants’, objects such that we can say of each one of them that once it had such and such properties and did and suffered such and such things, that now it – *the very same object* – has such and such other properties and does and suffers such and such other things, and in the future it – *the very same object* – will have different properties again, and do and suffer different things” (Prior, 1968, *Papers on Time and Tense*, 78).

⁴⁰ Cf. Prior, 1976, *Thank Goodness That’s Over*, 78-84.

⁴¹ “events are just what what things do and what happens to them. The truth that I once fell out of a boat is not a truth about a falling-out-of-a-boat, but a truth about me, and about a boat. To say that this event is no longer present but only past is simply to say that although I *was* once falling out of a boat, I am not falling out of one now. I am a real object, and I did really fall, but my falling is not an additional real object” (Prior, 1996, *A Statement of Temporal Realism*, 45).

⁴² Prior, 1968, *Papers on Time and Tense*, 10-11.

⁴³ Smart, 1949, *The River of Time*, 216.

aliás, um dos problemas no célebre argumento de McTaggart contra a realidade daquilo que ele considerou ser o 'tempo' – a saber, conceber os acontecimentos como os candidatos a serem os sujeitos da mudança. O segundo problema foi o de conceber o tempo, ele mesmo, como uma *propriedade*. McTaggart definiu a mudança como a posse sucessiva, por partes dos acontecimentos, dos próprios momentos temporais: o acontecimento *a* muda, porque *a* começa por *ser futuro*, depois *é presente* e finalmente *é passado*. O terceiro problema foi o acolhimento da perspectiva eternalista, segundo a qual os acontecimentos persistem e mudam, à medida que se deslocam ou devêm no 'rio' do tempo, desde o mais longínquo futuro, até chegarem ao presente e, depois, afastarem-se para um passado cada vez mais remoto. Na verdade, com o seu argumento, McTaggart apenas conseguiu provar a irrealidade das suas próprias noções de tempo e de mudança.

Mas de entre este grupo de problemas, deve sublinhar-se, a meu ver, a clássica indistinção (não exclusiva de McTaggart) entre tempo e devir – indistinção, essa, que explica a reiteração de equívocas expressões como '*a passagem do tempo*' e de ininteligíveis questões como '*o tempo flui?*', o '*tempo é real?*', "*o tempo é estático ou dinâmico?*", se e quando lidas num sentido literal, e não no seu sentido meramente figurativo ou metafórico. Com efeito, o tempo não está, ele mesmo, em devir. Se o tempo se movesse, poderíamos naturalmente colocar a hipótese de ele abrandar e acelerar o seu ritmo de passagem, e de perguntar a que velocidade se move o tempo. Mas isso seria atribuir ao tempo uma certa quantidade de movimento num certo período de tempo, necessariamente medido por relação a um outro tempo – uma espécie de *super-tempo*. Este problema não é, todavia, inédito. Já Aristóteles havia observado, na *Física*, que o tempo não passa, nem se move, porque tempo não é, ele mesmo, susceptível de ser medido pelo tempo⁴⁴. A meu ver, é por isso importante ter bem clara a distinção entre:

– o *devir*, concebido como *processo causal de mudanças*, através do qual se operam as substituições de propriedades contrárias que, sob a mesma relação, se configuram como incompatíveis, ou mutuamente exclusivas, num mesmo sujeito; e

⁴⁴ Cf. Prior, 1968, *Changes in events and changes in things*, 1-14; Prior, 1958, *Time after time*, 244-246; e Aristóteles, 1957-1960, *Physics*, IV, 220b e segs.

– o *tempo*, concebido como *estrutura* que permite relacionar qualitativamente (em termos das noções distinguidas nas séries temporais A e B) e medir (em termos numéricos) os momentos desse devir, em termos de durações, intervalos e taxas de variação⁴⁵.

2.1 | A forma lógica das frases temporais

É no contexto da possibilidade de construção de uma lógica temporal que se torna fundamental enfrentar o problema da determinação da ‘forma lógica’ das frases temporais singulares. Como é bom de ver, “não se trata de uma mera escolha entre notações”, mas de uma escolha “entre genuínas alternativas filosóficas”⁴⁶. Como observou Mark Johnston, o problema da *existência temporal*, tal como ele é *formalmente* representado, consiste em “explicar o papel da qualificação temporal nas nossas predicções de mudança, onde explicar não significa apenas optar por um certo estilo na introdução dos símbolos ‘*t*’ e ‘*t**’, mas defender as concepções que acerca das propriedades, da natureza do tempo e da natureza dos individuais persistentes, justificam esse estilo”⁴⁷.

Dentre as múltiplas *formas* de analisar as frases temporais singulares⁴⁸, merecerá a pena destacarmos os seguintes quatro tipos de determinação das formas lógicas, porque elas atribuem diferentes *funções* ao tempo no quadro do discurso proposicional, incidindo, ora no *sujeito*, ora no *predicado*, ora na própria operação lógica de *predicação*.

1. ‘*a-em-t* é F’ ou ‘F(*a-em-t*)’. Esta forma lógica é defendida pelos defensores da ontologia quadri-extensionalista e da teoria perdurabilista da persistência. O indicador temporal incide sobre o sujeito, denotando, não um

⁴⁵ Cf. Barreau, 1988, *Temps et devenir*, 5-36.

⁴⁶ Cf. Poidevin, 1991, *Change, Cause and Contradiction. A Defense of the Tenseless Theory of Time*, p. 62; e Hawley, 2001, *How Things Persist*, 9-24.

⁴⁷ Johnston, 1987, *Is There a Problem about Persistence?*, 115.

⁴⁸ Por exemplo, para além dos quatro tipos de análise aqui descritos, dois outros foram propostos: 1) a análise pela *datação*: o tempo é expresso por uma data específica, permitindo intemporalizar a operação de predicação (esta foi a opção tomada por Frege, no seu ensaio *Der Gedanke*); e 2) a análise ‘espécime-reflexiva’ (*‘token-reflexive analysis’*) do tempo gramatical: este tipo de análise também permite intemporalizar o tempo gramatical, introduzindo datas e uma referência ao tempo do acto locutório. Esta última abordagem foi originalmente proposta por Reichenbach, em *Elements of Symbolic Logic* (§§ 50 e segs.). Para uma crítica de Prior a estas duas abordagens, vide Prior, 1976, *Thank Goodness That’s Over*, 78-84; e Prior, 1968, *On spurious egocentricity*, 15-25.

objecto duradouro que no curso da sua evolução conserva a sua identidade numérica própria, mas uma ‘parte temporal’ de um objecto (*‘a-em-t’*);

2. *‘a é F-em-t’*. O tempo incide directamente sobre as propriedades significadas pelos **termos gerais** dos predicados lógicos, indexando-as a um momento temporal específico⁴⁹;

3. *‘R_F(a, t)’*. Os predicados são representados como *relações* diádicas entre tempos e objectos⁵⁰;

4. *‘a é-t-mente-F’*. O tempo é expresso sob a forma de um *advérbio* que incide sobre a operação de *predicação*⁵¹.

Mark Johnston, tal como Prior, defendeu esta última solução, apresentando-a, porém, como uma temporalização da *cópula*⁵². É importante referir esta perspectiva particular, porque ela permitir-nos-á compreender a especificidade da natureza da opção *adverbial*. Naturalmente, esta subtilidade não altera a natureza da opção; todavia, ela cria ambiguidades e problemas desnecessários. Em primeiro lugar, é indiscutível a existência de frases predicativas no qual o nexos proposicional não é fabricado por uma cópula, enquanto terceiro elemento distinto do sujeito e do termo geral que significa uma propriedade. Isto sucede em todos os casos em que o verbo desempenha, ele mesmo, a função de signo da propriedade (por ex.: ‘O Miguel corre’). Ora, esta é uma questão irrelevante. Como já Aristóteles observara, Mark Johnston nota que, nestes casos, bastará reescrever a frase, decompondo o verbo em *cópula* e *participio*⁵³. Porém, isto só demonstra que a presença de uma cópula não é absolutamente necessária para a realização do nexos predicativo. E esta realidade é por si só suficiente para deixarmos de ver na cópula o ‘segredo’ do nexos predicativo. Acresce que esta teoria da cópula vem desafiar (desnecessariamente) a análise bipartida da estrutura proposicional elementar, estabelecida na lógica contemporânea por Frege: função proposicional (estrutura insaturada) e argumento. Quando se diz que a

⁴⁹ Wilson, 1955, *Space, Time and Individuals*, 589-98; e Wilson, 1956, 46-8.

⁵⁰ Cf. Mellor, 1981, *Real Time*, 111-14; e Inwagen, 1990, 245-55.

⁵¹ Lowe, 1988, *The Problem of Intrinsic Change: Rejoinder to Lewis*, 72-77; Lowe, 1998, *The Possibility of Metaphysics*, 94-95 e 129-36; Haslanger, 1989, *Endurance and Temporal Intrinsics*, 119-25.

⁵² Cf. Johnston, 1987, *Is There a Problem about Persistence?*, 107-35.

⁵³ Johnston, 1987, *Is There a Problem about Persistence?*, 128. Como já observara Aristóteles, “there is no difference between saying that a man *walks* and saying that a man *is walking*.” (*Aristotle – Categories and De Interpretatione*, 21b 9-10, p. 60 – itálicos introduzidos).

expressão adverbial modifica o predicado, isto quer dizer que ela modifica a *função predicativa* do predicado, e *não* a função semântico-lexical do termo geral (que remete para uma propriedade).

Com efeito, por si mesmos, os termos gerais não são predicados, mas significam apenas conceitos. Os termos gerais só adquirem a natureza de predicados quando inseridos no contexto da conexão sintáctico-semântica do discurso predicativo. É neste contexto que os termos gerais ascendem do nível meramente *semântico-lexical* de significação para o nível *alético* do discurso proposicional susceptível de ser verdadeiro e falso. Na perspectiva adverbial de Arthur Prior, o tempo lógico é, pois, concebido como uma modalidade *interna* à própria *operação de predicação*⁵⁴.

De acordo com a primeira opção (1), em cada momento temporal existe uma nova entidade numericamente distinta da que lhe antecede e da que lhe sucederá. Não são, assim, os diferentes sujeitos das frases temporais que *mudam*, mas é antes o putativo objecto quadri-extensional que, como um todo, mudará: ' $F(a\text{-em-}t_1) \ \& \ G(a\text{-em-}t_2)$ '.

Segundo a opção (2), são as próprias propriedades a serem temporalizadas (*properties as time-indexed*). Neste caso, afirmar que um objecto possui uma certa propriedade em t é afirmar que o objecto tem *intemporalmente* a propriedade $F\text{-em-}t$. A mudança será assim expressa: ' $F\text{-em-}t_1(a) \ \& \ G\text{-em-}t_2(a)$ '.

A solução (3) é dita relacional, porque as propriedades aparentemente tidas como monádicas são, na verdade, relações cujos relata são os sujeitos e os momentos temporais. Para expressar a mudança de a ser F num tempo t , e G no tempo t' , escrever-se-á: ' $R_F(a, t_1) \ \& \ R_G(a, t_2)$ '.

De acordo com a última solução (4), o tempo lógico incide, como vimos, sobre a função predicativa do predicado lógico, concebido como um operador modal, sob a forma de um operador proposicional. A mudança poderá ser, assim, expressa: ' $Fp \ \& \ Pp$ '; ou melhor: ' $P(Fa) \ \& \ F(Ga)$ '; ou ainda: ' $a \text{ é-}t_1\text{-mente-}F \ \& \ a \text{ é-}t_2\text{-mente-}G$ '.

Este foi o tipo de análise lógica defendido por Arthur Prior. O tempo deve ser representado por um *operador proposicional*, interpretando a modalidade, não do ponto de vista meta-linguístico, mas

⁵⁴ Neste aspecto, Prior reactualiza a concepção aristotélica acerca da natureza da modalidade. Como escreve Claude Imbert, "Aristote attribue la modalité au rapport entre choses réelles et secondairement à une proposition" (Imbert, 1965, *Connaissance de la logique ancienne*, 350).

da linguagem-objecto: a modalidade incide directamente, não sobre o signo (*de dicto*), mas sobre o seu significado (*de re*)⁵⁵.

Formados a partir do *presente* – se o *presente* é o tempo em que vivemos e produzimos os nossos discursos, o tempo será sempre o ponto de partida das flexões temporais⁵⁶ –, Prior introduzirá quatro operadores proposicionais:

P: Foi o caso que (pelo menos uma vez) – *p*

H: (até agora) Tem sido sempre o caso que – *p*

F: Será o caso que (pelo menos uma vez) – *p*

G: (a partir de agora) Será sempre o caso que – *p*

Ao tempo não é, aqui, atribuído o estatuto de argumento que deveria completar a proposição temporal como se esta fosse uma função proposicional – i.e., como ela fosse uma proposição incompleta (como defenderam Frege e Russell). Pelo contrário, retomando a perspectiva de Aristóteles e dos Estóicos, Prior defende que as proposições temporais são proposições *completas* e, por isso, susceptíveis de receber valores de verdade.

Isso significa que, para Prior, o tempo lógico não deve também ser concebido como uma função, cujo argumento seria a proposição – pese embora a notação utilizada omita essa diferença –, uma vez que o tempo *incide* sobre, modificando, a operação de *predicação* interna a uma frase declarativa. Ou seja, o tempo não é um predicado da totalidade da frase, tomada enquanto entidade linguística. O tempo é um operador que exprime o *modo* (temporal) segundo o qual um dado sujeito possui ou

⁵⁵ É, assim, compreensível a crítica que Prior dirige a Wilson, segundo o qual o *t* deveria atravessar o sujeito e a cópula para se fixar no atributo: “The thing means, not ‘The leaf was green-in-August’, but ‘The leaf was-in-August green’. The *t* in fact has not ‘crossed the copula’, but stopped at the copula; though it seems to me that *this bridge theory of the copula* is wrong anyway. Write ‘The leaf is green’ in the *modern way* as ‘ ϕx ’; here ‘*x*’ denotes the leaf, ‘ ϕ ’ means ‘is-green’, and there is no copula needed, but if we want one we can put ‘*S*’ for ‘It is the case that –’ before the whole. Then we can think of ‘at *t*’ as neither moving across the copula to the predicate, nor staying with the copula on the way to the predicate, but as moving across the predicate to the copula, and changing ‘It is the case that –’ to ‘It was the case in August that’ (Prior, *Thank Goodness That’s Over*, 80). Cf. Prior, 1957, *Tenses and Truth in the History of Logic*, 104-122.

⁵⁶ Como observa Susanne Niemeier, “[t]ense situates the profiled relationship of an utterance with respect to the speaker’s current conception of reality. We live in ‘current reality’, therefore the base of the tense system is the present moment of speech, or ‘speech time’” (Niemeier, 2013, *A Cognitive Grammar perspective on tense and aspect*, 31).

está privado de uma dada propriedade. O tempo funciona, assim, como um *advérbio*, representando uma modalidade *de re*: “pôr um verbo no passado ou no futuro é exactamente o mesmo que adicionar um advérbio à frase”⁵⁷.

Aliás, segundo Prior, nós tendemos a formar o passado gramatical através da modificação do presente gramatical, como por exemplo, ‘A Rainha Ana morreu’, em vez de dizermos ‘Foi o caso que a Rainha Ana morreu’, porque é difícil autonomizarmo-nos da estrutura proposicional clássica ‘sujeito-predicado’ ou ‘nome-verbo’⁵⁸.

Incidindo sobre a operação de *predicação*, o tempo lógico tem, como seu correlato ontológico, a *posse* de uma dada propriedade por um dado objecto. O tempo lógico representa, desta forma, o carácter temporalmente *dinâmico* dos *modos de ser* dos objectos, em função da instanciação e da inerência temporal das propriedades.

É nesta óptica que a Lógica temporal poderá expressar o devir (o processo de mudanças) real e objectivo das entidades persistentes, tal como ele é representado e expresso pelo pensamento e pela linguagem natural⁵⁹.

⁵⁷ Prior, 1968, *Changes in events and changes in things*, 13. Segundo Prior, “[w]hat we need here is not to introduce the *adjectives* ‘past’, ‘present’, and ‘future’ (...) What we need rather is to introduce *tenses* into logic, and these are not adjectives but modifications of verbs, quite like the modification affected by a negation sign. Just as the truth that I am *not* falling out of a boat may be expressed by writing ‘It is not the case that –’ in front of ‘I am falling out of a boat’, so the truth of that I *was* falling out of a boat may be expressed by writing ‘It was the case that –’ in front of ‘I am falling out of a boat’.” (Prior, 1996, *A Statement of Temporal Realism*, 46). Aliás, Prior chega mesmo a dizer: “it is only an historical accident that we generally form the past tense by modifying the present tense, e.g. by changing ‘am’ to ‘was’, rather than by tacking on an adverb” (Prior, 1968, *Changes in events and changes in things*, 8).

⁵⁸ Prior, 1968, *Changes in events and changes in things*, 10. Cf. Lowe, 2002, *Tense and Persistence*, 47-48.

⁵⁹ Cf. Rescher & Urquhart, 1971, *Temporal Logic*; Øhrstrøm & Hasle, 1995, *Temporal Logic. From Ancient Ideas to Artificial Intelligence*, sobretudo, cap. 2.5, 167-179; Barringer, Fisher, Gabbay, Gough (eds.), 2000, *Advances in Temporal Logic*; e Øhrstrøm e Hasle, 2006, A. N. Prior’s *Logic*, 399-446.

3 | Uma lógica temporal flexionada relativista

Mas como pensar a relação desta Lógica temporal com a TRR? Regressemos, para concluir, ao problema das relações entre a Lógica, a Física e a Ontologia.

Segundo Gerald Massey, se a TRR rejeita a noção de uma simultaneidade absoluta, afirmando o carácter relativo de cada ordenação temporal de fenómenos, que sentido fará procurar construir uma lógica temporal? Nenhum, porque esta estará “fundada em má física”⁶⁰. Com efeito, como se justificará, nestes termos, a construção de uma lógica temporal?

A meu ver, existem três tipos de respostas principais que poderão justificar um tal projecto, se bem que eles não se excluam necessariamente.

1. A primeira opção será considerar obsoleta a própria questão. Tendo adquirido o estatuto de ‘disciplina científica autónoma’ no curso da sua evolução histórica, a Lógica assumir-se-á hoje como uma “teoria formalizante do *logos* humano”, detendo como “seu objecto próprio, as leis e as regras estruturadoras da coerência intrínseca do pensamento e do discurso”, e visando determinar e formalizar as estruturas subjacentes às operações do raciocínio, do ponto de vista da sua *forma* e da sua *validade* (i.e., mediante a abstracção do conteúdo e da questão da verdade).

Nesta óptica, à Lógica *não* cumprirá o desígnio de visar (ou de reflectir, como espelho) as formas *ontológicas* da realidade, nem tão-pouco procurar revelar as formas *comuns* à realidade (*modi essendi*), ao pensamento (*modi intelligendi*) e à linguagem (*modi significandi*), reabilitando, por essa via, o projecto de uma *grammatica speculativa*, arduamente prosseguido ao longo dos séculos XIII e XIV. Ao invés, a Lógica trabalha com certas *formas* de *representação* e de *expressão* da realidade no pensamento e no discurso⁶¹. Ora, equacionada nestes termos, porque razão deveria a Lógica restringir-se, dentre as diversas formas de expressão proposicional – como as modais: diacrónicas, aléticas, deônticas, epistémicas –, somente àquela ou àquelas que mais se adequariam aos tipos de discurso e de representação elaborados por certas ciências naturais ou, mesmo, às ontologias por elas estabelecidas?

⁶⁰ Massey, 1969, *Tense logic! Why bother?*, 31.

⁶¹ Cf. Sardo, 2000, *Logos e Racionalidade. Na génese e estrutura da Lógica de Aristóteles*, 58-62.

A questão propriamente *lógica* que, com pertinência, se colocará a propósito de uma lógica temporal será, por conseguinte, a possibilidade de se construir uma lógica apta a dar conta das características e das funções especificamente temporais do pensamento e das línguas naturais – como é o caso paradigmático do ‘aspecto’, que tem o poder de associar informações temporais a modos de acção⁶² –, tendo sempre presente que “o objectivo final de uma formalização do tempo gramatical – como observaram Robert Martin e Frédéric Nef – é escolher as *propriedades formais* do tempo que geram um conjunto de fórmulas *válidas*, e tão próximas quanto possível do conjunto das fórmulas temporais aceites como plausíveis pelos falantes de uma língua”⁶³.

Se bem que Prior tenha sustentado a ideia de que a Lógica não se debruça, primeiramente ou exclusivamente, sobre símbolos ou sobre a linguagem, mas sobre a própria realidade⁶⁴, Prior não deixará de defender que nenhum tipo de discurso pode ser excluído, *a priori*, do domínio da Lógica, pelo facto de ele parecer ser menos estruturado, ou menos apto a ser formalizado, até porque isto trata-se, sobretudo, de “*uma questão de graus*”.

Nada pode, pois, substituir-se ao estudo dos diferentes tipos de discurso e à própria tentativa de os formalizar: “*try it out and see what happens. You can’t settle the question a priori*”⁶⁵.

⁶² Nas línguas naturais, o ‘aspecto’ tanto pode significar ‘eventos’ (situações dinâmicas télicas: processos culminados e culminações), como ‘estados’ (situações estáticas), ‘processos’ (eventos atéticos), ou ainda ‘pontos’ (como eventos temporalmente indivisíveis). Cf. Oliveira, 2003, Tempo e aspecto, 134-138. Para uma abordagem do tempo linguístico e do aspecto, do ponto de vista da ‘gramática cognitiva’, vide Boogaart e Janssen, Tense and Aspect, 803-828.

⁶³ Martin e Nef, 1981, Temps linguistique et temps logique, 10 (itálicos introduzidos).

⁶⁴ “Philosophy, including Logic, is not primarily about language, but about the real world”; “Formalism, i.e. the theory that Logic is just about symbols and not about things, is false. Nevertheless, it is important to ‘formalise’ as much as we can, i.e., to state truths about things in a rigorous language with a known and explicit structure. It is also necessary to pay attention to the structure of our language in order to expose and eliminate philosophical ‘pseudo-problems’ and in order to distinguish real objects from mere ‘logical constructions’” (Prior, 1996, A Statement of Temporal Realism, 45).

⁶⁵ “some subjects do in fact have *more order, more structure, more form*, than others – that some subjects are more capable than others of being handled by means of a formal symbolic calculus – and in these cases it is more proper than in others to speak of a ‘logic’ of the thing. (...) In any case the important point is that these things are a matter of degree, and the only way to discover whether a given field can be handle as a logic, that is as the subject of a calculus, and how far it can be so handled, is to try it out and see what happens. You can’t settle the question *a priori*” (Prior, 1976, What is Logic?, 128-129). Outra via de pesquisa é protagonizada pelo desenvolvimento das ‘lógicas naturais’. Por exemplo, a lógica natural construída por Jean-Blaise Grize apresenta-se como o estudo das *operações lógico-discursivas*

2. A segunda justificação reside na constatação *científica* de que, não apenas a TRR não é uma teoria completa do mundo físico, como ela não esgota a própria concepção física de tempo, existindo outros domínios da física que, em outras escalas e relativamente a outros domínios de aplicação, subscrevem a noção objectiva do devir e da sua ordenação temporal. Como observa Hervé Barreau, substituir o 'mito newtoniano' pelo 'mito minkowskiano', não só é cientificamente indesejável (como qualquer mito o é em ciência), como carece de fundamento científico: "a própria física torna este espaço-tempo [da relatividade] limitado na sua pretensão ontológica"⁶⁶. Se o espaço-tempo relativista fosse o conceito único e derradeiro do espaço e do tempo físicos, o que fazer da actual teoria cosmológica padrão (derivada, aliás, da relatividade geral), onde se trabalha com um tempo cósmico absoluto? Ou, ainda, que sentido teria encerrar definitivamente a mecânica quântica no espaço-tempo relativista, quando este último parece não dar conta da realidade física ao nível quântico da sua organização?⁶⁷

3. A terceira alternativa é conciliatória: mesmo aceitando-se a pertinência das duas primeiras respostas, pode tentar-se construir uma lógica temporal flexionada que se adeque à TRR.

Para se compreender um tal projecto importa, todavia, distinguirmos as duas diferentes ordens de questões que, sobre a natureza das *determinações temporais* no quadro da TRR, podem ser colocadas.

1) Em primeiro lugar há a questão relativa à objectividade do devir e da ordenação temporal dos acontecimentos (i.e., das mudanças) que constroem causalmente a persistência ou a *biografia de cada entidade individual*. Ora, esta objectividade é universalmente garantida pela TRR, em

que intervêm na elaboração de *esquematizações* na prática argumentativa e, por conseguinte, no contexto de relações dialógicas e através da linguagem natural. Cf. Grize, 1996, *Logique naturelle et communications*; e Grize, 2004, *Le point de vue de la logique naturelle: démontrer, prouver, argumenter*, 35-43. Para uma apreciação das diferentes perspectivas (monistas, pluralistas global e local, e instrumentalistas) acerca das relações (desde a eliminação até à integração unificada) entre os diferentes 'sistemas lógicos' existentes, *vide* Haack, 1995, *Philosophy of Logics*, 221-232.

⁶⁶ Barreau, 1988, *Temps et Devenir*, 23.

⁶⁷ Barreau, 1988, *Temps et Devenir*, 24. Acresce que a comunidade científica encontra razões para questionar e, mesmo, para rever certos aspectos da TRR (pelo menos, tal como esta foi originalmente formulada), em virtude dos problemas internos à própria Teoria da Relatividade, bem como das dificuldades empíricas com que ela se confronta. Cf. Selleri (ed.), 1998, *Open Questions in Relativistic Physics*.

virtude da relação de *causalidade* ser um *invariante* relativista, e em função da noção relativista de ‘tempo próprio’ (*Eigenzeit*; representado pelo símbolo τ). Com efeito, este é o tempo – distinto do ‘tempo-coordenada’ (*Zeitkoordinate*; representado pela variável t) – apto a fornecer uma medida *independente* de qualquer referencial. Se, num dado sistema referencial, o acontecimento B tiver sido causado pelo acontecimento A , a ordem temporal entre A e B permanecerá a mesma para qualquer outro sistema referencial. A ordem temporal entre acontecimentos causalmente relacionados é invariante – i.e., é independente das ‘perspectivas’ locais dos diferentes sistemas de referência⁶⁸.

2) Em segundo lugar, há o problema de se reconhecer a existência de *diferentes* ordenações temporais entre acontecimentos *não* causalmente relacionados entre si, por parte dos diversos sistemas de referência. Porém, neste caso é possível correlacionar e traduzir entre si as diferentes medições temporais, através das regras de transformações de Lorentz (nas quais se expressam as invariâncias da velocidade da luz e das leis da Física), sendo assim possível *conhecer* a natureza *objectiva* de tais medições e, bem assim, das distinções associadas entre o que é presente, passado e futuro para cada sistema de referência⁶⁹.

O projecto de construção de uma lógica temporal flexionada que se adegue, sob este segundo aspecto, à TRR, foi desenvolvido por Thomas Müller⁷⁰, a partir de uma perspectiva já ensaiada por Arthur Prior, em “Tense logic and the logic of earlier and later”⁷¹.

Se bem que a TRR recuse o carácter universal da noção de simultaneidade e da ordenação temporal dos fenómenos físicos não

⁶⁸ A natureza *estática* do tempo e a *irrealidade* do *dever*, ambas frequentemente atribuídas à TRR, são fruto de uma falha em reconhecer a coexistência destas duas noções distintas de tempo no seio da TRR. Cf. Arthur, 2008, Time Lapse and the Degeneracy of Time: Gödel, Proper Time and Becoming in Relativity Theory, 207-227; Dieks, 2006, Becoming, relativity and locality, 157-176; Lévy-Leblond, 1996, *Aux Contraires – L'exercice de la pensée et la pratique de la science*, 137-8; Barreau, 1980, Pour le centenaire d'Albert Einstein: Einstein et les concepts d'espace et de temps, 363; Barreau, 1998, *Le Temps*, 99-101; Whitrow, 1972, *What Is Time?*, 118; e Whitrow, 1980, *The Natural Philosophy of Time*, 348 e 352-355.

⁶⁹ Como Richard Arthur observa, “[t]he relativity of simultaneity entails that no one of these relative times can be privileged as the ‘actual time’, just as Gödel had argued. Nevertheless, each measure of duration is consistently related to any of the others by the Lorentz transformation formulas.” (Arthur, 2008, 209-210).

⁷⁰ Müller, 2004, The Language of Special Relativity, 1-9.

⁷¹ Prior, 1968, Tense logic and the logic of earlier and later, 116-134.

causalmente relacionados, é possível construir-se uma lógica temporal flexionada apta a representar as múltiplas medições e ordenações temporais feitas pelos diferentes sistemas de referência. A 'lógica dos pontos de vista', ensaiada por Prior em 1968, tornar-se-á, assim, uma 'lógica temporal dos sistemas de referência': "as fórmulas não modalizadas φ da LPV [lógica dos pontos de vista] devem ser interpretadas como '*Neste sistema de referência φ* ' (isto assemelha-se à lógica temporal, onde o simples φ é interpretado como '*É agora o caso que φ* ')"⁷². Cada contexto locutório fornecerá, assim, o seu sistema de referência, como o seu parâmetro próprio de verdade.

Este projecto consiste, assim, em assumir as relações distinguidas pela série temporal A como relativas aos diferentes sistemas de referência. Todavia, com base nas regras de transição (transformações) de Lorentz para o tempo, qualquer observador pode conhecer o carácter objectivo do 'ponto de vista' de um outro observador, estabelecendo uma correlação entre as suas medições respectivas (a passagem de um sistema de referência para outro sistema de referência são mudanças gerais de perspectiva)⁷³:

"a objectividade, entendida como um acordo intersubjectivo entre diferentes sistemas de referência, é totalmente garantida: sabendo 'onde o meu interlocutor está', i.e., sabendo que transformação l conduzirá ao seu ponto de vista, o seu φ pode ser concebido como o meu $[l]\varphi$. Desta forma, poderá estabelecer-se um acordo universal sobre o que é presente, passado e futuro. Reconhecer a natureza indexical de uma determinada característica não significa que esta seja 'relativa'"⁷⁴.

Ao passo que a ordenação temporal de acontecimentos causalmente relacionados é absoluta e universalmente partilhável, no caso de acontecimentos *não* causalmente relacionados, a TRR oferece-nos um mundo de *múltiplas ordenações temporais* possíveis, se bem que objectivas,

⁷² Müller, 2004, 5 (itálicos introduzidos).

⁷³ "In relativity theory, points of view are naturally taken to be inertial frames, and changes between these frames are described by Lorentz transformations. Since we need to consider not only rotations and boosts, but also translations, we have to use the full inhomogeneous Lorentz group, which is also known as the Poincaré group. Thus, for every element (Lorentz transformation) l of the 10-parameter (proper orthochronic) Poincaré group P , we introduce a modal operator $[l]$." (Müller, 2004, 5).

⁷⁴ Müller, 2004, The Language of Special Relativity, 8; Cf., ainda, Müller, 2006, On the Problem of Defining the Present in Special Relativity: A Challenge for Tense Logic, 441-457; e Müller and Strobach, 2012, A letter on the present state of affairs. Prior, indeterminism and relativity 40 years later, 477.

porquanto ancoradas em diferentes sistemas de referência⁷⁵. Ora, a *legalidade física* dessas múltiplas objectividades, não apenas é ontologicamente afirmada, como é susceptível de ser universal e objectivamente *conhecida*, através das transformações de Lorentz.

É neste sentido que a lógica temporal flexionada dos ‘pontos de vista’ (dos diferentes ‘sistemas de referência’) poderá representar e expressar o tempo, tal como ele se manifesta na linguagem natural e em consonância com a TRR⁷⁶.

⁷⁵ Como já havia escrito Prior, “there is a logic of such functors as ‘It appears from a certain point of view that –’, and one could therefore give a good sense to talk about an infinity of different ‘apparent’ time-series. I suspect that the infinity of ‘local proper times’ which figure in relativistic physics amount simply to what appears from various points of view, or in various ‘frames of reference’, to be the course of events. And given how the course of events appears from a certain point of view, your relativistic physicist will be able to calculate how it will appear from certain other points of view.” (Prior, 1968, *Tense logic and the logic of earlier and later*, 133).

⁷⁶ Sobre as relações actuais entre a Lógica temporal flexionada e a Teoria da relatividade, bem como sobre o projecto particular de Thomas Müller, cf. Øhrstrøm e Hasle, 2006, *Modern Temporal Logic: The Philosophical Background*, cap. 8: *Tense Logic and Relativity*, 482-489.

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Cinco palavras para compreender Poincaré: Geometria, Caos, Simetria, Princípios, Convencionalismo

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Introdução: Porquê Cinco Palavras

Henri Poincaré (1854-1912), ao longo do seu percurso universalista na ciência e na filosofia estabeleceu ligações inesperadas e surpreendentes entre os seus temas de estudo e a geometria. Mesmo quando não estava a investigar nessa área a sua imaginação parecia conduzi-lo inelutavelmente a ideias geométricas. Assim, se fosse necessário eleger uma única palavra para compreender Poincaré, "geometria" seria certamente a escolhida. Na sua obra, "geometria" está associada a outras palavras tais como "caos", "simetria" ou "convencionalismo", que surgiram nalguns dos seus trabalhos com repercussões importantes na ciência e na filosofia. Nos próximos cinco parágrafos procura-se caracterizar o trabalho de Poincaré precisamente com base nessas palavras e nas suas relações com a geometria.

Geometria e Invenção

Poincaré afirmava que o cerne da sua filosofia do conhecimento era o tema da invenção científica, embora raramente se tivesse dedicado a analisar a génese das suas próprias ideias¹. No entanto, o matemático e filósofo francês tratou a invenção matemática em conferências e publicações destinadas ao grande público, talvez por pensar que essa seria uma forma de comunicar as suas ideias científicas àqueles que dificilmente entendem o significado e o alcance dos seus trabalhos matemáticos. A invenção matemática pode realmente ser um caso particularmente

¹ Paty, 1999, 251.

interessante de invenção, dadas as numerosas ligações da matemática com outras formas de conhecimento científico ou artístico e que permitem o acesso a conceitos difíceis de comunicar a quem não conhece o formalismo matemático. Talvez porque o tema da invenção é aliciante para os que refletem sobre o conhecimento no âmbito da filosofia ou da psicologia, Poincaré falou dele aos psicólogos na conferência proferida no *Institut Général Psychologique*, em 1908, publicada no *Bulletin de l'Institut Général de Psychologie*² e julgada suficientemente representativa para ser também publicada em *Science et méthode*³.

Embora “não se dedicasse frequentemente a analisar a gênese das suas ideias”⁴, ao falar sobre invenção matemática Poincaré descreveu com bastante pormenor a descoberta das funções “fuchsianas” ocorrida trinta anos antes⁵, logo no início da sua carreira. O relato descreve as várias fases do trabalho de investigação, até que se detém num passo crucial do processo criativo, aqui transcrito na língua original:

« Au moment où je mettais le pied sur le marchepied, l'idée me vint, sans que rien dans mes pensées antérieures parût m'y avoir préparé, que les transformations dont j'avais fait usage pour définir les fonctions fuchsiennes étaient identiques à celles de la géométrie non euclidienne. »⁶

“O momento em que põe o pé no degrau do autocarro” é pois apresentado por Poincaré como o de uma “revelação” que ocorreu após várias semanas de reflexão e de cálculo matemático. Como ele próprio afirmou sobre a sua descoberta, “o que primeiro o impressionou foi a aparência de iluminação súbita, manifestamente sinal de um longo trabalho inconsciente anterior cujo papel parece incontestável na invenção matemática”⁷.

Dada a importância da invenção de Poincaré ocorrida no tal momento crucial “em que põe o pé no degrau” este incidente tornou-se um objecto de relato citado em vários artigos da história da matemática⁸,

² Poincaré, 1908-a.

³ Poincaré, 1908-b.

⁴ Paty, 1999, 251.

⁵ Gray & Walter, 1997, 1-25.

⁶ Poincaré, 1908-b, 43-63.

⁷ Poincaré, 1908-b, 43-63.

⁸ Gray, 1981, 85.

embora seja apenas um pormenor, relevante para a questão da invenção, mas não propriamente para a história. Mas talvez se possa dizer que os historiadores são tocados por esse relato simples mas pertinente sobre a invenção científica feito por um grande matemático.

Resta saber o que aconteceu de tão fundamental para a matemática no momento em que Poincaré pôs o pé no degrau, ou seja, qual a importância de que as transformações usadas por Poincaré fossem idênticas às da geometria não euclidiana? Sem exagerar, pode dizer-se que a partir desse momento a geometria não euclidiana começou a sair da obscuridade para vir ocupar um grande papel. De facto, até então a geometria não euclidiana não era usada em quaisquer aplicações, constituindo essencialmente “uma curiosidade matemática”. O próprio Poincaré o afirma: “Essa geometria (não euclidiana) que parecia apenas um jogo da mente interessante para um filósofo, mas sem utilidade para um matemático. Percebeu-se que não era assim...”⁹

O que desencadeou a mudança de estatuto da geometria não euclidiana foi, como afirma Scott Walter, “o trabalho de um jovem matemático assistente na faculdade de ciências de Caen, Henri Poincaré”¹⁰. Como consequência da descoberta, o interesse dos matemáticos pela área aumenta consideravelmente no início da década de 1890, pois o trabalho de Poincaré havia evidenciado as possibilidades da geometria não euclidiana, considerada até então interessante apenas do ponto de vista filosófico¹¹.

No seu artigo sobre invenção, Poincaré dedicou várias páginas a descrever os progressos na investigação matemática do problema de equações diferenciais que lhe havia sido proposto pelo seu professor Charles Hermite (1822-1901) e que levou às funções fuchsianas e à geometria não euclidiana. Esse trabalho destinava-se a concorrer ao prémio oferecido pela Academia das Ciências de Paris em 1878, prémio que naturalmente seria o que de melhor podia desejar um jovem matemático de vinte e quatro anos que, como o próprio relata em *Science et Méthode*, fazia ainda o seu serviço militar. No entanto, o júri decidiu atribuir apenas o segundo prémio¹² ao trabalho que haveria de tornar-se famoso não só

⁹ Poincaré, 1928, I, X – XI.

¹⁰ Walter, 1996, 95.

¹¹ Gray & Walter, 1997, 1–25.

¹² Gray & Walter, 1997, 18.

pelas equações diferenciais mas sobretudo pela ligação que estabeleceu entre elas e as geometrias não euclidianas. Esta ligação entre Análise Matemática e Geometria permitiu desde logo a Poincaré, segundo Gray e Walter, “avançar rapidamente na sua investigação”¹³.

A importância da descoberta de Poincaré fica perfeitamente justificada com os artigos de Scott Walter e Jeremy Gray que têm uma extensa e prestigiada obra na história da matemática. Mas do ponto de vista da invenção é indispensável fazer uma observação relativamente à ligação estabelecida por Poincaré entre as equações diferenciais e a geometria logo no seu primeiro trabalho de investigação. Essa capacidade de estabelecer conexões entre diferentes domínios viria a confirmar-se ao longo da sua carreira de investigador, sendo que a geometria foi indiscutivelmente uma área privilegiada do ponto de vista dessas conexões. Sendo assim, ao tratar a invenção em Poincaré, tem todo o sentido começar pela palavra que aparece ao longo de todo o seu trabalho de investigação: “geometria”¹⁴

Simetria e a Noção de Grupo

Quando se fala de geometria a propósito de Poincaré surgem outras palavras obrigatórias, tal como “simetria”, que por sua vez está associada à teoria de grupos uma área pujante da matemática que tem relações com a teoria de números, a análise matemática e a geometria. A noção de simetria é usada também noutras disciplinas, como por exemplo em arte¹⁵, em física¹⁶ ou em cristalografia¹⁷. A importância da descoberta das funções fuchsianas por Poincaré deve-se em grande parte, tal como foi referido, à relação que esse trabalho permitiu estabelecer entre essas funções e a geometria não euclidiana, mas também ao impacto da descoberta noutras áreas, em particular em teoria de grupos¹⁸.

Embora muitas das questões de teoria de grupos sejam complexas, é possível dar exemplos de aplicação simples¹⁹, como por exemplo o da

¹³ Gray & Walter, 1997, 15.

¹⁴ Serra, 2014, 153-166.

¹⁵ <http://cfcul.fc.ul.pt/equipa/docs/iserra/simetria.pdf>.

¹⁶ Sternberg, 1994.

¹⁷ Hiller, 1986.

¹⁸ Gray, 1984, 10.

¹⁹ <http://cfcul.fc.ul.pt/equipa/docs/iserra/simetria.pdf>.

simetria de um cubo ou de um tetraedro²⁰. Mas os casos elementares que ilustram visualmente o estudo da simetria não permitem explicar o trabalho de Poincaré com os grupos. No entanto, o grande divulgador da ciência que foi Richard Feynman (1918-1988) em *O que é uma lei física* deu um exemplo simples da ligação estabelecida por Poincaré entre os grupos e a física: "Foi Poincaré quem teve a ideia de analisar o que se pode fazer às equações da física sem as modificar. Foi o primeiro a reparar na simetria das leis da física"²¹.

Feynman foi também o primeiro a sintetizar, usando uma só frase, esta grande contribuição da matemática para a física: a análise das leis da física do ponto de vista da sua simetria.

A simetria das leis da física de que fala Feynman pode traduzir-se pela ideia de invariância: as leis do movimento ou as equações que traduzem o movimento, são as mesmas em dois referenciais, em que um deles se desloca com movimento retilíneo e uniforme em relação ao outro, ou seja, são invariantes em todos os "referenciais de inércia". Se, por exemplo, viajarmos num comboio em movimento retilíneo e uniforme (ou seja, um comboio que constitui um "referencial de inércia" relativamente ao solo) podemos garantir que os copos e os pratos colocados sobre a mesa da carruagem restaurante se comportam como se estivessem na nossa casa, ou seja, não caem da mesa se ninguém lhes tocar. Sabemos que já não é assim, por exemplo, quando o comboio trava, sobretudo se travar bruscamente. Nesse caso, o comboio deixa de ser um referencial de inércia. Para traduzir esta realidade dizemos que as leis do movimento são invariantes em referenciais de inércia, ou usando outra terminologia, as leis do movimento são simétricas. A simetria no espaço, no tempo, ou noutras variáveis, está normalmente associada a uma propriedade invariante. As transformações que preservam essa propriedade definem um grupo, chamado grupo de simetria.

A noção de grupo na física, onde foi usada pela primeira vez por Poincaré, tornou-se extraordinariamente importante²². Mas, para além da física, o trabalho de Poincaré tem inúmeras intersecções com a teoria de grupos, como Jeremy Gray pôs em evidência em *Poincaré e a ideia de grupo*, onde começa por afirmar que apesar da ideia de grupo estar quase sempre

²⁰ Armstrong, 2014.

²¹ Feynman, 1989, 121.

²² Sternberg, 1994.

presente e ser fundamental nas suas investigações, Poincaré nunca trabalhou propriamente em teoria de grupos²³. No entanto, as suas ideias sobre os grupos eram já bem definidas quando da invenção das funções fuchsianas. No artigo que submeteu ao já referido concurso da Academia das Ciências de Paris onde apresentou esse estudo, ele observa: “O que é, de facto uma geometria? É o estudo de um grupo de operações formado pelos deslocamentos que uma dada figura pode sofrer sem se deformar”²⁴. Esta ideia é fácil de visualizar num plano: uma figura geométrica, por exemplo um triângulo, pode deslocar-se no plano através de rotações e translações mantendo a sua forma. Essas operações de rotação e de translação formam um grupo, o que quer dizer que obedecem a certas propriedades, como por exemplo a “propriedade associativa”, tal como acontece com as operações de cálculo elementar, a adição e a multiplicação. No exemplo das rotações ou translações de uma figura geométrica a ideia de grupo está pois associada ao espaço onde vivemos que é o “espaço geométrico”. Mas o conceito de grupo é mais geral²⁵ e viria a ser aplicado a diversas situações na ciência, em particular às equações e à física, aplicação onde Poincaré teve um papel decisivo. A noção de grupo na geometria foi estabelecida em 1872 através do *Erlangen program*²⁶ de Felix Klein (1849-1925) que o considerou como o princípio organizador da geometria, o que significava, em particular, que cada geometria euclidiana ou não euclidiana tinha um grupo associado. Poincaré tinha adquirido essa noção já em 1880, quando publicou o seu trabalho sobre as funções fuchsianas, embora não se saiba qual a origem desse conhecimento²⁷.

Procurar qual a origem das noções que Poincaré talvez não seja muito importante do ponto de vista da história da matemática mas pode contribuir para esclarecer a questão invenção em Poincaré. Nessa perspectiva não é possível deixar de reparar que o artigo *Poincaré and the idea of a group*²⁸ caracteriza muito bem precisamente a inventividade de Poincaré no que se refere ao “grupo”. De acordo com Jeremy Gray, embora não se interessasse pela teoria de grupos, Poincaré soube encontrar as grandes potencialidades desses objetos matemáticos e usá-los de forma

²³ Gray, 2012, 178.

²⁴ Gray & Walter, 1997, 15.

²⁵ [https://pt.wikipedia.org/wiki/Grupo_\(matematica\)](https://pt.wikipedia.org/wiki/Grupo_(matematica)).

²⁶ https://en.wikipedia.org/wiki/Felix_Klein

²⁷ Gray & Walter, 1997, 15.

²⁸ Gray, 2012.

inovadora e até surpreendente para os que sabiam mais do que ele sobre a questão. Os relatos de Gray²⁹ e de Gray e Walter³⁰ põem claramente em evidência a forma ousada como Poincaré usou os grupos, tendo em conta que essa utilização estava associada a alguma ignorância na área da matemática correspondente³¹. Mas apesar dessa suposta ignorância, as ideias geométricas de Poincaré foram extremamente originais e deram ao conceito de simetria um novo sentido. Simetria é pois uma palavra que representa bem todo o seu trabalho que se relaciona com os grupos e mais uma vez traduz, tal como a palavra geometria, o cruzamento de diferentes saberes.

Convencionalismo, Geometria e “Princípios” da Física

O “convencionalismo geométrico”, tal como é chamada a filosofia da geometria de Henri Poincaré, surgiu provavelmente como resultado da reflexão filosófica em torno dos seus primeiros trabalhos de investigação, em que estabeleceu a relação entre equações diferenciais e geometria não euclidiana³². Mas as suas ideias sobre a geometria, que desde a sua elaboração até ao presente têm desencadeado tão diferentes interpretações e tantas questões³³, vieram a ser publicadas só alguns anos depois, em 1887 no *Bulletin de la Société Mathématique de France*³⁴ e em 1891 na *Revue générale des Sciences Pures et Appliquées*³⁵, artigo que foi reproduzido posteriormente em *Science et Hypothèse*³⁶.

A ideia que está na base do convencionalismo geométrico de Poincaré é o da “equivalência” entre geometrias, por exemplo, entre euclidiana e não euclidiana. A equivalência entre diferentes geometrias fundamenta-se no trabalho de vários matemáticos, começando pelo dos inventores das geometrias não euclidianas³⁷, por volta de 1830. Félix Klein

²⁹ Gray, 1981.

³⁰ Gray & Walter, 1997.

³¹ Gray, 1981, 298.

³² Rougier, 1920.

³³ Giedymin, 1992, 423-443.

³⁴ Poincaré, 1887, 203-216.

³⁵ Poincaré, 1891, 769-774.

³⁶ Poincaré, 1902.

³⁷ https://en.wikipedia.org/wiki/Non-Euclidean_geometry

no chamado *Erlangen program*³⁸ (1872) apresenta a geometria como o estudo das propriedades de um espaço que é invariante para um dado grupo de transformações, uma concepção que viria a influenciar profundamente a evolução da matemática. Klein mostrou que as propriedades essenciais de uma dada geometria são representadas por esse grupo de transformações, um conhecimento que Poincaré adquiriu desde muito cedo na sua vida de investigador e em que, precisamente, se fundamentaria para elaborar a sua filosofia da geometria.

Para Poincaré, as diferentes geometrias são “equivalentes” pois distinguem-se apenas pelo grupo de transformações associado, sendo que é possível obter os mesmos resultados em matemática usando duas geometrias diferentes. A equivalência entre geometrias levou Poincaré a afirmar que a escolha de uma dada geometria para tratar um problema é apenas uma questão de “convenção” e que todos os cálculos que se fazem com geometria euclidiana podem ser feitos com a não euclidiana embora possa ser menos “cómodo”. O pensamento geométrico de Poincaré ficou sempre associado à palavra “convencionalismo” um termo que no entanto ele nunca usou, e que tem em geral sentido e contexto muito diferentes dos do seu convencionalismo geométrico³⁹.

O convencionalismo de Poincaré foi mal aceite pela comunidade de físicos e matemáticos, pois a ideia da equivalência das geometrias é incompatível com a existência de uma “verdadeira” geometria do espaço físico. De facto, se a escolha da geometria fosse uma questão de convenção e de comodidade de cálculo, essa escolha não estaria relacionada com a natureza dos fenómenos que ocorrem no espaço, ou seja, seria independente do mundo físico. Sendo assim, deixava sequer de ter sentido procurar estabelecer empiricamente qual a estrutura geométrica do espaço. Esta consequência das ideias de Poincaré não agradava nem a físicos nem a matemáticos cuja expectativa era a de poder estabelecer qual a geometria própria do espaço onde vivemos e onde ocorrem os acontecimentos humanos e científicos⁴⁰.

Na ciência, desde o Renascimento e até ao século dezanove, a ideia de que o espaço é euclidiano era tida como uma verdade inabalável. De facto, a geometria de Euclides e o método de Descartes que estavam

³⁸ https://en.wikipedia.org/wiki/Felix_Klein

³⁹ <https://en.wikipedia.org/wiki/Conventionalism>

⁴⁰ Walter, 1996, 90.

associados na física matemática, foram usados com sucesso para descrever matematicamente os problemas da mecânica e também os de outras áreas da física. Essa prática determinou decisivamente as concepções de espaço e geometria da maioria dos físicos e matemáticos do século XIX⁴¹. Até então, na ciência, o mundo era considerado como uma realização do espaço euclidiano, que por sua vez era concebido como uma entidade real descrita adequadamente pelo método cartesiano.

Neste quadro científico que implicava a física e a matemática, o convencionalismo geométrico de Poincaré não podia ser bem-vindo, pois desde logo era incompatível com a ideia de que o mundo é necessariamente euclidiano. No entanto, essa ideia que aparecia como constituindo a base da física matemática, a da existência de um mundo perfeito regido pela geometria euclidiana e pelo método cartesiano, viria a ser abalada por grandes transformações na física. Nesse contexto, instala-se uma profunda crise na física nos finais do século XIX, identificada aliás por Poincaré⁴² e por ele analisada no quadro do seu trabalho filosófico⁴³. Na sua análise, que podemos chamar estruturalista⁴⁴, Poincaré descreve a evolução da física matemática desde os tempos da sua emergência, no Renascimento, onde é dominada pela “física das forças centrais”, sendo depois substituída pela “física dos princípios”. Para além da sua interpretação original da evolução da física, que traduz não só um conhecimento aprofundado dessa matéria mas também espírito filosófico, Poincaré contribuiu para a evolução da própria física com ideias e resultados fundamentais, em particular na relatividade restrita. A questão não será no entanto aqui desenvolvida, embora seja importante na obra de Poincaré e a sua contribuição para a relatividade tenha desencadeado na história e filosofia das ciências uma enorme controvérsia relativamente à prioridade entre ele e Albert Einstein (1879-1955) no enunciado do princípio de relatividade⁴⁵. No entanto é imprescindível dizer que no seu trabalho sobre relatividade os grupos estão mais uma vez presentes e que a ideia inaugurada por Poincaré de associar a física com os grupos e a simetria frutificou de tal maneira que originou novos ramos na disciplina⁴⁶.

⁴¹ Torreti, 1978, 34.

⁴² Darrigol, 1995, 1.

⁴³ Poincaré, 1905, 187-205.

⁴⁴ Laydman, 2016.

⁴⁵ Darrigol, 2004, 614-626

⁴⁶ [https://en.wikipedia.org/wiki/Symmetry_\(physics\)](https://en.wikipedia.org/wiki/Symmetry_(physics))

A física aparece na obra de Poincaré também através da “convenção”, que justificava a existência da “física dos princípios”, e que se revelou um conceito adequado para combater a posição do “realismo ingênuo” de alguns cientistas segundo a qual a ciência descreve a realidade tal como ela é. Distanciando-se desse realismo ingênuo, vários “cientistas-filósofos” dos séculos XIX e XX⁴⁷, entre os quais Poincaré, construíram novas perspectivas epistemológicas sobre a natureza e a função das teorias científicas⁴⁸: positivismo, pragmatismo, convencionalismo ou instrumentalismo.

Em *Science et Hypothèse* Poincaré argumenta a favor do convencionalismo a propósito dos postulados da mecânica que “se possuem uma generalidade e uma certeza, que não existem nas verdades experimentais de onde eles provêm, é porque eles se reduzem em última análise a uma convenção que temos o direito de fazer porque estamos certos que nada a irá contradizer”⁴⁹. No entanto, para Poincaré, esta convenção não é arbitrária, como acontece noutras convenções, por exemplo as sociais. De facto, uma dada convenção é adotada “porque certas experiências mostraram que ela é cómoda”. Pode dizer-se que é um convencionalismo de origem empirista mesmo que esta afirmação pareça paradoxal.

Tal como no convencionalismo geométrico, também no físico aparece a palavra “comodidade” para justificar as convenções adotadas. Vários autores consideram que é a investigação de Poincaré dos fundamentos da geometria que está na origem do seu convencionalismo físico⁵⁰. Por exemplo, Heinzmann afirma que “Poincaré utiliza na mecânica o mesmo procedimento que na geometria para passar das leis empíricas, tomadas como hipóteses gerais, aos princípios, integrando elementos explicitamente convencionais”⁵¹. A noção de convenção na física permite também, em Poincaré, justificar a substituição de uma teoria física por outra⁵².

⁴⁷ Wagner, 2002, 37.

⁴⁸ Serra, 2012, 415-426.

⁴⁹ Poincaré, 1902, 136.

⁵⁰ Giedymin, 1977, 271.

⁵¹ Heinzmann, 2017, 9.

⁵² Poincaré, 1905, 198-215

No contexto das “cinco palavras para compreender Poincaré” é pertinente sublinhar que a generalização do convencionalismo geométrico à física veio estabelecer uma ponte entre geometria e física através da filosofia. Na obra de Poincaré, a geometria aparece então ligada não só a outras áreas da matemática, mas também à física, e de duas formas: por um lado através dos grupos e da simetria e por outro através do convencionalismo onde os “princípios” encontraram um lugar e uma justificação.

Caos e Geometria

Caos, a última das cinco palavras aqui escolhidas “para compreender Poincaré”, é também um caso importante das ligações da geometria. Entre todas as questões fundamentais que Poincaré trabalhou, o caos, cronologicamente, foi também a última a revelar-se importante.

De facto, a sua contribuição para essa questão tão fundamental na ciência de hoje, só foi compreendida e utilizada a partir dos anos 1960, cerca de oitenta anos depois das publicações de Poincaré. A descoberta do caos ocorreu quando o matemático estudava a possibilidade de resolver o chamado “problema dos três corpos”⁵³ que, embora complicado do ponto de vista das equações matemáticas, pode ser enunciado de uma forma simples: qual o movimento de três corpos no espaço, (por exemplo, o sol, a terra e a lua) sabendo que eles se movem por ação das forças de gravidade, tal como se considerava desde Newton. O problema é estudado matematicamente à custa de equações diferenciais cuja solução não é evidente. Para contornar as dificuldades, Poincaré usou a geometria e essa estratégia conduziu a resultados importantes que foram sendo publicados no *Journal des Mathématiques Pures et Appliquées* e reproduzidas nas suas *Obras Completas*⁵⁴.

A aplicação dos seus resultados de equações diferenciais ao problemas dos três corpos permitiu-lhe estudar a estabilidade do sistema solar e assim, em 1887, ganhar o prémio estabelecido com essa finalidade pelo rei Oscar II da Suécia. Mas muito antes de lhe ser atribuído o prémio, Poincaré tinha consciência de que os seus resultados “abriam novos

⁵³ https://en.wikipedia.org/wiki/Three-body_problem

⁵⁴ Poincaré, 1916-1956.

campos de trabalho para a geometria”⁵⁵. A definição do que mais tarde se chamou “caos” apareceu no *Journal des Mathématiques Pures et Appliquées*⁵⁶ e foi reproduzida em *Science et Méthode*:

“Pequenas diferenças nas condições iniciais (das equações diferenciais que estava a usar) engendram grandes diferenças nos resultados finais. Pequenos erros introduzidos nessas condições iniciais produzem grandes erros nos resultados finais. Torna-se então impossível fazer previsões”⁵⁷.

A chamada “sensibilidade às condições iniciais” é uma das características dos fenómenos caóticos, que se tornaram tão universais como o próprio Poincaré. O que possibilitou essa universalização foi o desenvolvimento, após a II Guerra, de computadores com grandes capacidades de cálculo. A sensibilidade às condições iniciais foram redescobertas pelo meteorologista Edward Lorenz (1917-2008) em 1963 ao fazer simulações computacionais com equações diferenciais⁵⁸ por ele estabelecidas com a finalidade de estudar os movimentos das massas de ar atmosféricas. Lorenz obteve à custa de cálculo computacional os resultados que Poincaré tinha deduzido a partir de um estudo teórico (ou “analítico”, como se diz em matemática). A descoberta de Lorenz, que é considerado o momento fundador da teoria do caos, mostra que os modelos computacionais de comportamento da atmosfera não permitem fazer previsões rigorosas a longo prazo. A teoria do caos é atualmente utilizada em inúmeros problemas de diversas áreas⁵⁹, o que teria certamente agradado a Poincaré, criador de vários objetos matemáticos com aplicações na física. Como filósofo, ele procurou justificar essa capacidade matemática com a pergunta- resposta: “Quem nos ensinou a conhecer as verdadeiras e profundas analogias, aquelas que os olhos não veem e que a razão adivinha?” -, “Foi o espírito matemático, que despreza a matéria para se unir apenas à forma pura”⁶⁰.

⁵⁵ Poincaré, 1881, p. 377.

⁵⁶ Poincaré, 1916-1956.

⁵⁷ Poincaré, 1908, 62.

⁵⁸ Gleick 1987, 17.

⁵⁹ https://en.wikipedia.org/wiki/Chaos_theory.

⁶⁰ https://en.wikipedia.org/wiki/Chaos_theory.

Conclusão

A geometria esteve sempre presente ao longo dos trabalhos de Poincaré aqui referidos, e nalguns casos foram mesmo as descobertas de Poincaré que revelaram as relações profundas que se podem estabelecer entre a geometria e outras áreas. Os exemplos das equações diferenciais e das leis física são particularmente importantes pois abriram novos métodos e novos horizontes nesses campos. As ligações que Poincaré estabeleceu entre geometria e outras áreas científicas, por vezes surpreendente para os seus contemporâneos, ainda hoje causam admiração a cientistas e filósofos. Mas as cinco palavras escolhidas não esgotam no entanto o tema das ligações da geometria, que podia incluir ainda outros exemplos e também uma abordagem mais especializada e profunda. Aqui procurou-se apenas sobrevoar os diversos casos tratados sem mergulhar em cálculos matemáticos e fugindo o mais possível de descrições demasiado técnicas.

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El papel que juegan las definiciones matemáticas en la práctica del investigador: estudio de un caso

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Introducción

A lo largo de la historia han predominado en Occidente dos visiones acerca de la naturaleza de la matemática. Una considera que la característica central del quehacer del matemático consiste en la demostración de teoremas, en tanto que la otra visión considera a la matemática fundamentalmente como una actividad de resolución de problemas. Estas dos visiones de la matemática encuentran expresión en la filosofía de las matemáticas contemporánea que se comienza a desarrollar en los albores del siglo XX. Mientras que en el enfoque estándar la formalización axiomática, basada exclusivamente en razonamientos deductivos, suele considerarse la forma de representación más rigurosa, algunas voces críticas subrayan el papel central del razonamiento plausible en el contexto de la práctica matemática, donde el empleo de diferentes tipos de razonamientos, como el inductivo y el analógico entre otros, resulta indispensable para alcanzar nuevos resultados. Una pregunta interesante que surge a partir de este planteo es qué papel juegan las definiciones en el contexto de la investigación matemática. Mientras que la filosofía de las matemáticas estándar abordó el estudio de las definiciones atendiendo principalmente a la función que desempeñan en la formalización axiomática, recientemente el rol de las definiciones en matemáticas comenzó a ser reconsiderado junto con un renovado interés

por el estudio de los procesos de búsqueda y resolución de problemas vis-à-vis la construcción de la prueba axiomática considerada de mayor rigor. En este trabajo nos proponemos considerar el papel de las definiciones en el contexto específico de la práctica de resolución de problemas matemáticos a través del estudio de un caso perteneciente a la matemática del siglo XVII. Focalizaremos en la definición de "infinitamente pequeño" que Leibniz propone en *De Quadratura Arithmetica*, un texto de 1675/76 publicado póstumamente. En este texto, Leibniz presenta un método general de resolución de problemas de cuadratura de figuras curvilíneas que partiendo de la resolución de una serie de instancias culmina en el tratamiento aritmético de la cuadratura del círculo. El objetivo de nuestro estudio consiste en mostrar a través del caso analizado el valor heurístico que posee la definición en el contexto de resolución de problemas. A fin de enmarcar el caso de la práctica matemática de Leibniz en el debate reciente acerca del rol de las definiciones, presentaremos primeramente la concepción de las definiciones propuesta por Carlo Cellucci (2013, 2017).

La concepción heurística de las definiciones según Cellucci

Desde el punto de vista de la filosofía de la matemática estándar definir no es más que estipular arbitrariamente el significado de un término, el cual se introduce como una herramienta para garantizar el rigor formal. Con cada nueva definición lo único que se pretende es abreviar una expresión compleja mediante su reemplazo por una expresión más simple. Dado que no se trata más que de meras estipulaciones, las definiciones matemáticas no pueden ser incorrectas, de manera que siempre se podrá sustituir el término definido o *definiens* por el *definiendum* sin que ningún paso en la estructura de la prueba deductiva se vea afectado. Es por este motivo que en la concepción estipulativa las definiciones se consideran dispensables, y en consecuencia, eliminables. De acuerdo con uno de sus críticos contemporáneos (Cellucci 2013, 2017), esta caracterización sería aceptable sólo en caso que la práctica matemática pudiera ser entendida exclusivamente en términos de la actividad de formalización axiomática. Sin embargo, si nos proponemos concebir a la matemática como una actividad centrada fundamentalmente en la resolución de problemas, la concepción estipulativa carecería de sentido dado que no logra de ningún modo

capturar los rasgos fundamentales de las definiciones efectivamente empleadas en la práctica matemática, la cual se desarrolla justamente de acuerdo con un orden inverso al establecido por el modelo estándar que parte de axiomas y definiciones. Aquí, el punto de partida es el problema mismo para cuya resolución el matemático elabora una hipótesis a partir de la información provista por el propio problema y alguna otra información disponible. Cada hipótesis formulada plantea a su vez un nuevo problema que requiere la elaboración de nuevas hipótesis, generando así un movimiento abierto y ascendente mediante el cual se amplía el conocimiento matemático. En este marco de referencia, una definición se presenta como una hipótesis que el matemático formula en el proceso de resolución de un problema, como Carlo Cellucci explica en un artículo reciente: "a definition is the hypothesis that there is something which satisfies the condition stated in the definiendum" (CC 2017, 9). Nos preguntamos entonces, qué papel juegan las definiciones en la investigación matemática puesto que los matemáticos por lo general dedican buena parte de su tiempo a esclarecer la naturaleza de sus objetos de estudio en función de los problemas que se proponen resolver.

Una concepción adecuada de la definición debe dar cuenta del hecho de que las definiciones en matemáticas con frecuencia desempeñan un rol fundamental para la extensión del conocimiento matemático. Ellas son el fruto de un análisis que revela aspectos del objeto de estudio bajo consideración que aún no habían sido considerados o que lo eran pero - hasta el momento - de una manera imprecisa y/o incompleta. Así, una definición puede contribuir a la extensión del conocimiento en la medida en que permite aclarar la naturaleza del objeto de estudio, delimitando claramente qué propiedades le corresponden y cuáles no, e incluso en ocasiones especificando bajo qué condiciones tal objeto es concebible, como lo destaca Cellucci:

"When mathematicians give a definition of a concept that had previously been used without a precise definition, the definition generally contains an analysis of the concept and may therefore extend our knowledge". (Cellucci 2013, 37)

En el contexto del enfoque sobre el conocimiento matemático defendido por Cellucci, que el autor vincula fundamentalmente con la metodología del análisis, la concepción estipulativa de las definiciones

plantea fuertes limitaciones dado que no toma en cuenta numerosos aspectos centrales de la producción del conocimiento matemático. Entre estas objeciones podemos mencionar las siguientes. Primero, la resolución de un problema muchas veces depende de la elaboración de una definición adecuada; segundo, un concepto matemático frecuentemente es utilizado por mucho tiempo, a veces siglos, aun sin poseer una definición precisa; tercero, una definición aceptada en un contexto determinado puede ser cuestionada a la luz de análisis posteriores. Estas y otras limitaciones que plantea Cellucci lo llevan a proponer una concepción de la definición alternativa, concepción que denomina *heurística*.

En nuestro estudio de caso hay dos aspectos de la caracterización de Cellucci que nos interesa considerar puntualmente. Por un lado, hay definiciones que expresan un análisis de conceptos ya en uso en las investigaciones matemáticas hasta ese momento, como es el caso de la definición del concepto "indivisible". Por otra parte, el análisis de conceptos puede contribuir a la resolución de problemas, como es el caso del concepto "infinitamente pequeño", que Leibniz propone en lugar de la noción de "indivisible" empleada no sin frecuentes cuestionamientos hasta ese momento. Mientras que Cellucci rechaza radicalmente la concepción estipulativa de la definición, es de notar que según Leibniz las definiciones estipulativas desempeñan también un papel importante en la práctica de resolución de problemas matemáticos. En este contexto, sostenemos que para plantear una concepción adecuada de la definición es necesario llevar a cabo un cuidadoso estudio de casos históricos de resolución de problemas.

La práctica matemática de Leibniz y el recurso a los indivisibles en *De Quadratura Arithmetica* (1675-76)

En *De Quadratura Arithmetica* (DQA) Leibniz presenta un método general de resolución de problemas de cuadratura para figuras curvilíneas. Basándose en gran medida en el método de determinación de tangentes expuesto por Pascal en *Traité des sinus du quart de cercle* (Pascal 1965, IX), el procedimiento ideado por Leibniz consigue generar, a partir de una curva de área desconocida, una nueva curva proporcional cuya área es posible calcular en virtud de una reelaboración de los métodos aritméticos

desarrollados por sus predecesores (Saint Vincent 1647, Wallis 1656, Mercator 1668). Este procedimiento consta de una serie de pasos, entre los cuales se destaca la hipótesis de que es posible aproximar una figura curvilínea y un polígono inscrito compuesto de un número finito de figuras elementales, de modo tal que la diferencia entre sus áreas sea menor que cualquier diferencia dada. (Cf. Leibniz 2004, 47) La posibilidad de realizar tal aproximación había sido asumida sin demostración por los matemáticos más destacados de ese período. (Cf. Breger, 2008) En el caso de los métodos precedentes, Leibniz consideró que las figuras consisten de infinitos indivisibles y a partir de la comparación de los indivisibles de la figura curvilínea con los del polígono que la contiene se establecía la aproximación entre ambas figuras. Muchos investigadores no se mostraban satisfechos con estos métodos que implicaban el recurso a elementos indivisibles. Tales elementos, como se ha señalado en diversos estudios sobre la matemática de ese periodo, eran objeto de numerosas polémicas. (Cf. Mancosu 1996, Goldenbaum & Jesseph 2008). Si bien se reconocía que el empleo de indivisibles en los razonamientos geométricos agilizaba la investigación, algunos críticos sostenían que al mismo tiempo se aumentaba el riesgo de que surjan contradicciones en un análisis posterior. Los críticos a los métodos que recurren a indivisibles, a quienes Leibniz llama “escrupulosos”, exigían una demostración de la posibilidad de aproximar una figura curvilínea y un polígono inscrito. La demostración exigida por los “escrupulosos” debía responder a los estándares de rigor establecidos por la geometría clásica, la cual procedía para estos casos mediante el empleo del denominado *método de exhausción* (*Elementos*, XII) que no requería postular elementos indivisibles. Para satisfacer las demandas de rigor de los escrupulosos, Leibniz desarrolla en la proposición 6 de su tratado una demostración de carácter general del hecho geométrico que se daba por supuesto – a pesar de que en el escolio, inmediatamente después de presentar su demostración, expresa una cierta actitud de fastidio ante una exigencia de rigor que considera obstructiva para el progreso de la investigación.¹

A medida que avanzamos en *De Quadratura Arithmetica*, y una vez que fueron presentadas todas las proposiciones relativas al método de

¹ Afirma Leibniz en dicho escolio: “Je me serais volontiers dispensé de cette proposition car rien ne m’est moins naturel que de suivre certains auteurs dans leurs peu féconds mais ostentatoires scrupules de détail”. (Leibniz, 2004, 63)

resolución, Leibniz inicia el estudio de casos concretos: la cuadratura de la cicloide, de la parábola, de la hipérbola y del círculo. De acuerdo con el requisito de rigor exigido por los escrupulosos, para el abordaje de cada uno de estos casos se debía recurrir al procedimiento empleado para demostrar la proposición 6, y con ello, a la construcción de figuras inscritas y circunscritas y al recurso a demostraciones por reducción al absurdo. A fin de evitar para cada caso el pesado circuito de la proposición 6, nuestro autor decide recurrir, tal como procedían muchos de sus colegas – Pascal y Huygens, entre otros –, al empleo de los polémicos indivisibles. Por su intermedio será posible ofrecer una demostración directa de los resultados enunciados en las proposiciones relativas a los problemas de cuadratura de la cicloide, la parábola, la hipérbola y el círculo. En ese punto exacto del tratado Leibniz lleva a cabo, a fin de clarificar el objeto de estudio de la investigación, un análisis de la noción de indivisible, y hacia el final del extenso escolio a la proposición 11, ofrece su propia definición. Veremos que tal definición introduce una noción nueva que reemplaza la de indivisible: la de “infinitamente pequeño”. Sigamos entonces paso por paso en análisis que realiza Leibniz.

Leibniz y el papel de las definiciones en la práctica de resolución de problemas

En la proposición 11 de DQA Leibniz demuestra que a partir del área bajo un sector arbitrariamente pequeño de una curva es posible construir una figura de longitud infinita cuya área sea igual al doble del área de la curva que le dio origen. En el extenso escolio a esta misma proposición el autor empieza destacando la relevancia que posee el estudio de las magnitudes y espacios no finitos. Ya en la primera afirmación queda planteado el tema que será objeto de su análisis: “L’étude des espaces infinis en longueur bien que de grandeur finie mérite d’être mentionné”. (Leibniz 2004, 97) De acuerdo con esta proposición, que en una primera lectura puede generar desconcierto, es posible medir un *espacio* (el área bajo una curva, por caso) aun si éste se halla delimitado por *líneas de longitud infinita*; tal espacio será, de acuerdo con lo enunciado en la proposición 11, el doble de un espacio finito dado. Esta proposición le resulta de gran utilidad a Leibniz para emprender el estudio de aquellas curvas que poseen asíntotas como, por ejemplo, el caso de la hipérbola.

Seguidamente Leibniz comenta que los geómetras antiguos, a pesar de haber tenido conocimiento de la existencia de rectas asíntotas, no llevaron a cabo un estudio pormenorizado del tema. Tal estudio, según el autor, empieza con los trabajos de Torricelli y Saint Vincent en torno a la hipérbola, así como el de Wallis y Huygens sobre la cuadratura del círculo. Luego, se aboca directamente al análisis de sus objetos de estudio. La definición se presenta aquí como una herramienta para la investigación que permite clarificar las nociones relativas al dominio de las magnitudes y espacios no finitos. Nociones como las de "indivisible" e "infinito", que habían sido excluidas del análisis geométrico de los antiguos y aun por Descartes, son tratadas ahora a partir de las herramientas formales empleadas en la investigación de las magnitudes y espacios finitos. En este contexto, propone entonces distinguir entre la noción de "indivisible" y la de "infinitamente pequeño", lo mismo que entre la noción de "ilimitado" (non-bornée, DQA 101) y la de "infinito":

"Au demeurant il semblera à certains paradoxal de parler de Lignes infinies qui soient cependant bornées, je dois donc faire remarquer qu'il existe une grande différence entre indivisible et infiniment petit et de la même manière une grande différence entre infini et non borné". (Leibniz 2004, 97)

La diferencia que existe entre una magnitud "indivisible" y un "infinitamente pequeño", como explica a continuación, radica en que este último, a diferencia de un indivisible, admite seguir siendo dividido. Leibniz advierte que asumir en la investigación elementos indivisibles puede resultar riesgoso. Aquí el carácter fructífero de la definición incluye la búsqueda de rigor formal, por lo que ambos aspectos de la investigación matemática no pueden concebirse ni entenderse independientemente:

"La Géométrie des indivisibles est trompeuse si on n'y adjoint quelques précisions au sujet des infiniment petits; utiliser des points véritablement indivisibles comporte un certain risque et il faut au contraire employer des lignes qui, pour être infiniment petites, n'en sont pas moins des lignes et sont, par conséquent, divisibles..." (Ibid., 99)

Luego aplica un criterio análogo para diferenciar una magnitud ilimitada de una infinita. A una magnitud considerada infinita siempre se le podrá añadir una magnitud del mismo tipo. Notemos aquí que el análisis

de Leibniz procura establecer claramente cuáles son los objetos de estudio de su investigación: un indivisible entendido como la magnitud más pequeña, lo mismo que, por caso, la magnitud de una línea ilimitada entendida como la línea más grande, no son objetos que puedan ser tratados por la geometría:

"(...) une quantité non bornée diffère d'une quantité infinie. La grandeur d'une ligne non bornée, tout comme celle d'un point, sort du champ des considérations Géométriques." (*Ibid.*)

Así concebidas las cantidades "no finitas", es decir, infinitamente pequeñas e infinitas, las reglas que permiten operar con ellas no difieren de aquellas empleadas para las cantidades que denominamos "clásicas". Recordemos que de acuerdo con el libro V de los Elementos de Euclides, existe una razón entre dos cantidades siempre que se trate de cantidades del mismo tipo (homogéneas). La idea de "línea más pequeña" tanto como la de "línea más grande" no pueden ser tenidas en cuenta en los razonamientos geométricos puesto que, a diferencia de las magnitudes de la geometría clásica, no son conmensurables. En este punto cabe destacar entonces que el análisis de Leibniz procura caracterizar las nociones de "infinito" e "infinitamente pequeño" dentro del marco de la geometría clásica. En este sentido, el papel de la definición en su tratado se muestra a través del registro de la reconciliación entre el ámbito de lo finito y el de lo no finito, antes radicalmente divorciados:

"Et tout comme une ligne infinie bornée est constituée de lignes finies, une ligne finie est composée de lignes infiniment petites mais néanmoins divisibles. On ne peut donc pas dire qu'une Ligne bornée soit une moyenne proportionnelle entre un point, soit la plus petite ligne, et une ligne non bornée, soit la plus grande ligne. Mais on peut dire qu'une ligne finie est une moyenne proportionnelle, dans un sens non pas approximatif, mais véritable et exact, entre une ligne infiniment petite et une ligne infinie." (*Ibid.*, 101)

Hacia el final del esolio 11 y como resultado del análisis de estas nociones, Leibniz presenta la siguiente definición del concepto "infinito":

(...) "j'appelle infinie une quantité, bornée ou non, des que nous la concevons comme plus grande que toute quantité assignable, c'est-à-dire numériquement exprimable." (*Ibid.*)

La misma definición, pero en sentido inverso, se aplica para las cantidades infinitamente pequeñas; una cantidad infinitamente pequeña será definida por Leibniz como "una cantidad menor a cualquier cantidad que pueda ser asignada".

En la última línea del mencionado escolio Leibniz aclara que no es necesario en el contexto de la práctica matemática especular acerca de la naturaleza metafísica de las nociones que acaba de definir: "Déterminer si la nature souffre des quantités de ce genre est l'affaire du Métaphysicien; il suffit au Géomètre de démontrer ce qui résulte de leur supposition". (*Ibid.*) A lo largo de DQA también se pueden encontrar referencias similares del autor acerca de la naturaleza de este tipo de cantidades.² La despreocupación que Leibniz expresa aquí acerca del problema ontológico de la existencia o no de tales cantidades, tiene su contrapeso en la insistencia del autor por destacar fuertemente las facilidades que ofrecen en la práctica al momento de trabajar con ellas una vez que son supuestas. En particular, el empleo de cantidades infinitas e infinitamente pequeñas permite ahorrar todos los pasos implicados en la proposición 6, abriendo de esa manera el camino de la invención:

"Peu importe que de telles quantités soient ou non naturelles, on peut se contenter de les introduire par le biais d'une fiction dans la mesure où elles offrent bien des commodités dans les formulations, dans la pensée, et finalement dans l'invention aussi bien que dans la démonstration, en rendant inutiles l'usage des figures inscrites et circonscrites, les raisonnements par l'absurde et la démonstration qu'une erreur est plus petit que toute erreur assignable." (*Ibid.*, 183-185)

La función de abreviar o *compendiar* que cumple en este caso la noción de "infinitamente pequeño" va para Leibniz de la mano del valor heurístico que posee la definición propuesta. A esta propiedad de compendiar información que posee la noción de infinitamente pequeño

² Casi treinta años más tarde, en el contexto de los debates en torno a los fundamentos del Cálculo al interior de la Academia de Ciencias de París (*Académie Royale des Sciences*), durante el extenso intercambio epistolar con Varignon (1702-1716), Leibniz insiste sobre la misma idea: "(...) mon dessein a été de marquer, qu'on'a point besoin de faire dépendre l'analyse Mathématique des controverses métaphysiques, ny d'asseurer qu'il y a dans la nature des lignes infiniment petites à la rigueur...". (GM IV, 91 - Carta a Varignon)

refiere casi treinta años más tarde, en el contexto del intercambio epistolar con Varignon. En esa oportunidad Leibniz afirma:

"D'où il s'ensuit, que si quelqu'un n'admet point des lignes infinies et infiniment petites à la rigueur métaphysique et comme des choses réelles, il peut s'en servir sûrement comme des notions idéales qui abrègent le raisonnement, semblables à ce qu'on nomme racines imaginaires dans l'analyse commune (comme par exemple $\sqrt{-2}$), lesquelles toutes imaginaires qu'on les appelle, ne laissent pas d'estre utiles, et même nécessaires à exprimer analytiquement des grandeurs reelles." (GM IV, 92 - Carta a Varignon)

En pocas palabras, la definición de "infinitamente pequeño" se propone en el contexto de resolución de problemas de cuadratura compendiar en un término una serie de engorrosos pasos geométricos que Leibniz había concedido hacer explícitos por única vez en la proposición 6 de su tratado (DQA). Pero, el estatus ontológico del objeto definido no es materia de análisis para el geómetra sino, en todo caso, "un asunto de los metafísicos". El geómetra asume su existencia a modo de hipótesis de trabajo, y a partir de ello, extrae consecuencias. Así, la definición propuesta garantiza operar de manera ágil y concisa contribuyendo de ese modo con el proceso de resolución de problemas.

Conclusiones

Nuestro estudio de caso se centra en el concepto "indivisible", un concepto ampliamente utilizado en la resolución de problemas de cuadratura, tanto por Leibniz como por la comunidad de expertos del momento, aun cuando no se disponía de una definición precisa de dicho concepto. A raíz de las objeciones que suscitaba el empleo de indivisibles, Leibniz emprende el análisis de este concepto en el marco del problema planteado en la proposición 11 de DQA con el objetivo de clarificar la naturaleza de los objetos involucrados en el proceso de resolución. El resultado de dicho análisis se cristaliza en la definición de "infinitamente pequeño", un concepto introducido por Leibniz para reemplazar al de "indivisible" y que permitirá la resolución de los problemas que Leibniz se propone resolver. A partir de estas observaciones acerca del papel que juegan las definiciones en la práctica matemática, podemos concluir que la

caracterización de la definición como mera estipulación resulta insuficiente. En primer lugar, nuestra conclusión coincide con el planteo general de Cellucci. No obstante, las reflexiones de Leibniz sobre su propia práctica indican claramente que también debemos reservar un lugar importante a las definiciones "estipulativas". En nuestro trabajo notamos que, según Leibniz, el concepto "infinitamente pequeño" juega un rol importante para compendiar o abreviar una gran cantidad de información en un solo término, permitiendo de este modo agilizar el razonamiento, un aspecto que resulta decisivo para la resolución efectiva de problemas. Una lección que podemos extraer de nuestro estudio de la resolución de los problemas de cuadratura abordados por Leibniz es que, para formular una concepción adecuada de la definición matemática, que dé cuenta del rol esencial que la definición posee en la práctica de resolución de problemas, es necesario contemplar algunos casos de resolución de problemas pertenecientes a diferentes períodos y culturas matemáticas. Una concepción adecuada de la definición no debería estar divorciada de la práctica matemática misma, la cual, como nos propusimos mostrar a través del caso analizado, de ningún modo puede ser reducida a un marco conceptual único.

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Revisiting the Bohr Model from an artefactual perspective

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Abstract

Our main objective in this work is to review the notion of atom in science, focusing mainly on Niels Bohr's proposal of 1913, analyzing it as the final product of the evolution process of what was considered as the most basic constituent of the matter. We believe that the notion of atom fits well with a non-representative approach of modeling in science that supports the vision of model as epistemic tools or mediators, mainly maintain by Tarja Knuuttila. This approach to modeling process affirms that models are autonomous agents, theoretically "hybrids", and it emphasizes the function of interface between theory and phenomenon that models have in science. The Bohr's atom is a clear example of the perspective of Knuuttila, it is a model that besides having a spatial structure of objects, has a description that is an integrated and inseparable part of the model itself, and not just a "model description" that is separate from and secondary to imagined entities. We also suggest that the notion of the atom evolves over years, losing its realistic commitment, until arriving at the current model as application or instantiation of axioms of Quantum Mechanics.

1 | Models As Mediators

The scientific practice oriented to know the world surrounding us has shown in recent years that scientific models are very useful. In particular, when we focus on a portion of reality that manifests itself unattainable due to its great complexity or its dimensions (macro or microscopic). The models take the form of mockups, miniatures, drawings, maps, equations, digital 3D volumes (via *occulus reef*), among others. That is, models would be tools that would enable us to know the portion of reality that we want to know. In this sense, the models fulfill an interface function between the act of knowledge and the reality, or between the theory and phenomena, both in the case of concrete and abstract models.

For the classical perspective, the syntactic view of theories, that considers science as an interpreted axiomatic system, the models are identified with semantic rules that interpret sets of sentences in a given logical domain language [Winther, 2016]. In this view, models are particular cases within the framework of a theory and they are irrelevant to science and have only a heuristic or pedagogical purpose. That is, the models interpret the general terms of a theory in a particular case that makes what the theory dictates. In this perspective, a model can “represent a theory” in the sense that it interprets the laws and axioms of that theory in a particular instance. For example, it is the case of the model of the Solar System or the model of the double helix representing the DNA. An exception to this way of understanding models, but always within the syntactic view, is Nagel who gives them a more leading role between the postulates of theory and rules of correspondence [Nagel, 1961].

The semantic view of theories appears as opposed to the classical perspective about scientific models [van Fraassen, 1980; Giere, 1988; Suppe, 1989]. Although different versions of the semantic view assume a different notion of model, they all agree that models are the central element of scientific theorizing and view scientific theories as families of models. In this approach, the theory itself is specified by its class of models, which are constitutive of the theory and not dispensable as in the classical perspective. Models are autonomous agents, in other words, they are relatively independent of theories [Morrison, 1999]. This independence appears in terms of how they are constructed and which are their functions. The semantic view argues that models are constructed independently of

theories and that their construction is an art rather than a mechanical process. With respect to the functions of models, this view claims that they have functions which they could not perform if they were a part of, or strongly dependent on, theories.

A third approach is that models are thought to mediate between phenomena and theories that attempt to account for phenomena [Morrison, 1999; Morgan and Morrison, 1999; Cartwright, 1999]. In this approach models are autonomous agents, theoretically “hybrids”, working like instruments for acquiring knowledge, with a representative function as long as they are idealizations or simplifications of phenomena. It should be added that while the representative function is an essential feature in this perspective, this function in our day does not have a theory agreed upon by the philosophers of science. This view of the models is contrasted to the syntactic and semantic perspectives.

In all three perspectives, the interface function between theory and reality is present with a different degree of importance on the model concept. This function that models fulfill establishes two types of specific relations: one, with theory, either of submission or relative dependence; on the other hand, a relation of representation with phenomena.

For our article we will focus on the third perspective, that is, on models as mediators and their specific function as interface that establishes a relationship between theory and phenomena.

The hybrid character of the models emphasizes an essential feature of the models, which is particularly appreciable in models discussed in this work: models as mediators have no special commitment to any theoretical perspective. In the constitution of a model can participate elements of different theories (sometimes incompatible) or even of a single theory but other principles or laws of the same theory are rejected. Morgan and Morrison [Morgan and Morrison, 1999] think that a model is not only constructed using hypotheses belonging to one or several different theories but also appealing to diverse empirical data of different types. Nancy Cartwright [Cartwright, 1999], on the other hand, argues that physical theories do not represent what happens in the world and therefore it is necessary to have models because they have this representative capacity. But Cartwright argues that models are not part of any specific theory and therefore must be understood independently.

With regard to the representative function, models as mediators represent the phenomena. Empiricists like van Fraassen [van Fraassen, 1980] only allow for observables to qualify as such, while realists like Bogen and Woodward [Bogen and Woodward, 1988] do not impose any such restrictions. Here, it is understood by phenomenon all relatively stable and general features of the world that are interesting from a scientific point of view.

How phenomena are represented by models is a difficult question. Following to Frigg [Frigg and Hartmann, 2017], we can affirm that there are mainly two problems in this question. The first problem is to explain in virtue of what a model is a representation of something else. If the model is a mere description, this leads back to the old problem of the relationship between language and reality of the philosophy of language. If the models are objects, the question now is how an object represents the phenomena, which is answered according to the ontology that is adopted. The second problem refers to the modes or styles of representation of models, there is more than one style of representation and each one represent the phenomenon in a different way. This can happen even for a single phenomenon, for example, Weizsäcker's liquid drop model represents the nucleus of an atom in a manner very different from the shell model.

In general, according to the style of representation adopted by models, the following notions not mutually exclusive are considered: scale models, idealized models, analogical models, phenomenological models and mathematical models. The latter category is important in the semantic perspective of theories where the model-phenomena interface is established in terms of isomorphism [van Fraassen, 1980; Suppes, 2002] or partial isomorphism [Da Costa and French, 2003]. The distinctions are far from clear for the first four categories.

We will now discuss some recent perspectives that consider more productive for science to construct a modeling approach that does not have the commitment of representation, without renouncing to the mediators character of models, given the lack of clarity surrounding the concept of representation and focus on the surrogate nature of model-based reasoning.

Indeed, some authors [Callendar and Cohen, 2006; Knuuttila 2011, 2017] have seen the opportunity to go beyond the notion of representation through the postulation of models as artifacts. We will dwell especially on

the latest work by Knuuttila [Knuuttila, 2017] as it considers an interesting criticism to the most widespread way of understanding models as fictions: the theory of make-believe [Walton, 1990].

The artefactual account of Knuuttila treats models as purposefully created artefacts that are constructed for the study of certain scientific problems, models are *epistemic tools*. Knuuttila considers that the representational approach to scientific models is too limiting, and it is better to capture the epistemic qualities of the models if we consider them from a productive point of view. Approaching models as concretely constructed artefacts addresses the actual representational tools that are used in model construction. These representational tools — mathematical, diagrammatic, pictorial, 3-D, and so on — form an integrated and inseparable part of the model itself, and not just a “model description” that is separate from and secondary to imagined entities.

The point of Knuuttila is to put forward an account of models that preserves the independence of model systems from real-world systems and to emphasize on the surrogate nature of models as pivotal feature of model-based reasoning. Knuuttila is against Walton’s make-believe approach that propose/concerns the separation of internal representation from external representation (imagined-object from the so-called object-description).

For Knuuttila the separation of imagined objects (models) from their descriptions (model descriptions), and the roles assigned to them in scientific reasoning, raises difficult questions concerning how imagined entities are related to actual representational tools and with the real-world phenomena. She argues that such account tend to face coordination problems with the target-system, especially if Walton’s account prioritize imagined models as the vehicles of representation. The artefactual account she put forward, and that we consider more appropriate, preserves the insight of Walton concerning the relative independence of model systems from real-world systems and emphasizes that models are materialized through their representation.

Such an artefactual turn proposed by Knuuttila [Knuuttila, 2011] is close to a dialogical approach to fictions as literary artefacts [Thomasson, 1999]. In effect, it was developed a dialogically pragmatic approach of the artefactual theory of Amie Thomasson [Redmond, 2010; Rahman, 2011;

Fontaine and Rahman, 2014] that allows argumentations between existent and non-existent beings in a dialogical frame. On the basis of this work, we develop general insights for an understanding of models as artefacts in Thomasson's perspective [Valladares and Redmond, 2014]. In this perspective, the models (certain type of them) are existing abstract objects that are present in the world through their particular form of representation that function as support. The relationship between these objects and the supports is dependent. They also have a dependency relationship with the authors who created it.

Artefacts are in both cases external entities. What are the models made of? For Knuuttila the representational instruments are the material from which the model is made as an artefact (a material embodiment is needed for a model to function as a vehicle for communication). For the dialogical approach of Thomasson, the instruments of representation are the material support that makes possible that they exist in the world and with which they maintain a dependency relationship that Thomasson calls "constant". If this constant relationship is interrupted by the destruction of the support, for example, the artefact may disappear forever. We believe that both the materiality of Knuuttila and the support function of the material in Thomasson are not incompatible and enrich the study of the models.

Regarding the imagination, Thomasson considers it as the faculty that allows the creation of artefacts (intentionality). But once they are created, they keep independence with her even though the imagination can function as another of the tools of representation. In Knuuttila a kind of Copernican Turn operates: it is not the imagination that expresses itself through means of representation, but the tools of representation that determine the way we imagine:

"The imaginary accounts focusing on imagined-objects are bound to approach modeling as to primarily mental activity, and the activity of using external representational means as that of merely (partially) describing the mental content of scientists. In contrast, the artefactual account renders visible the characteristics of the various representational tools used in model construction and scientific shape, also delimiting what can be thought of." [Knuuttila, 2017:3]

Next, we review the notion of atom and its evolution, in particular we discuss how the Bohr's atom model fits the perspective of Knuuttila. That is, the view that holds that models are existing entities built for scientific purposes (gain new knowledge), embodied (Knuuttila) or supported (Thomasson) by means of representation tools and used as means for surrogate reasoning [Knuuttila, 2017; Frigg and Nguyen, 2016, 2017; Godfrey-Smith, 2006; Weisberg, 2007, Suárez, 2009; Mäki, 2011].

2 | Historical antecedents of Bohr's atomic model

As we know, an important number of philosophers in ancient Greek develop a natural philosophy, the atomism that accounts for the origins of everything from the interaction of indivisible bodies: atoms. It means that in the action of dividing or separating bodies (matter) into parts, we would find a last portion of matter that is no longer divisible. It is clear then that, according to this approach, the ultimate portion of matter is represented as a portion of itself that was no longer divisible into parts. The idea of the atom as the basic constituent of matter has survived to this day, founded on empirical evidence and losing the indivisibility as the distinctive feature. The evolution of idea of atom is in part proper and partly accompanying the historical development of science. In the current notion of what is the atom, it is presented (not always explicitly) as a "model" of application of the Quantum Physics theory.

Early ideas about the existence of atoms are usually attributed to philosophers of Ancient Greece. Around 440 BC, Leucippus of Miletus probably first called "atoms" to the very small indivisible corpuscles that compose the matter. Democritus of Abdera (c. 460-371 BC) elaborates a systematic view of the world based on the concept of the atom. The concept of atom emerging in Ancient Greece was a metaphysical thesis more than an experimental result. In effect, Atomism purports to establish claims about the ultimate nature of material reality by philosophical arguments. The objective of Greek atomism was to explain the Greek world, and then, it was applied to the material world and to the spiritual world [Berryman, 2008].

The Greek atomists considered that all the properties all properties of world objects are reducible to and arise as a consequence of the size, shape and motion of underlying atoms; whether an object is hot, cold,

or of a certain color, is determined by the type of atom and the quantity of void in it. Some core ideas of the Greek atomism were persistent for many centuries:

- all matter is composed of small corpuscles, which are bits of matter too small to be seen,
- atoms are solid and they cannot be further split into smaller portions (in Greek, the prefix "a" means "not" and the word "tomos" means cut),
- the space between atoms is empty, there is a void in this space.
- atoms have not internal structure, they are homogeneous.
- atoms have a few basic properties: shape, size, and motion. They have all sorts of shapes and different sizes: some are rough, some hook-shaped, some concave, some convex and some have other innumerable variations.

At first glance, the idea that Greek atomism was based on metaphysical arguments rather than empirical evidence, seems to go against calling it modeling. However, if we consider a model as an artefact that mediates for the surrogates reasoning about phenomenon it seems to fit smoothly. In effect, the idea of the atom, as we pointed out above, allowed them to solve the problem of the origin of things and to explain the Greek world (or at least they tried from this idea).

During the seventeenth and eighteenth centuries, most of the mechanical philosophers accept the ideas of Greek atomism, with the important difference that atomism is only applied to the material world. Not all the philosophers of mechanicism shared the same ideas about the atom, but of all of them, it was Robert Boyle (1627-1691) who developed the atomism more elaborately. Its most famous result is the so-called "Boyle's Law", which states that if the volume of a gas is decreased, the pressure increases proportionally. Boyle understands that this law could only be explained if all gases were made of tiny particles (atoms). Furthermore, Boyle argued that atoms have an immense variety of shapes and that they are provided with tips, hooks, and ramifications. These appendices make possible the combination of atoms, for instance, the dissolution of a metal in an acid is due to the fact that the tips of the atoms of the acid penetrate the pores of the atoms of the metal. If on the contrary, an acid was unable to dissolve a metal, it was because the pores of the metal atom were too narrow for the tips of the acid atom to penetrate into

them. The union of the atoms through their hooks, points, and pores explained the metamorphosis of all substances, such was the faith of the mechanistic philosophers.

The ideas of Isaac Newton (1643-1727) allowed explaining the cohesion of the substances without resorting to the variety of the forms of the atoms and their mechanical connection. Newton stated that matter is formed of solid, massy impenetrable particles, and that the bond that holds these particles together and groups them into macroscopic bodies is an immaterial force analogous to the force of gravity. The Newton atom did not have a great impact on the science of the eighteenth century given the impossibility of specifying and quantifying the immaterial forces of cohesion through philosophical arguments or experimental evidence.

It was only in the nineteenth century that the idea of atom as a basic constituent of matter began to bear significant fruit in science, mainly due to Dalton's ideas. For Dalton the atoms remained unchanged during chemical processes, and according to its structure, atoms can be simple (elements), compound (simple molecules), and complex (complex molecules). Also, using a deductive reasoning from the experimental evidence known at the time, he concludes that weight is the distinctive property of atoms and deduces the law of multiple proportions. Dalton was the first to publish a table of relative atomic weights and in his work, there is one of the earliest depictions of molecular structure, in which the atoms in a molecule are arranged in space in a specific fashion. It is important to note that Dalton did not propose any internal structure of the atom. From Dalton, the knowledge of atoms is not established by a priori philosophical argument but by appeal to quite specific experimental results.

Dalton's atomic theory did not explain much experimental evidence accumulated during the second half of the nineteenth century, such as the determination of specific heats, and it remained controversial throughout the 19th century. While the law of multiple proportions was accepted, the hypothesis that this was due to atoms was not so widely accepted. This prompted the scientists to propose models that could explain it.

The most surprising property of atoms was their permanence in chemical reactions. What kind of physical entity could not be broken in chemical reactions, and maintains its identity even at high temperatures? Lord Kelvin (W. Thomson) in 1867, inspired by a work of Helmholtz on vortices, conceived the idea that atoms could be vortices in the ether.

A vortice or vortex line is an imaginary line whose tangent is everywhere parallel to the local vorticity vector inside a fluid. The main idea guiding this analogy between atoms and vortices was that vortices, as well as atoms, remained unchanged: vortices in an ideal fluid is ever composed of the same particles. Also, vortices can participate in an analogy of a chemical reaction, a vortex can trap another and then continue their evolution together. It is interesting to note that the author of the first model of the internal structure of the atom, J. J. Thomson (from now on, "Thomson" refers to J. J. Thomson), proposes in 1883 a theory to explain the nature of matter and its electrical and chemical properties from a hydrodynamic point of view, based on the "atom vortex model" [Thomson, 1883a].

Until the appearance of the vortex model, there seems to be a strong realistic commitment to the different proposals on the atom, being difficult to affirm that the different proposals were "models" or artifacts destined to a surrogate reasoning. In the case of the vortex model, its consideration as a model is directly linked to the question of considering ether as an existing object or simply as an epistemic tool to explain the phenomenon of the propagation of light. We believe that the evolution of the idea of atom from the proposal of the Thomson atom (the "plum pudding model", see below) is linked more directly to the notion of atom as a scientific model or epistemic tool. From this point in the evolution of atom idea, the characteristic explanatory purpose of the models is palpable in the different proposals.

The need for a new atom model is clear as a result of the discovery of the electron in 1897 and the suggestion that the electron may be a constituent part of atoms. Thomson, in 1897, based on the results of his experiments to determine the nature of cathode rays, proposes that these rays are a new type of particle, which he calls "corpuscles", suggesting the existence of the subatomic particle known today as the electron. Thomson believed that the corpuscles are charges of negative electricity carried by particles of matter and that they emerged from the gas atoms in the vicinity of the cathode ray tubes. Thus, he concluded that atoms were divisible and that electrons were their building blocks. The Thomson atom abandoned his earlier hypothesis in which atoms were composed of vortices in the ether. Thomson's discovery of the electron posed a new theoretical problem for physicists because atoms as a whole are electrically neutral. Where was the neutralizing positive charge and what held it in place?

In 1903, Kelvin proposes that the atom is a sphere of homogenous positive charge within which the electrons are embedded, this sphere being the full size of the atom. The sphere must be held together by unknown non-electrical forces. Following in this line of thought, Thompson in 1904 proposed which is considered the first model of the internal structure of the atom: the atom is formed by electrons distributed inside a cloud of positive charges; this distribution reminds many scientists how plums are distributed in the common English dessert, plum pudding, hence the name of "plum pudding model" [Thomson, 1904]. The charge distribution proposed in the model was in agreement with the neutrality of the atoms and the electrons did not have fixed positions but were distributed in rotating rings.

The Thomson model was the most popular atom model among physicists in the period 1904-1910. Two other models contemporaneous to that of Thomson that did not have so much influence were the model of Lewis or cubical atom in which electrons were positioned at the eight corners of a cube [Lewis, 1916], and the model proposed by the Japanese physicist Nagaoka Hantaro, in which the atom resembled the planet Saturn, with rings of electrons surrounding a concentrated positive charge [Hentschel K, 2009].

One problem with the Thomson model was that the atom would collapse by losing energy by the energy emitted by the accelerated electrons. Thomson was able to show that the radiation of accelerated electrons decreases with the number of electrons in the rings, and it could therefore be ignored. Then, Thomson's model seemed to be stable. Other reasons made the model rather problematic. One of them was the nature of the cloud of positive charge surrounding the electrons, it should be frictionless and massless, and finally be a manifestation of the electrons themselves. Thomson hoped to improve the model by disregarding the positive charge and replacing its effects with some property of the electrons. Finally, while Thomson was able to explain the results of atomic scattering experiments of beta particles as a multiple scattering case, he could not explain atomic scattering of alpha particles with his model [Kragh, 1999]. The experiments of scattering were not the death certificate of the Thomson model, but the model was losing little to a little adept, already in 1911 when the results of Rutherford's experiences and its proposal appeared, Thomson's model was not had adepts.

In 1911, Rutherford proposes a new model of the atom that allows interpreting the results of scattering experiments of alpha and beta particles by atoms [Rutherford, 1911]. As an intriguing result, the experiments show that around 1 in 8000 alpha particles were deflected by very large angles, while the rest passed through with little deflection. This result could not be explained by the Thomson model, as Rutherford points out in the paper.

In the model proposed by Rutherford, the atom is assumed to consist of a central charge surrounded by a spherical distribution of charges having opposite sign to the central charge. The central charge is the modern atomic nucleus, though Rutherford did not use the term "nucleus" in his paper. The large deflexions of particles would be mainly due to their passage through the strong electric field of the central charge. Further, the nucleus is assumed to be the seat of the essential part of the mass of the atom and to have linear dimensions exceedingly small compared with the linear dimensions of the whole atom. The Rutherford model proposes that the atom was essentially formed by empty space, unlike the compact conglomerate of Thomson or Dalton atom. The predictions of Rutherford's model are independent of the sign of the central charge, it could be positive or negative. Rutherford finds no definite evidence to determine whether the charge of the nucleus is positive or negative, and adopts the positive sign for a matter of convenience, presenting some arguments to justify this choice. It is clear here how works the model: it allows explain all the multiplicity of information coming from experimentation. The scientist presupposes that information coming from an unitary object and propose an explanation of that object in the form of a model. That model should allow making inferences (surrogate reasoning) in the place of the supposed existing object.

It is important to note that the issue of atom stability is simply not considered by Rutherford in the paper. This stability will depend upon the minute structure of the atom, and on the motion of the constituent charged parts, neither of them are considered in the discussion of the atom model. The model is the best way to interpret the scattering experiment data, concentrating the mass of the atom in a small nucleus creating a central electric field, but didn't attribute any structure to the remaining electrons and remaining atomic mass. Rutherford refers to the atom model of H. Nagaoka as a possible internal structure of the atom [Nagaoka, 1904]. Nagaoka considered the properties of a "Saturnian" atom consisting of a

large, massive, positively charged sphere surrounded by many rings of rotating electrons analogous to Saturn's ring, and showed that such model was stable if the attractive force was large. Rutherford points out that from the point of view considered in the paper, the chance of large deflexion would practically be unaltered, whether the atom is considered to be a disk or a sphere. The atom model of Nagaoka is the earliest published quasi-planetary model of the atom [Hentschel, 2009].

3 | The Bohr Model

In this section we describe Bohr's atomic model for the hydrogen atom as it was originally presented in the paper "*Part I - On the Constitution of Atoms and Molecules*", this is the first of three papers published by him in 1913 [Bohr 1913, a]. While in this article, Bohr also discusses the applicability of his proposal to multielectronic atoms and molecules, we will only refer here to the proposed model for atoms with one electron. The other two articles of the year 1913 deal with the application of Bohr's proposal to atoms with more than an electron and systems containing several nuclei [Bohr 1913, b and c].

Bohr presents his paper as an attempt to show that the application of the ideas about energy quantization of Planck to Rutherford's atom-model affords a basis for a theory of the constitutions of atoms. Before to develop his proposal, Bohr expresses his conviction about the inadequacy of the classical electrodynamics in accounting in describing the behavior of system of atomic size, taking into account the predictions of the theory in its application to the Rutherford's atom-model. According to classical electrodynamics, an electron orbiting around a central nucleus can never possess an orbit that remains unchanged over time, in other words the electron can no longer describe stationary orbits. The reason for this is that the accelerated electron ought to radiate energy, then, it will approach to the nucleus describing orbits of smaller and smaller dimensions until the orbit has the dimensions of the nucleus. As the electron falls to the nucleus, the emission would rapidly increase in frequency as the orbit got smaller and faster, distributing the radiation frequencies in a continuous range. So, according to classical electrodynamics, the Rutherford's atomic model would collapse, it is unstable.

Contrary to predictions of Rutherford's atomic model, the experimental evidence shows that any atoms occurring in nature has a permanent (stable) state with absolutely fixed dimensions, and that when an atom emits radiation, it will settle down in a state of equilibrium (stable state) in which atom dimension has the same order of magnitude as before the process. Also, experiments show that atoms will only emit light (that is, electromagnetic radiation) at certain discrete frequencies. It is important to note that the experimental results disagree with the Rutherford's model by taking into account classical electromagnetism.

In view of above considerations, Bohr express that whatever the alteration in the laws of motion of the electrons may be, it seems necessary to introduce the Planck's radiation law about the quantization of energy radiated by oscillators. The same ideas had been used by Einstein to explain the photoelectric effect [Einstein, 1905].

In order to explain the spectral-energy distribution of radiation emitted by a blackbody, Planck proposes an empirically fitting function of thermal radiation data emitted by a blackbody, and then he constructed a physical derivation of this function (now called the Planck's law). To support the empirical function, Planck proposes a blackbody model from which it is possible to derive it. The model consists of a cavity with perfectly reflective walls containing many identical oscillatory bodies, several such oscillators at each of many characteristic frequencies. Planck's radical assumption is that the energy of a oscillator of frequency ν can not have any value, but it can only have any of a series of discrete values, which are integer multiples of a definite physical unit of energy $e=h.\nu$ (h is now known as the Plank's constant). For Planck, the hypothetical oscillators does not exist in nature but are mathematical devices that enabled him to derive a single expression for the black body spectrum that matched the empirical data.

In 1905, Einstein proposed an explanation of the photoelectric effect using the Planck ideas [Einstein 1905]. For this purpose, Einstein derives an equation for the photoelectric effect that agreed with experimental results based of idea that a beam of light could be described as a collection of discrete wave packets (now called photons), each with energy $e=h.f$, being f the light frequency.

To explain experimental results related with the atom and propose his model, Bohr uses the same spatial structure of Rutherford's atom. Thus, Bohr assumes that the positive nucleus has very small dimension (according

to Rutherford findings), that the mass of the electron is negligibly small in comparison with that of the nucleus, and that the velocity of the electron is small compared with the speed of light. Also, Bohr assumes that the electrostatic interaction electron-nucleus can be described by a law of classical electromagnetism, the Coulomb's inverse-square law. Possibly in the election of spatial structure of Rutherford atom, there is no a realistic commitment to the representation of the minimum portion of nature but a commitment to the best explanation of the experimental data.

To adjust the atom model of Rutherford to natural facts, Bohr proposes the following mechanism to describe the binding of electrons by positive nucleus. Bohr assumes that the electron at the beginning of the interaction with the nucleus is at a great distance apart from the nucleus, and it has no velocity relative to nucleus. After the interaction, the electron is settled in a stable circular orbit around the nucleus. During the binding of the electron, it emits an electromagnetic radiation field (electromagnetic wave) which is homogeneous and it has frequency ν equal to the half of frequency of revolution of electron in the final orbit. Then taken into account the Planck's law, Bohr proposes that the amount of energy radiated by the system during the binding process must be quantized according to the Planck law. The energy radiated should be equal to the binding energy W of electron in the atom model. Then, accepting that the orbital movement of the electron in any stable orbit can be described by classical laws of mechanics and electromagnetism, Bohr derives a mathematical expression of the binding energy

$$W = \frac{2\pi^2 m^2 e^2 E^2}{n^2 h^2} \quad (1)$$

where n is an integer, m is the electron mass, e and E are the electron and nucleus charge, respectively.

What is the binding energy W ? The energy W is interpreted as the ionization energy of hydrogen atom, in other words, the energy that the electron would radiate out when it is attached in any stable orbit. This is an important prediction derived of Bohr model. Note that W according to the expression in Eq. 1 is a discrete magnitude, i.e. W is quantized, since it depends on the integer n . The greatest value of W is obtained for $n=1$, then $n=1$ correspond to most stable configuration of electron-nucleus system. Different values of n would correspond to configurations of hydrogen atom

where there is no radiation of energy, and consequently, they will be stationary states as long as atom is not disturbed from outside.

In addition to the expression of binding energy W , Bohr derives an expression for the electron revolution frequency (ω) and for orbit dimension of electron in his model (d). All these magnitudes are important since they are predictions of the model that can be contrasted with experimental evidence. The values of frequency ω for different values of n correspond to spectrum frequencies of the emission spectrum of atoms, and the diameter of more stable orbit in the model ($n=1$) correspond to the linear dimension of atom. Bohr shows that the predictions for d , ω and W are of the same order of magnitude as the linear dimensions of the atoms, the optical frequencies, and the ionization-potentials, respectively.

Perhaps the most striking success of the Bohr model is its explanation of the emission spectrum of the hydrogen atom. When it is said that Bohr explained the spectrum of hydrogen, what is claimed is usually that from Bohr model, the Balmer-Rydberg formula can be derived. Bohr interprets each line of the emission spectrum as the radiation emitted by the electron when it makes a transition from a high energy orbit to a lower energy orbit and manages to derive the Balmer-Rydberg formula

$$\nu = R \left(\frac{1}{n} - \frac{1}{m} \right) \quad (2)$$

where ν is the frequency of emitted radiation, R is the Rydberg constant, n and m are integer. Balmer had given this formula in 1885 as a phenomenological model for certain lines of the hydrogen spectrum. Bohr was able to derive not only this formula, but a definition of R constant. When the values of the electron parameters were replaced in the expression found for R , the value of R agrees with the observed value to three significant figures.

Let us here summarize the Bohr proposal. The Bohr model – according to our point of view – shows how the various representational tools used in its construction allow and configure scientific imagination and allows predictions to be made. In fact, the model of Bohr can be seen as an artifact delimitation what can be thought of the minimal portion of reality. The electrons, the particles with positive charge (nucleus), the orbits are constituents of this artefact that obtain their ontological status between

other existing ones through the representational tools of which they are made of. The model also includes rules that determine it. The rules are:

Rules about stationary or stable states.

The electron can move in different closed orbits in turn to nucleus, each orbit correspond to a stable state of electron-nucleus system. Only in these stable states or dynamic equilibrium states, the atomic systems could be described by classical mechanics laws and partially by classical electromagnetism laws. Why the model only partially obeys the laws of classical electromagnetism? Because while the interaction electron-nucleus is described by Coulomb force law, a particular notion of classical electromagnetism is violated by the model: the charged particle do not radiate when it is accelerated in the orbit corresponding to a stationary state. In other words, according to the model, the atom can be partially described by the classical physics when the atom is in a stable state (only in this situation). Each stable state is characterized by an energy value, which represent the amount of energy required to remove the electron from corresponding orbit. There is a state of the model with the lowest energy, this stable state corresponds to the stable state of hydrogen atom.

Rules about transitions between states.

The electromagnetic radiative emission of the electron in the model only occur when the electron jump between stable states. In addition, the rule states that the emitted energy during the transition is a discrete magnitude, i.e. it is quantized. These transitions can not be described by theories of classical physics in any way. The transitions are followed by the emission of energy in a form of a homogeneous electromagnetic radiation field of frequency ν . This frequency is related with the emitted energy W by the Planck relation: $W = h \nu$. The frequency ν is equal to half the frequency of revolution of the electron in its final orbit.

The rules are an inseparable part of the Bohr's atom model and they are imposed with the aim of reformulating the Rutherford model by creating a new model with two objectives. On the one hand, to make the Rutherford atom model compatible with the observed stability of atoms. On the other hand, to give meaning within the model to the different empirical terms of an successful empirical rule or mathematical model: the Balmer-Rydberg formula (Eq. 2).

In the literature, it is understood that the Bohr model is a model of transition between theories. That is to say, there is no theory that is the "text" in terms of which the model is specified, since it takes some laws of Classical Mechanics and Electromagnetism and disregards others explicitly. From our point of view, we believe that this would be a weakness of the model only in case we attribute some degree of reality to it. The priority would be its function for surrogative reasoning beyond the ontological commitments of each theory used. In other words, the model does not need to be essentially a representation of reality, but a mean to interpret phenomena and make verifiable predictions.

The atom is a hypothetical corpuscle that build all material things of cosmos, it is the smallest unit that defines all the characteristics of a chemical element. Unlike Greek atomism, the Bohr model has an internal structure as in the Thomson and Rutherford models. The target has lost the distinctive feature of the proposal of Greek atomism: the atom is not indivisible. The atom could be "cut" and now, the atom itself is mostly empty space. The existence of the new target is in agreement with results of previous experiences to 1913, experiences of analyzing and synthesizing chemical substances during the 19th century, the experiences of J.J. Thomson considered as the discovery of the electron and the particular dispersion pattern of alpha particles by golden sheets of Rutherford experiments suggesting the existence of the atomic nucleus.

From the point of view of representational style of the Bohr model, it could be classified as a phenomenological model. Phenomenological models have been defined in different, though related, ways. According to a traditional definition, phenomenological type is a model that only represent observable properties of their target and refrain from postulating hidden mechanisms as "causes" of these properties. Certainly, the Bohr model is built as a spatial structure (electron-nucleus system) that it can account for of Rutherford experimental results on the existence of a nucleus that concentrates almost all the mass of the atom. The rules added to this structure are introduced without any theoretical justification, except that they have been successfully used in previous models and they appear to be necessary in order to account for experimental facts. At the time of Bohr's proposal, there was no theory from which the model with its rules could be derived or whose axioms acted as a justification of the rules of the model. McMullin [McMullin, 1968] defines phenomenological models as models

that are independent of theories. From this point of view, the Bohr's model is a phenomenological model since it incorporates principles and laws associated with accepted theories and it reject part of them. The Bohr model accepts describing the electron orbit in stationary states by means of classical mechanics principles and the interaction electron-nucleus with the Coulomb force law from the classical electrodynamics, but the model rejects that the electron (charged particle) can emit an electromagnetic field.

Bohr model is no one of the possible models of the atom that coexists with other models with partial explanatory capabilities, as is the case of the different atomic nucleus models (nuclear shell model, liquid drop model, standard model). It is assigned within a limited time frame to a few years after 1913, the exclusivity in the explanation of available experimental evidence related to the atom. This is despite the fact that at the time of its presentation, its possible failure to represent atoms with more than one electron is recognized.

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Beliefs: from inconsistent to consistent

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1 | Introduction

The notion of belief has proven to be one of the most interesting in the epistemological literature.¹ One might even say that belief is more important than knowledge, as in most ‘real’ scenarios the involved agents might not have complete information about the real world, and therefore can be completely certain about very few things. If agents had to act based only on what they know, they would have very little manoeuvring space; luckily, agents can also act based on what they believe.

Within formal epistemology, the notion of belief has been represented in several different ways: syntactically, as either a plain set of formulas or else a deductively closed one; semantically, by means of probability distributions, ordinal functions, a system of spheres or a plausibility relation.² In particular, the *plausibility framework* that surrounds the latter (and in which beliefs are defined as what holds in the most plausible –maximum– epistemic possibilities) has proven to be a flexible setting that allows the representation not only of several other epistemic attitudes (as knowledge and safe, strong and conditional beliefs) but also of processes through which beliefs change (more famously, that of *belief revision*³).

¹ See, e.g., Schwitzgebel, 2015.

² Rott, 2001; Popper, 1955; Rényi, 1955; Spohn, 1988; Grove, 1988; van Benthem, 2007; Baltag and Smets, 2008, respectively.

³ Gärdenfors, 1992; Gärdenfors and Rott, 1994; Williams and Rott, 2001; Olsson and Enqvist, 2011; Tennant, 2012; van Benthem, 2014.

The present work is a rather technical proposal that explores a small variation on the plausibility framework. In a plausibility frame, beliefs are consistent, but ‘real life’ agents might not always satisfy this requirement. Still, this is not necessarily a problem, as in most cases agents are capable to restore consistency by revising their beliefs. This proposal’s contribution is precisely in these two directions. First, it presents a generalisation of (finite) plausibility frames that allows for situations in which the agent has inconsistent beliefs (and, as a consequence, allows the definition of weaker notions of belief); second, it provides strategies through which beliefs can be revised in order to reach consistency. The first is achieved by dropping a requirement from the plausibility relation: instead of asking for it to be a total preorder, the relation is only asked to be a preorder. Then, beliefs might be inconsistent, as maximum worlds might not always exist. The second is achieved by following the *dynamic epistemic logic* approach (*DEL*⁴), where changes in the agent’s epistemic attitudes are semantically described as operations that change the structure representing them. In this case, the provided operations change the plausibility relation in such a way that the existence of maximum worlds is guaranteed, and thus consistency is obtained.

The structure of the paper is as follows. Section 2 recalls the basic definitions of the plausibility setting: a plausibility frame, and the notions of knowledge and beliefs in such structures. Then, Section 3 allows the plausibility relation to be non-total (and thus beliefs to be inconsistent), discussing alternative definitions of knowledge and beliefs, and also two epistemic notions that emerge in this new setting. Section 4 presents operations that make inconsistent beliefs consistent. Finally, Section 5 closes summarizing the work and describing lines for further research.

⁴ van Ditmarsch et al., 2008; van Benthem, 2011.

2 | The plausibility framework

Definition 2.1 (Plausibility frame) A (finite) *plausibility frame* is a tuple $(W, \leq)$ with $W \neq \emptyset$ a finite set of worlds and $\leq \subseteq (W \times W)$ a total preorder (i.e., a total, reflexive and transitive relation, the agent's *plausibility relation* over W , with $u \leq v$ read as “ u is at most as plausible as v ”).^{5 6}

Intuitively, a plausibility frame represents the order the agent assigns to her epistemic possibilities. Thus, while her knowledge is given by what holds in all possible worlds, her beliefs are given by what holds in the most plausible ones. More precisely,

Definition 2.2 (Knowledge and belief) Let $(W, \leq)$ be a plausibility frame. Given $U \subseteq W$, define the set of $\leq$ -maximum elements in U in the standard way:

$$Mxm_{\leq}(U) := \{v \in U \mid \forall u \in U. u \leq v\}$$

Then, given a proposition $P \subseteq W$ and a world $w \in W$, the notions of knowledge (K) and beliefs (B) are defined in the following way.

$K P$ holds at w iff_{def} $W \subseteq P$ (equivalent to $W = P$, as W is the domain)

$B P$ holds at w iff_{def} $Mxm_{\leq}(W) \subseteq P$

In words, the agent knows P at world w ($K P$) if and only if all worlds of the domain are P -worlds, and she believes P ($B P$) if and only if all her $\leq$ -maximum worlds are P -worlds.

Here are two important details to notice. First, the *evaluation point* w is irrelevant when defining the two notions. As $\leq$ is a total preorder, all worlds in W are comparable (i.e., $u \sim v$ holds for all $u, v \in W$); then, knowledge is defined as what holds in all worlds in W (i.e., all the agent's epistemic possibilities), and beliefs is defined as what holds in the $\leq$ -maximum worlds across the whole frame (the most plausible ones).

⁵ As usual, define the *strict plausibility* relation $< \subseteq (W \times W)$ as $u < v$ iff_{def} $u \leq v$ and $v \not\leq u$, the *comparability* relation $\sim \subseteq (W \times W)$ as $u \sim v$ iff_{def} $u \leq v$ or $v \leq u$, and the *equal-comparability* relation $\approx \subseteq (W \times W)$ as $u \approx v$ iff_{def} $u \leq v$ and $v \leq u$.

⁶ In the original Baltag and Smets, 2008, the plausibility relation is also conversely well-founded, forbidding infinite strictly ascending $\leq$ -chains. Here the domain is finite, so this requirement is satisfied automatically.

Second, as W is non-empty and $\leq$ is a total preorder, the set $M_{x m_{\leq}}(W)$ is never empty: thus, beliefs are always consistent (that is, $\mathcal{B} \emptyset$ is never the case).

Finally, the notions of knowledge and belief have been defined for ‘basic’ propositions $P \subseteq W$. But, of course, the agent might have knowledge and beliefs about more complex propositions (as, e.g., their Boolean combination). Here are the (standard) short-cuts that will be used for talking about them:

$$\neg P := W \setminus P, \quad P \wedge Q := P \cap Q.$$

3 | Allowing incomparability (thus inconsistency)

According to Definition 2.2, the agent believes what holds in all her maximum epistemic possibilities; thus, beliefs would be inconsistent if the set of such elements were empty. This is not possible within plausibility models: the plausibility relation is required to be total. But, of course, such requirement can be lifted.

Definition 3.1 (Quasi plausibility frame) A *quasi plausibility frame* is a plausibility frame $(W, \leq)$ in which the plausibility relation $\leq$ is only required to be a preorder.

When dropping totality, some worlds of the domain might not be relevant anymore for defining the notions of knowledge and belief. More precisely, these notions cannot be given by what holds in all worlds in the frame and in the most plausible worlds in it, respectively, as some worlds might not be comparable to the current evaluation point. Thus, the definitions of these epistemic concepts should change.

Definition 3.2 (Knowledge and belief) Let $(W, \leq)$ be a quasi plausibility frame. Given a world $w \in W$, define the *$\leq$ -comparability class of w* (i.e., the set of those worlds that can be $\leq$ -compared to w) as:

$$C_{\leq}(w) := \{ u \in W \mid w \leq u \text{ or } u \leq w \}$$

Then, given a proposition $P \subseteq W$ and a world $w \in W$, the notions of knowledge ($\mathcal{K}$) and beliefs ($\mathcal{B}$) are defined in the following way.

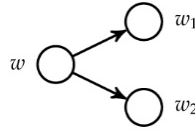
$$\mathcal{K} P \text{ holds at } w \quad \text{iff}_{def} \quad C_{\leq}(w) \subseteq P$$

$$\mathcal{B} P \text{ holds at } w \quad \text{iff}_{def} \quad Mxm_{\leq}(C_{\leq}(w)) \subseteq P.$$

Thus, the agent's knowledge at $w \in W$ is defined as what holds in all the epistemic possibilities that can be compared to w , and her beliefs are given by the most plausible ones among them.

Now, observe how, even though knowledge is still consistent ($C_{\leq}(w)$ is never empty, as $\leq$ is reflexive), beliefs might not have this property ($\leq$ is not total, and thus the set $Mxm_{\leq}(C_{\leq}(w))$ might not have elements).

Example 3.1 Consider the frame below (reflexive edges omitted).



Note how $C_{\leq}(w) = \{w, w_1, w_2\}$ but still $Mxm_{\leq}(C_{\leq}(w)) = Mxm_{\leq}(\{w, w_1, w_2\}) = \emptyset$.

The problem is that the epistemic possibilities might 'evolve' into incomparable branches (in the previous example, w 'evolves' into the incomparable w_1 and w_2). Thus, even though each branch must have maximum elements (the domain is finite), the maximum elements of different branches might be different, and thus there will be no maximum worlds in the comparability class. In these cases, the concept that makes sense is not that of maximum elements, but rather that of *maximals* ones.

Definition 3.3 Let $(W, \leq)$ be a quasi plausibility frame. Given $U \subseteq W$, define the set of $\leq$ -maximal elements in U in the standard way:

$$Mxl_{\leq}(U) := \{ v \in U \mid \neg \exists u \in U. v < u \}$$

Thus, $Mxl_{\leq}(U)$ returns all those worlds that are end-points of branches in U (in Example 3.1, $Mxl_{\leq}(\{w, w_1, w_2\}) = \{w_1, w_2\}$). Note how, if U is non-empty, then $Mxl_{\leq}(U)$ is non-empty too, as otherwise there would

be an infinite $<$ -ascending chain, contradicting W 's finiteness. Hence any non-empty $U \subseteq W$ always has maximal elements.

Clearly these maximal worlds are epistemically significant, as they are the 'most plausible ones' from the agent's point of view. Thus, one can still define a (different) notion of belief in terms of them.

Definition 3.4 (Weak belief) Let $(W, \leq)$ be a quasi plausibility frame. Given a proposition $P \subseteq W$ and a world $w \in W$, the notion of *weak belief* ($\mathcal{B}^w$) is defined in the following way.

$$\mathcal{B}^w P \text{ holds at } w \text{ iff}_{def} MxI_{\leq}(C_{\leq}(w)) \subseteq P.$$

Thus, the agent weakly believes what holds in all her maximal epistemic possibilities within $C_{\leq}(w)$. An alternative would be to define weak beliefs as what hold not in the maximals among all worlds that are $\leq$ -comparable to w (the set $C_{\leq}(w)$), but rather in the maximals among all worlds that are $\leq$ -reachable from w . As the following proposition shows, and due to the reflexivity and transitivity of $\leq$, these two options produce the same result.

Proposition 3.1 Let $(W, \leq)$ be a quasi plausibility frame. Given any relation $S \subseteq (W \times W)$ and any $w \in W$, define the set of worlds that are S -reachable from w as

$$R_S(w) := \{u \in W \mid Swu\}$$

Then, for any $w \in W$,

$$MxI_{\leq}(C_{\leq}(w)) = MxI_{\leq}(R_{\leq}(w)).$$

Proof. ($\subseteq$) By definition, $u \in MxI_{\leq}(C_{\leq}(w))$ implies both that $u \in C_{\leq}(w)$ and that no world in $C_{\leq}(w)$ is strictly $\leq$ -above u . Now, $u \in MxI_{\leq}(C_{\leq}(w))$ implies $u \in R_{\leq}(w)$ (otherwise, u would not be a maximal in $C_{\leq}(w)$, as $u \in C_{\leq}(w)$ but $u \notin R_{\leq}(w)$ would imply $u < w$ with $w \in C_{\leq}(w)$ –the latter due to reflexivity–). Moreover, if no world in $C_{\leq}(w)$ is strictly $\leq$ -above u , then no world in the potentially smaller $R_{\leq}(w)$ (clearly $R_{\leq}(w) \subseteq C_{\leq}(w)$) is strictly $\leq$ -above u . Hence, $u \in MxI_{\leq}(R_{\leq}(w))$. ($\supseteq$) Take $u \in MxI_{\leq}(R_{\leq}(w))$; then $u \in R_{\leq}(w)$ (and thus $w \leq u$) and

no world in $R_{\leq}(w)$ is strictly $\leq$ -above u . Note how every $v \in C_{\leq}(w) \setminus R_{\leq}(w)$ is such that $v < w$; thus, by transitivity, $v < u$. Hence, no v in $C_{\leq}(w) \setminus R_{\leq}(w)$ can be strictly $\leq$ -above u , so no world in $C_{\leq}(w)$ can be strictly $\leq$ -above u , and thus $u \in MxI_{\leq}(C_{\leq}(w))$.

As $C_{\leq}(w)$ is never empty, $MxI_{\leq}(C_{\leq}(w))$ is never empty, and thus weak beliefs are consistent (in symbols, $\mathcal{B}^w \emptyset$ is never the case).

It is possible to define an even weaker notion of belief. Instead of focussing on what holds at *all* maximal worlds, one can only focus on what holds in some of them. Of course, not every subset of maximal worlds make sense (it might contain worlds belonging to different branches); here is a way to select meaningful ones.

Definition 3.5 Let $(W, \leq)$ be a quasi plausibility frame. For any $U \subseteq W$, define its partition into $\simeq$ -equivalence classes⁷ as

$$Pt_{\simeq}(U) := \{R_{\simeq}(u) \subseteq W \mid u \in U\}$$

Then,

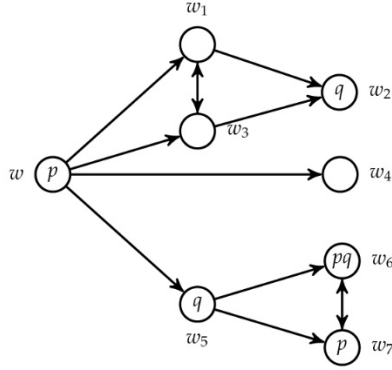
Definition 3.6 (Very weak belief) Let $(W, \leq)$ be a quasi plausibility frame. Given a proposition $P \subseteq W$ and a world $w \in W$, the notion of *very weak belief* ($\mathcal{B}^{vw}$) is defined in the following way.

$$\mathcal{B}^{vw} P \text{ holds at } w \quad \text{iff}_{def} \quad \text{there is } U \in Pt_{\simeq}(MxI_{\leq}(C_{\leq}(w))) \text{ such that } U \subseteq P$$

In words, while the weak belief of Definition 3.4 looks at what holds in all maximal worlds within w 's comparability class (that is, at what holds in *all* end-points of *all* branches from w), the very weak belief of Definition 3.6 looks at what holds in some $\simeq$ -maximal subset of such maximal worlds within w 's comparability class (that is, at what holds in *all* end-points of *some* branch from w). Note how, since $Pt_{\simeq}(MxI_{\leq}(C_{\leq}(w)))$ defines a partition of $MxI_{\leq}(C_{\leq}(w))$, it never contains $\emptyset$.

⁷ That $Pt_{\simeq}(U)$ indeed defines a partition of U follows from the fact that $\simeq$ is an equivalence relation.

Example 3.2 Consider the frame below (reflexive and transitive edges omitted) and the propositions $P := \{w, w_6, w_7\}$ and $Q := \{w_2, w_5, w_6\}$.



Note how $C_{\leq}(w) = \{w, w_1, w_2, w_3, w_4, w_5, w_6, w_7\}$. Moreover,

$$Mxm_{\leq}(C_{\leq}(w)) = \emptyset \quad \text{but} \quad Mxl_{\leq}(C_{\leq}(w)) = \{w_2, w_4, w_6, w_7\}$$

Finally,

$$Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w))) := \{\{w_2\}, \{w_4\}, \{w_6, w_7\}\}$$

Thus, for example,

- $\mathcal{B} \emptyset$ holds at w (as $Mxm_{\leq}(C_{\leq}(w)) = \emptyset$), but nevertheless
- both $\mathcal{B}^w \emptyset$ and $\mathcal{B}^{vw} \emptyset$ fail at w (since both $Mxm_{\leq}(C_{\leq}(w)) \neq \emptyset$ and $\emptyset \notin Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$, respectively).

Maybe more interesting, while $\mathcal{B}^w P$, $\mathcal{B}^w Q$ and $\mathcal{B}^w(\bar{P} \cap \bar{Q})$ all fail (witness, for example, w_4 for the first two, and w_2 for the last),

- $\mathcal{B}^{vw} P$ holds at w (witness $\{w_6, w_7\} \in Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$),
- $\mathcal{B}^{vw} Q$ holds at w (witness $\{w_2\} \in Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$) and
- $\mathcal{B}^{vw}(\neg P \wedge \neg Q)$ holds at w (witness $\{w_4\} \in Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$).

More generally, given the quantification pattern of each notion of belief and the relationship between their definitions,

Proposition 3.2

- $\mathcal{B} \emptyset$ holds for some worlds of some frames (beliefs might be inconsistent).
- $\mathcal{B}^w \emptyset$ fails for every world of every frame (weak beliefs are always consistent).
- $\mathcal{B}^{vw} \emptyset$ fails for every world of every frame (very weak beliefs are always consistent).

Moreover, given $P, Q \subseteq W$,

- $\mathcal{B}^w(P \cap Q)$ and $(\mathcal{B}^w P \cap \mathcal{B}^w Q)$ coincide for every world of every frame (weak beliefs are closed under conjunction elimination and conjunction introduction), but
- although $\mathcal{B}^{vw}(P \cap Q)$ implies $(\mathcal{B}^{vw} P \cap \mathcal{B}^{vw} Q)$ for every world of every frame (very weak beliefs are closed under conjunction elimination), $(\mathcal{B}^{vw} P \cap \mathcal{B}^{vw} Q)$ does not imply $\mathcal{B}^{vw}(P \cap Q)$ for some worlds of some frames (very weak beliefs are not closed under conjunction introduction).

Finally, it is useful to emphasise that this weaker quasi plausibility setting is indeed a generalisation of the plausibility one, not only because it relaxes a requirement in the latter, but also because the epistemic notions defined in the former (knowledge, belief, weak belief and very weak belief) coincide with those defined in Section 2 when $\leq$ is total. More precisely,

Proposition 3.3 *Let $(W, \leq)$ be a quasi plausibility frame; take any $w \in W$ and any $P \subseteq W$. If $\leq$ is total, then*

1. $K P$ and $\mathcal{K} P$ coincide,
2. $\mathcal{B} P, \mathcal{B}^w P, \mathcal{B}^{vw} P$ and $\mathcal{B}^{vw} P$ all coincide.

Proof. Suppose $\leq$ is total. Then,

1. By Definition 3.2, $\mathcal{K} P$ iff $C_{\leq}(w) \subseteq P$. But $\leq$ is total; thus, $C_{\leq}(w) = W$ so $\mathcal{K} P$ iff $W \subseteq P$. Hence, by Definition 2.2, $\mathcal{K} P$ iff $K P$.
2. By Definition 3.6, $\mathcal{B}^{vw} P$ iff there is $U \in Pt_{\approx}(MxI_{\leq}(C_{\leq}(w)))$ such that $U \subseteq P$. Since $\leq$ is total, $MxI_{\leq}(V)$ and $Mxm_{\leq}(V)$ coincide for all $V \subseteq W$; thus, $\mathcal{B}^{vw} P$ iff there is $U \in Pt_{\approx}(Mxm_{\leq}(C_{\leq}(w)))$ such that $U \subseteq P$. But then, by definition, all elements of $Mxm_{\leq}(C_{\leq}(w))$ are pairwise $\approx$ -related. Hence, $Pt_{\approx}(Mxm_{\leq}(C_{\leq}(w)))$ has a single element: $Mxm_{\leq}(C_{\leq}(w))$ itself. Therefore,

$\mathcal{B}^{vw} P$ iff $Mxm_{\leq}(C_{\leq}(w)) \subseteq P$ iff $Mxl_{\leq}(C_{\leq}(w)) \subseteq P$; thus, by Definitions 3.2 and 3.4, $\mathcal{B}^{vw} P$ iff $\mathcal{B} P$ iff $\mathcal{B}^w P$.

Finally, as $\leq$ is total, $C_{\leq}(w) = W$, so $\mathcal{B}^{vw} P$ iff $Mxm_{\leq}(W) \subseteq P$. Hence, by Definition 2.2, $\mathcal{B}^{vw} P$ iff $\mathcal{B} P$.

Caveat: neighbourhood semantics The reader familiar with neighbourhood ‘minimal’ frames/models⁸ might have recognized the similarity between the semantic definition of *very weak belief* and the ‘subset’ semantic interpretation of the modal operator $\Box$ in a neighbourhood setting. For those not acquainted, a neighbourhood frame (W, N) is a generalisation of a relational frame in which the so-called *neighborhood function* $N : W \rightarrow \wp(\wp(W))$ assigns a family of subsets of the domain to each possible world.⁹ In such setting, the modal operator $\Box$ can be semantically interpreted in at least two different ways. Take a proposition $P \subseteq W$. The ‘membership’ semantic interpretation states that $\Box P$ holds at w iff P is in the neighbourhood of w :

$$\Box P \text{ holds at } w \quad \text{iff}_{def} \quad P \in N(w).$$

In contrast, the ‘subset’ semantic interpretation states that $\Box P$ holds at w iff the neighbourhood of w contains a subset of P :

$$\Box P \text{ holds at } w \quad \text{iff}_{def} \quad \exists U \in N(w) . U \subseteq P.$$

It is not difficult to see the correspondence between the very weak belief of above ($\mathcal{B}^{vw}$; Definition 3.6) and the subset case of $\Box$ in a neighbourhood setting. Indeed, given a quasi plausibility frame $(W, \leq)$, one can define a neighbourhood frame $(W, N_{\leq})$ by taking $N_{\leq}(w) := Pt_{\leq}(Mxl_{\leq}(C_{\leq}(w)))$ for each $w \in W$; under such definition, $\mathcal{B}^{vw}$ and $\Box$ coincide. However, it should be noted that a quasi plausibility frame still provides more information: while the neighbourhood function only indicates the end points of the branches, the plausibility relation describes the full ordering, including the relative position of the intermediate points.

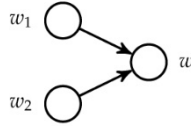
⁸ Scott, 1970; Montague, 1970.

⁹ In contrast, in a relational frame (W, R) , the relation $R \subseteq (W \times W)$ can be understood as a function $R : W \rightarrow \wp(W)$ assigning a subset of the domain to each possible world ($w \rightarrow^R \{u \in W \mid Rwu\}$).

4 | From inconsistency to consistency

As it has been mentioned, while beliefs in a plausibility frame are consistent, beliefs in a quasi plausibility frames might not have such property (though weak beliefs and very weak beliefs do). However, being inconsistent does not mean that one will always be.

From a technical point of view, what allow inconsistent beliefs is the dropping of the totality requirement on the plausibility relation; thus, a way to recover consistency is to make the relation total. However, although totality indeed guarantees consistent beliefs, consistent beliefs do not require totality. Consider, for example, the frame below (reflexive edges omitted). The plausibility relation is not total (w_1 and w_2 are not comparable), and nevertheless the beliefs of the agent from any world are consistent, as $Mxm_{\leq}(C_{\leq}(w_1)) = Mxm_{\leq}(C_{\leq}(w_2)) = Mxm_{\leq}(C_{\leq}(w)) = \{w\}$.



Then, the goal should not be to make the relation total, but rather to guarantee that, for any world $w \in W$, the set $Mxm_{\leq}(C_{\leq}(w))$ is not empty. This can be achieved simply by connecting all maximal worlds.

Still, note how there is more than one way to do this. Take a possible world w and consider the set $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$. As discussed above, this set contains the end-points of all branches from w grouped by equal plausibility (see Example 3.2, where $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w))) := \{\{w_2\}, \{w_4\}, \{w_6, w_7\}\}$). Then, one can arrange all these worlds in either one single layer (all worlds in $\bigcup Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$ become equally plausible), or else in two layers (partition $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$ in two components, making the worlds within each component equally plausible, and then decide which one will be at the top), or else in three layers (partition $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$ in three components, making the worlds within each component equally plausible, and then decide the components' order), and so on. If $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$ has n elements, then there are

$$\sum_{k=1}^n S(n, k) \cdot k!$$

different ways to arrange such elements in plausibility layers, with $S(n, k)$ the number of ways to partition a set of n objects into k non-empty subsets (the number of layers),¹⁰ and $k!$ the number of ways to order such subsets (the order among the layers). Here the focus will be the two extreme cases: creating a single layer or creating n of them.

One final detail: relating the worlds in the members of $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$ is not enough. This is because the plausibility relation in a quasi plausibility frame is required to be a preorder, and thus so should be the one that will result from the to-be-defined operation. Reflexivity is satisfied automatically, as the approach is to add edges, and the original relation already contain the reflexive ones. On the other hand, transitivity might require further arrangements.¹¹

4.1 | Creating a single layer

Conceptually, this operation represents an action through which the agent decides that all her 'divergent' epistemic alternatives are equally relevant, and thus she revises her beliefs by simply merging the end-points of all of them.

If the goal is to create a single layer containing all worlds in the members of $Pt_{\approx}(Mxl_{\leq}(C_{\leq}(w)))$, then one can work directly with the set $Mxl_{\leq}(C_{\leq}(w))$, as its $\approx$ -partition is not important anymore. The definition below follows this spirit, taking additional care of not adding edges across different *weakly $\leq$ -connected components* of the frame. For this, first some definitions.

Definition 4.1 (Weak $\leq$ -connectedness) Let $(W, \leq)$ be a quasi plausibility frame. Given $w_1, w_2 \in W$, world w_1 is *weakly $\leq$ -connected* to world w_2 when there is a non-empty sequence of edges in $\leq \cup \geq$ from w_1 to w_2 .¹² The notion of weak $\leq$ -connectedness is an equivalence relation. First, the reflexivity of $\leq$ makes this notion reflexive; then, if w_1 is weakly $\leq$ -connected to w_2 , then clearly w_2 is weakly $\leq$ -connected to w_1 (take the same sequence of edges);

¹⁰ Such numbers are called *Stirling numbers of the second kind*, and also *Stirling set numbers*.

¹¹ Consider the frame in Example 3.2. If an edge from w_2 to w_4 is added then, in order to preserve transitivity, one should also add an edge to w_4 from w_1 and w_3 .

¹² For contrast, w_1 is $\leq$ -connected to world w_2 when there is a non-empty sequence of edges in $\leq$ from w_1 to w_2 .

finally, the weak $\leq$ -connectedness of w_1, w_2 and w_2, w_3 imply that of w_1, w_3 (put the sequence of edges together). Thus, the relation of weak $\leq$ -connectedness defines a partition among W ; denote such partition by $WCC_{\leq}(W)$.

Note how a weakly $\leq$ -connected component is not the same as a comparability class. In Example 3.1, the whole domain $\{w, w_1, w_2\}$ is a single weakly $\leq$ -connected component (there is a non-empty sequence in $\leq_U$ between any pair of worlds), and nevertheless the comparability classes of w_1 and w_2 are $\{w_1, w\}$ and $\{w_2, w\}$, respectively. Note also how elements of different weakly $\leq$ -connected components are, by definition, incomparable (i.e., $U_1, U_2 \in WCC_{\leq}(W)$ with $U_1 \neq U_2$ imply $u_1 \not\sim u_2$ for any $u_1 \in U_1$ and $u_2 \in U_2$).

Here is, then, the definition of the new plausibility relation. It works separately with each weakly $\leq$ -connected component of the given frame, then putting all the pieces together.

Definition 4.2 Let $(W, \leq)$ be a quasi plausibility frame. The set of maximals of each weakly $\leq$ -connected component $U \in WCC_{\leq}(W)$, denoted by $MC_{\leq}(U)$, is given by the union of the maximals of each element of U :

$$MC_{\leq}(U) := \bigcup_{u \in U} MxI_{\leq}(C_{\leq}(u)). \quad 13$$

Then, for each $U \in WCC_{\leq}(W)$, define its new plausibility ordering as

$$\leq'_U := (\leq|_U \cup (MC_{\leq}(U) \times MC_{\leq}(U)))^+. \quad 14$$

Thus, the new plausibility ordering within U , denoted by $\leq'_U$, will be the transitive closure (in symbols, $^+$) of the relation that extends the old plausibility ordering within U (in symbols, $\leq|_U$) with edges between any pair of worlds within $MC_{\leq}(U)$ (in symbols, $MC_{\leq}(U) \times MC_{\leq}(U)$).

Finally, the new plausibility ordering is just the union of the new ones for each weakly $\leq$ -connected component:

$$\leq := \bigcup_{U \in WCC_{\leq}(W)} \leq'_U$$

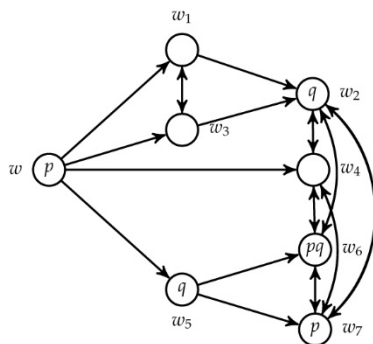
¹³ As illustration, consider Example 3.1, where $MC_{\leq}(W) = \bigcup_{u \in \{w, w_1, w_2\}} MxI_{\leq}(C_{\leq}(u)) = MxI_{\leq}(C_{\leq}(w)) \cup MxI_{\leq}(C_{\leq}(w_1)) \cup MxI_{\leq}(C_{\leq}(w_2)) = \{w_1, w_2\} \cup \{w_1\} \cup \{w_2\} = \{w_1, w_2\}$.

¹⁴ With $\leq|_U := \leq \cap (U \times U)$ the restriction of $\leq$ to U , and S^+ the transitive closure of a relation S .

The fact that the elements of each weakly $\leq$ -connected component are incomparable allow us to work independently with each one of them. With respect to the transitivity issue, in general the union of transitive relations might not be transitive. But the property is preserved when their domains are mutually exclusive, as it happens in this case. Hence,

Proposition 4.1 *Let $F = (W, \leq)$ be a quasi plausibility frame. Then the structure $F_- = (W, \leq')$, with $\leq'$ as in Definition 4.2, is a quasi plausibility frame, as $\leq'$ is indeed reflexive and transitive.*

Example 4.1 Consider the frame of Example 3.2, which describes the epistemic state of an agent with inconsistent beliefs, and which contains a single weakly $\leq$ -connected component W . Note how $MC_{\leq}(W) = \{w_2, w_4, w_6, w_7\}$ and thus, after applying the operation of Definition 4.2, the resulting frame is the following (again, reflexive and transitive edges omitted):



As it is clear from the diagram (recall, transitivity is omitted), $Mxm_{\leq}(C_{\leq}(w)) = \{w_2, w_4, w_6, w_7\}$ and therefore $\mathcal{B} \emptyset$ is no longer the case: by means of this process, the agent's beliefs are now consistent. As examples, and by taking $P := \{w, w_6, w_7\}$, note how $\mathcal{B}(P \vee \neg P)$ holds but $\mathcal{B} P$ fails.

4.2 | Creating n layers

Instead of deciding that all her 'divergent' epistemic alternatives are equally relevant, the agent might rank each one of them differently. Thus, when revising her beliefs to make them consistent, each set of end-points will become a different layer. Here is the definition of an

operation following such idea, working again separately with each weak $\leq$ -connected component.

Definition 4.3 Let $(W, \leq)$ be a quasi plausibility frame. For each $U \in WCC_{\leq}(W)$, take the union of the maximals, $MC_{\leq}(U)$ (Definition 4.2), and then its partition in $\simeq$ -equivalence classes, $Pt_{\simeq}(MC_{\leq}(U))$ (Definition 3.5); let $|Pt_{\simeq}(MC_{\leq}(U))| = n$. Assume the priority of the elements of $Pt_{\simeq}(MC_{\leq}(U))$ is given by a permutation π , with $\pi(1)$ returning the element of $Pt_{\simeq}(MC_{\leq}(U))$ with the lowest priority (and thus the one that will be at the bottom among them) and $\pi(n)$ returning the element of $Pt_{\simeq}(MC_{\leq}(U))$ with the highest priority (and thus the one that will be at the top among them).

For each $U \in WCC_{\leq}(W)$, define its new plausibility ordering as

$$\leq'_U := (\leq|_U \cup \bigcup_{i=1}^{n-1} \bigcup_{j=i+1}^n (\pi(i) \times \pi(j)))^+$$

Thus, the new plausibility ordering within U , denoted by $\leq'_U$, will be the transitive closure (in symbols, $^+$) of the relation that extends the old plausibility ordering within U (in symbols, $\leq|_U$) with edges from the elements of $\pi(i)$ to the elements of $\pi(j)$, for each $i < j$ (in symbols, $\bigcup_{i=1}^{n-1} \bigcup_{j=i+1}^n \pi(i) \times \pi(j)$).

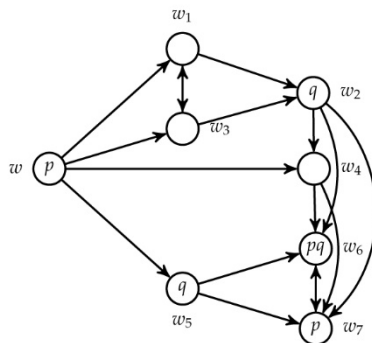
Finally, the new plausibility ordering is just the union of the new ones for each weakly $\leq$ -connected component:

$$\leq' := \bigcup_{U \in WCC_{\leq}(W)} \leq'_U$$

Proposition 4.2 Let $F = (W, \leq)$ be a quasi plausibility frame. Then the structure $F_{\equiv} = (W, \leq')$, with $\leq'$ as in Definition 4.3, is a quasi plausibility frame, as $\leq'$ is indeed reflexive and transitive.

Here is an example of this operation in action.

Example 4.2 Consider again the inconsistent-beliefs frame of Example 3.2; recall that $MC_{\leq}(W) = \{w_2, w_4, w_6, w_7\}$, so $Pt_{\simeq}(MC_{\leq}(U)) = \{\{w_2\}, \{w_4\}, \{w_6, w_7\}\}$. Suppose the priority ordering is given by $\pi = [\{w_2\}, \{w_4\}, \{w_6, w_7\}]$. Then, after applying the operation of Definition 4.3, the resulting frame is the following (again, reflexive and transitive edges omitted):



As it is clear from the diagram, $Mxm_{\leq}(C_{\leq}(w)) = \{w_6, w_7\}$ and therefore $\mathcal{B} \emptyset$ is no longer the case: by means of this process, the agent's beliefs are now consistent. As examples, and by taking $P := \{w, w_6, w_7\}$ again, observe how $\mathcal{B}(P \vee \neg P)$ still holds, but now so does $\mathcal{B} P$.

5 | Conclusions and further work

The present work proposes a generalisation of the plausibility framework for representing beliefs. The generalisation consists on not requiring for the plausibility relation to be total, which makes it possible for the agent to have inconsistent beliefs, and thus allow a more faithful representation of 'real life' situations. But, in real life, being inconsistent does not mean that one will always be. In order to allow the agent to revise her beliefs, this work also introduces two model operations, thus providing the agent epistemic actions to reach consistency.

There are several aspects of this proposal that can be further explored; here we mention two of them. The first concerns a formal language to describe the studied structures and to define the proposed epistemic concepts. This paper's presentation has been entirely semantic, but the introduction of formal (modal) language will increase the interest of the proposed setting, as it will be possible to talk about the beliefs the agent might or might not have about her own beliefs (that is, high-order beliefs), including the concepts of positive and negative introspection. A formal language also raises questions on *definability*: which kind of modal operators are needed in order to characterise syntactically the discussed epistemic notions? Which is a sound and complete axiomatisation for

such language? In case the language is a non-standard one, is it possible to characterise its expressivity semantically by means of some form of bisimulation? And, when extending the language with modalities for the proposed belief revision operators (in order to talk about the changes the revision process induces), does the expressivity remain the same? Is it increased? Which is a sound and complete axiom system for this extended language?

The second concerns the jump to a multi-agent setting, which brings forward *multi-agent* epistemic notions (as common knowledge and common belief) and *private* forms of belief revision.

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Belief Revision Revisited

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Abstract

We review the well-known AGM approach to belief revision and critically evaluate some of its fundamental assumptions invoking some principle from philosophy of science.

The view that machines cannot give rise to surprises is due, I believe, to a fallacy to which philosophers and mathematicians are particularly subject. This is the assumption that as soon as a fact is presented to a mind all consequences of that fact spring into the mind simultaneously with it. It is a very useful assumption under many circumstances, but one too easily forgets that it is false.

Alan M. Turing¹

1 | Introduction

The postulates for belief contraction and revision introduced by Carlos Alchourrón, Peter Gärdenfors, and David Makinson in 1985² have turned out to be an extremely fruitful starting point for abstract accounts to belief revision. Although single postulates were criticised and sometimes replaced by alternative formulations, the general framework, labelled by the acronym AGM, is today established as the standard approach to belief revision.

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¹ Turing, 1950, p. 451; cited also in Aaronson, 2013, p. 261.

² Alchourrón, Gärdenfors, and Makinson, 1985.

The aim of this paper is to have a critical view to the rational behind AGM, using partly a metatheoretical perspective, invoking, for instance, Hilbert's axiomatic method or by put it in context with Kuhn's concept of scientific revolution.

In the course of this review we will question some fundamental assumptions of AGM. These critics are not new in the sense that one can find them discussed, in one or the other way, within the AGM community. Often they are just taken as starting points for local refinements; we, however, would like to take the critics much further and challenge not only the applicability of AGM-like theories to a person's belief, but even the rationality standards of the fundamental postulates.

As conclusion, and despite of all merits of AGM, we ask for a fresh account to belief revision addressing these challenges.

2 | A brief review of AGM

In this section, we give a review of the presentation of AGM in the survey article AGM 25 Years, a paper celebrating *Twenty-Five Years of Research in Belief Change*.³ This source is complement to the excellent entry of Hansson in the Stanford Encyclopedia of Philosophy.⁴ We cite first the compact presentation of AGM given by Fermé and Hansson.⁵

In the basic AGM framework a person's belief state is represented by a set K of sentences that is closed under the operation Cn of logical consequence, i.e. $K = Cn(K)$. In order to contract K by a sentence p we consider the set $K \perp p$ of all maximal subsets of K not implying p . A selection function γ selects the most plausible elements of $K \perp p$. Whenever $K \perp p$ is non-empty, $\gamma(K \perp p)$ is a non-empty subset of $K \perp p$. The *partial meet contraction* $\sim\gamma$ based on γ is defined as $K \sim\gamma p = \cap \gamma(K \perp p)$. In the limiting case when p is a tautology, $K \sim\gamma p = K$. If $\gamma(K \perp p) = K \perp p$ for all p , then $\sim\gamma$ is the *full meet contraction*, also denoted $\sim$. If $\gamma(K \perp p)$ is always a singleton when $K \perp p$ is non-empty, then $\sim\gamma$ is a *maxichoice contraction*. If there is a transitive relation $\sqsubseteq$

³ Fermé and Hansson, 2011.

⁴ Hansson, 2011.

⁵ Fermé and Hansson, 2011, §2.1 AGM Briefly Summarized.

such that $\gamma(K \perp p) = \{X \in K \perp p \mid (\forall Y \in K \perp p)(Y \sqsubseteq X)\}$ for all nonempty $K \perp p$, then $\sim\gamma$ is transitively relational. There are two operations that add beliefs to a belief set: $K + p = \text{Cn}(K \cup \{p\})$ (expansion) and $K * p = \text{Cn}[(K \sim_\gamma \neg p) \cup \{p\}]$ (revision). In the AGM article, axiomatic characterizations were provided for all these operations. These axioms are listed in Table 1. Subsequently, several other equivalent characterizations have been developed.

Table 1 | The Standard AGM postulates

Basic contraction postulates	Basic revision postulates
$K \div p = \text{Cn}(K \div p)$ (closure)	$K * p = \text{Cn}(K * p)$ (closure)
$K \div p \subseteq K$ (inclusion)	$K * p \subseteq K + p$ (inclusion)
If $K \not\vdash p$ then $K \div p = K$ (vacuity)	If $K \not\vdash \neg p$ then $K + p \subseteq K * p$ (vacuity)
If $\not\vdash p$ then $p \notin K \div p$ (success)	$p \in K * p$ (success)
If $\vdash p \leftrightarrow q$ then $K \div p = K \div q$ (extensionality)	If $\vdash p \leftrightarrow q$ then $K * p = K * q$ (extensionality)
$K \subseteq (K \div p) + p$ (recovery)	If $p \not\vdash \perp$ then $K * p \not\vdash \perp$ (consistency)
Supplementary contraction postulates	Supplementary revision postulates
$(K \div p) \cap (K \div q) \subseteq K \div (p \& q)$ (conjunctive overlap)	$K * (p \& q) \subseteq (K * p) + q$ (superexpansion)
If $p \notin K \div (p \& q)$ then $K \div (p \& q) \subseteq K \div p$ (conjunctive inclusion)	If $K * p \not\vdash \neg q$ then $(K * p) + q \subseteq K * (p \& q)$ (subexpansion)

The six basic contraction postulates exactly characterize partial meet contraction, and all eight contraction postulates exactly characterize transitively relational partial meet contraction. Similarly, the six basic revision postulates exactly characterize partial meet revision, and all eight revision postulates exactly characterize transitively relational partial meet revision. $X \vdash p$ is an alternative notation for $p \in \text{Cn}(X)$ and $\vdash p$ for $p \in \text{Cn}(\emptyset)$

The paper goes on with presenting refined concepts *as safe and kernel contraction, epistemic entrenchment, Grove's spheres, distance models, and specified meet contraction*. After a short but informative section on *criticism of the model* an extensive discussion of modification of the AGM model follows, including *extended representations of belief states, iterated change, and alternative operations of change*. The final section concerns *applications and connections* to other areas of research and is followed by an impressive list of 276 references.

The state of the art of AGM-like belief revision provides one with an overwhelming amount of sophisticated concepts and operations which try to take into account countless examples (and counter-examples) of belief changes. Even though some approaches reject some of the original AGM postulates, all of them are elaborated in the spirit of the original approach. And this spirit is in excellent company as we will discuss in the next section.

3 | AGM as instance of Hilbert's axiomatic method

At the turn of the 19th/20th century, David Hilbert advocated what he called *axiomatic method* as a tool to clarify the “position and logical relation of theorems” within a “system of known theorems”.⁶ This method was to invoke whenever we encounter problems in a scientific realm, asking for a logical reconstruction of certain theories. Hilbert himself used it first in his seminal new axiomatization of Geometry⁷. This method was also used by Ernst Zermelo to axiomatize set theory.⁸ In his problem list at the International Congress of Mathematicians in Paris in 1900, Hilbert posed as sixth problem the one of *axiomatization of physics*.⁹ While this problem is today considered as too vague, it is notable that Hilbert included probability theory under physics, and Kolmogorov's axiomatization from 1933¹⁰ serves as another example for the success of the axiomatic method.

A specific feature of the axiomatic method is that it does not need to be concerned with a concrete example, but with the general (abstract) relations which, in a first approximation, are just the clear and uncontroversal principles of the “the intended model”.¹¹ It is only the second step to aim for a complete axiomatization. This step may not even be applicable and in such cases the study of the different possible structures for a given (partial) axiomatization can be very fruitful.

Belief revision attracted the attention of analytic philosophy in the XXth century as a specific topic of epistemology. As the intensional structure

⁶ Hilbert, 1902/03, p. 50; cited in Ebbinghaus, 2007, p. 40.

⁷ Hilbert, 1899.

⁸ Zermelo, 1908.

⁹ Hilbert, 1900.

¹⁰ Kolmogoroff, 1933.

¹¹ We put “the intended model” in quotes, since this nowadays familiar notion came up only after the axiomatic method was successfully installed in mathematical logic.

of beliefs gives rise to problems when defining concrete belief revision operators, it is the merit of Alchourrón, Gärdenfors, and Makinson to have applied here Hilbert's axiomatic method to provide an abstract theory. This theory does not give us any concrete belief revision operation, but only specifies the (minimal?) properties any such operation should fulfill. We may leave open, whether the AGM postulates "characterize" the introduced operations (as emphasized in the caption of Table 1). But, as a matter of fact, these postulates—their discussion and even their modification—gave rise to interesting studies of possible alternatives in the concrete design of different belief revision operations.

In this perspective, AGM is fully in line with the axiomatic method. However, critics are to come.

4 | Logical Omniscience

The postulates of AGM are explicitly considered as "rationality postulates".¹² The understanding of rationality includes, in particular, that beliefs are supposed to be deductively closed: $K = \text{Cn}(K)$.

As natural this condition seems to be for "rationality", it has devastating consequences. These consequences are related to a conceptional pitfall of the axiomatic method: it involves *idealizations*. They might be useful (or even necessary) sometimes as, for instance, in the case of a straight line in Geometry. But one runs, however, into the risk of "overridealization", i.e., presupposing an idealization which, in practice, disables the formal framework when it comes to applications. This is, unfortunately, the case for the demand of deductively closed beliefs, where AGM ignored Turing's warning given at the beginning of this paper.

4.1| Mathematics

AGM disqualifies itself for beliefs including Mathematics. It is not only an empirical fact that our beliefs don't cover all mathematical truths, Gödel's first incompleteness theorem even puts a definite end on any hope to have a recursive representation of deductively closed rational beliefs. But we do

¹² Hansson, 2011.

not need to invoke Gödel (or just Church and the undecidability of first-order logic): any deductive closure whose computational complexity exceeds polynomial time is generally considered as intractable. This is somehow admitted by Fermé and Hansson when they cite¹³ Nebel with the following conclusion:¹⁴

The general revision problem for propositional logic appears to be hopelessly infeasible from a computational point of view because they are located on the second level of the polynomial hierarchy.

4.2 | Consistency

The second problem is of another nature: to avoid triviality, deductively closed beliefs need to be *consistent*. As desirable this property is for rational beliefs, it is simply a matter of fact that beliefs are often inconsistent. Excluding these beliefs as “irrational” would—in our view—limit the applicability of AGM to rather uninteresting cases of beliefs. The subtle issue is here that, very often, the inconsistency of a new belief with the existing ones is simply not realized. Coming back to mathematical examples, it doesn’t seem to be irrational at all to *believe* in a certain, so far unproven, mathematical hypothesis, even if it might turn out that its negation belongs to the deductive closure of our mathematical axioms. Or, to put it differently, it is easy to imagine to find two “perfectly rational” mathematicians, each of them believing in the negation of a hypothesis in which the other one believes. But even outside mathematics, the possibility to have inconsistent beliefs seems to us one of the essential features of beliefs (in contrast to knowledge, for instance). This second problem will be discussed in more detail below in §5.1.

¹³ See the end of the paper Fermé and Hansson, 2011.

¹⁴ Nebel, 1998. When AGM is successfully applied in Artificial Intelligence, it often comes with an analysis of the computational complexity of the belief revision operations—which is supposed to be polynomial; in consequence, these applications cannot even cover full propositional logic.

4.3 | Disjunctions

A known problem of (the logical omniscience of) belief sets is concerned with “silly” disjunctions which enter automatically in any belief set: With a belief ϕ and for an arbitrary sentence, ψ , $\phi \vee \psi$ is, of course, in $\text{Cn}(\phi)$. It does not seem to be very rational to demand the belief in such disjunctions. The problem becomes evident when such a belief is revised by $\neg\phi$. It is, of course, the hope that with ϕ also $\phi \vee \psi$ will leave the belief set; otherwise one would be doomed to start believing in ψ no matter how absurd it is. Problems of this kind are known from *maxichoice contraction* and AGM was, at least in part, set up to address such problems.¹⁵

No matter how we define a belief revision operation to avoid that “silly” disjuncts end up in our belief set after a revision, we consider it important that not every disjunction expanding a believed formula will enter in the belief set in the first place. Apparently, in ordinary language there is an easy test to check whether a particular disjunction should be believed or not: assuming, again, that ψ is believed; so one should be committed to believe in $\phi \vee \psi$ if and only if one would subscribe the statement: “If ϕ weren’t the case, then I would believe ψ .” In such a case, a belief revision by $\neg\phi$ would correctly add ψ to the belief set.

It appears to be odd, that a theory of beliefs should, in one way or the other, depend on a preceeding theory of counterfactuals¹⁶; but, in our view, the proposed test provides a reasonable crosscheck for the adequacy of a combined belief and counterfactual theory.¹⁷

4.4 | First conclusion

From our critics we draw the first conclusion:

An adequate formalization of beliefs should not require that beliefs are closed under logical consequence.

Otherwise, any belief would supposed to contain all mathematical truths which is obviously inadequate. This conclusion hits the very heart of

¹⁵ Alchourrón, Gärdenfors, and Makinson, 1985, §1.

¹⁶ In a more literal designation, we would require here a theory of “counterbeliefs”.

¹⁷ We are not very happy with the current formal accounts to counterfactuals, due to Stanaker and Lewis, and have proposed an alternative approach in Kahle, 2017.

the AGM approach, as it does not only assume that any belief set is deductively closed, but *a fortiori* also the belief operations include the closure operation.

We may add that, Levi's distinction of "beliefs" and "commitments to believe"¹⁸ is of no help, as the belief revision operations in AGM, presupposing the deductive closure, work on these "commitments" and not on the "actual beliefs".¹⁹

5 | What triggers a belief revision?

The AGM postulates treat belief revision as a function which has as arguments a set of sentences (the current beliefs) and one sentence (the new belief) and returns the revised set of sentences.²⁰ Thus, belief revision is an operation processing a stream of incoming new beliefs *one after the other*.

5.1 | Realizing inconsistencies

As already said, very often the inconsistency of new belief with the existing beliefs is not (immediately) realized. This, of course, makes sense only in the case that a belief set is not deductively closed.²¹ To realize an inconsistent belief we would say that it is needed to have explicitly two contradictory beliefs p and $\neg p$ in the set. In this sense, the inconsistency of p with the $\{q, q \rightarrow \neg p\}$ is, for instance, not realized until the derivation of $\neg p$ from q and $q \rightarrow \neg p$ was actually performed. In the line of this argument we would say, quite generally that:

A belief revision is only triggered when an inconsistency of a belief set is realized.

¹⁸ "Isaac Levi (1991) has clarified the nature of this idealization by pointing out that a belief set consists of the sentences that someone is committed to believe, not those that she actually believes in." Hansson, 2011, §1.2. "Belief set" is, in this context, the term for the "committed beliefs".

¹⁹ The only way out seems to be working only with "belief bases" which are not supposed to be deductively closed, see Hansson, 2011, §5 and Kahle, 2002—but this is not the route, AGM is following.

²⁰ Hansson, 2011, end of §1.2.

²¹ Otherwise it would be the set of all sentences which, indeed, nobody believes.

We addressed this issue elsewhere²² and we will not discuss it here further. But we may even question whether a realized inconsistencies should (immediately) trigger a belief revision.

5.2 | Belief revisions as “small scientific revolutions”

Let us consider two new beliefs p and q coming in one after the other. According to AGM one needs to process p in $K * p$ before q which leads to $(K * p) * q$. Now, let us think of two new beliefs p and q where p “heavily” contradicts our “theory” K (one may say “falsifies” K in Popperian terms), while q supports K again. If the update with p was rigorously performed, $K * p$ might be already so reduced that q ’s support for the original K is essentially without effect. Experience from natural science shows that it is actually *irrational* to throw away an established theory by just one counterexample. It seems to be more rational to “take stock” of single counterexamples and to say:

A belief revision does not need to be performed with a single new income, but only if there is a “critical mass” of counterexamples.

We like to say that belief revisions have some characteristics of Kuhn’s famous *scientific revolutions*: they don’t come from single events, which might be just *anomalies*, but from accumulated contradicting information which requires the substantial rearrangement of the beliefs.²³

6 | The intensional structure of belief

Our last criticism concerns the intensional structure of beliefs. The extensionality postulates of AGM, being in line with the supposed logical omniscience of a belief set, demand that logically equivalent formulas as new incoming beliefs will produce the same result. Having abandoned

²² Kahle, 2002

²³ In this comparison we focus only on one and maybe even rather marginal aspect of Kuhn’s concept of scientific revolution, that of critical mass of anomalies. Of course, Kuhn’s concept goes far beyond this particular aspect. It focus, for instance, on the fact that theories after a scientific revolutions are no longer comparable—they are *incommensurable* in Kuhn’s terminology. If this would be transferable to beliefs, one would even run into problems of the domain and co-domain of the belief operations as they would have to link incommensurable belief sets.

logical omniscience there is no space for extensionality any longer, as it would (together with the success postulate) reintroduce the problems logical omniscience had caused. But it is not only the extensionality postulate we would like to ban, but even claim that the intensional structure of beliefs is the key ingredient of an adequate theory of belief revision, as it provides us with the reasons for a belief.

6.1 | The question, why do you believe something

A crucial property of beliefs is the possibility to equip a belief with the reasons for it, in fact, even maybe with several different reasons. This aspect is completely neglected by AGM. Again, mathematics provides a good example to illustrate this issue. A mathematician does not believe a theorem, simply because it is true, but, in first place, because (s)he is aware of a proof of it. But this is not the only possible reason. Probably most of the living mathematician believe in Fermat's Last Theorem, not because they know Wiles's proof, but because they trust the news given in the scholarly circles that "competent experts" verified this proof. And, of course, many mathematicians believe in one or the other still unproven hypothesis, like the famous Riemann hypothesis.²⁴ Such a belief is, obviously, not founded on a proof, but on other reasons.

The interesting point is that such reasons are easily to determine: one can simply ask for them! All what we need is to enlarge our theory of beliefs by including the reasons, or better: *justifications*, one has for a belief. *Truth maintenance systems*, as introduced by Doyle,²⁵ were an attempt to incorporate such justifications in the representations of derived formulas. As this approach was mainly studied in the context of computer assisted problem solving, it fall victim to implementation and complexity problems. Thus, a detailed study of AGM in combinations with justifications seems to be still a desideratum. We proposed an account in this direction, called *structured belief bases*²⁶, but it diverges already significantly from AGM.

²⁴ You can find this explicitly stated by David Hilbert in Hilbert, 1922, p. 160: "Nothing prevents us from taking as axioms propositions which are provable, or which we believe are provable. Indeed, as history shows, this procedure is perfectly in order: [...] Riemann's conjecture about the zeroes of $\zeta(s)$, [...]".

²⁵ Doyle, 1979.

²⁶ Kahle, 2002.

And we promote the inclusion of justifications—or: proofs—in theories for intensional phenomena also in other contexts.²⁷

7 | Conclusion

As an attempt to make good use of the axiomatic method, AGM has definitely its merits. But as AGM is based on a doubtful idealization—the deductive closure of beliefs—, it thwarts its own aim for rationality. As we have argued, it is not rational to expect that beliefs are deductively closed. In addition, belief revisions as they happen, for instance, in scientific progress, seems to work entirely different than AGM operations suggest. Instead of naive falsificationism, one faces a subtle interplay of pros and contras of current beliefs which may not result in simple revisions, but could even result in revolutions. The underlying paradigm changes seem to be out of reach for AGM. In consequence, we call for a search of a new theory for belief revision, abandoning the deductive closure, but taking into account the justifications for beliefs.

²⁷ Kahle, 2006; 2016; 2017.

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Modal Reasoning and Implicit Contextual Constraints

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This article presents a peculiar type of logical fallacy. I have decided to call it the *modal* fallacy because it emerges from an unjustified disregard for the accessibility relations in modal logic. A closely related fallacy (also known as the modal fallacy or, sometimes, modal scope fallacy¹), involves an inappropriate change in the scope of a modal operator and, therefore, amounts to a syntactical error analogous to the invalid inference of $\exists y \forall x Pxy$ from $\forall x \exists y Pxy$. In contrast, the fallacy considered here constitutes a valid argument from the point of view of logical syntax, but involves an implicit shift in the accessibility relations. This marks the problem as a characteristically modal phenomenon. Towards the end of this paper, we will turn to two simple applications of the insight gained from our diagnosis.

Incorrectly deducing necessity from truth

We begin with a simple theologically-flavoured version of the modal fallacy:

*Because Jesus died on the cross, if he was not crucified by the Romans, then (necessarily) he must have been crucified by someone else. This implies, then, that His death by crucifixion was necessary.*²

¹ Swartz, 2011.

² Readers familiar with Swartz's analysis of modal fallacies (Swartz, 2011) might argue that here we have a simple modal scope fallacy. A reconstruction of this point of view is given below in the historian's position and the derivation (A-D). I will argue that a different type of analysis leads to interesting conclusions.

We could of course colour the argument differently to argue for the inevitability of any other historical event. Interestingly, a view such as this differs from causal determinism because it accepts the contingency of any given causal mechanism leading up to an event. The point of the argument is that *if* an event took place then, necessarily, there was *some* causal mechanism leading up to it, and this fact makes the outcome necessary.

A historically minded person would probably accept the premise but reject the conclusion. Jesus was crucified, true, and this leads us to conclude that if he was not crucified by the Romans, then he must have been crucified by someone else. But this does not mean that had he not been crucified by Roman authorities, then he *would have* been crucified by someone else. Notwithstanding the Gospel of Peter's claim that the Jews were responsible for nailing Jesus to the cross³, we recognise crucifixion as a distinctively Roman form of judicial execution. If someone else had killed Jesus, why would they have chosen the cross?

What went wrong in the argument? Let us give a formalised version below to try to analyse it more carefully. Let j stand for 'Jesus', r for 'the Romans', Tx for 'x was crucified' and Cxy for 'x crucified y'. We proceed by applying the standard rules of the system T, with L representing the necessity operator⁴.

- | | |
|---|--|
| 1. $Tj \rightarrow L(\neg Crj \rightarrow \exists x Cxj)$ | (Premise) |
| 2. Tj | (Premise) |
| 3. $L(\exists x Cxj \rightarrow Tj)$ | (Premise) |
| 4. $L(\neg Crj \rightarrow \exists x Cxj)$ | (1, 2, Modus Ponens) |
| 5. $Crj \rightarrow \exists x Cxj$ | (Theorem, $Pa \rightarrow \exists x Px$) |
| 6. $L(Crj \rightarrow \exists x Cxj)$ | (5, Rule of Necessitation) |
| 7. $L(\neg Crj \rightarrow \exists x Cxj) \wedge L(Crj \rightarrow \exists x Cxj)$ | (4, 6, Law of Composition) |
| 8. $(L(\neg Crj \rightarrow \exists x Cxj) \wedge L(Crj \rightarrow \exists x Cxj)) \rightarrow L((\neg Crj \rightarrow \exists x Cxj) \wedge (Crj \rightarrow \exists x Cxj))$ | (Theorem, $(Lp \wedge Lq) \rightarrow L(p \wedge q)$) |
| 9. $L((\neg Crj \rightarrow \exists x Cxj) \wedge (Crj \rightarrow \exists x Cxj))$ | (7, 8, Modus Ponens) |

³ Swete, 2001, 25.

⁴ Hughes and Cresswell, 1966, 30.

10. $((\neg Crj \rightarrow \exists xCxj) \wedge (Crj \rightarrow \exists xCxj)) \rightarrow \exists xCxj$
(Theorem, $((p \rightarrow q) \wedge (\neg p \rightarrow q)) \rightarrow q$)
11. $L(((\neg Crj \rightarrow \exists xCxj) \wedge (Crj \rightarrow \exists xCxj)) \rightarrow \exists xCxj)$ (10, Rule of Neccesitation)
12. $L(((\neg Crj \rightarrow \exists xCxj) \wedge (Crj \rightarrow \exists xCxj)) \rightarrow \exists xCxj) \rightarrow (L((\neg Crj \rightarrow \exists xCxj) \wedge (Crj \rightarrow \exists xCxj)) \rightarrow L\exists xCxj)$
(Axiom, $L(p \rightarrow q) \rightarrow (Lp \rightarrow Lq)$)
13. $L((\neg Crj \rightarrow \exists xCxj) \wedge (Crj \rightarrow \exists xCxj)) \rightarrow L\exists xCxj$ (11, 12, Modus Ponens)
14. $L\exists xCxj$ (9, 13, Modus Ponens)
15. $L(\exists xCxj \rightarrow Tj) \rightarrow (L\exists xCxj \rightarrow LTj)$ (Axiom, $L(p \rightarrow q) \rightarrow (Lp \rightarrow Lq)$)
16. $L\exists xCxj \rightarrow LTj$ (3, 15, Modus Ponens)
17. LTj (14, 16, Modus Ponens, Conclusion)

From the syntactical point of view, the argument is flawless. It essentially recasts the formally correct

- i. $p \rightarrow Lq$ (Premise)
- ii. p (Premise)
- iii. Lq (1, 2, Modus Ponens, Conclusion)

in a slightly more elaborate form, so a historian might feel tempted to attack the first premise instead. What you mean, she might say to the theologian, is $L(Tj \rightarrow (\neg Crj \rightarrow \exists xCxj))$, which is to say that, necessarily, *if* Jesus was crucified, then if he was not crucified by the Romans, he was crucified by someone else. In that case, we would get the following valid argument.

- A) $L(Tj \rightarrow (\neg Crj \rightarrow \exists xCxj))$ (Premise)
- B) Tj (Premise)
- C) $Tj \rightarrow (\neg Crj \rightarrow \exists xCxj)$ (Axiom of Necessity)
- D) $\neg Crj \rightarrow \exists xCxj$ (B, C, Modus Ponens, Conclusion)

From here we can easily derive $\exists xCxj$, but we cannot get to the necessity of Tj even if we include premise (3). Having restated her point in more formal terms, the historian would probably consider the argument settled, but the puzzled theologian would be left mulling over a nagging

subtlety. Does premise (A) really represent what he meant to say with premise (1)? Surely, the death on the cross does not merely imply the contingent fact that *someone* was responsible for the crucifixion, rather we should be able to conclude that *it is impossible that no one was!*

In short, if we accept premises (1–3) the conclusion (17) *LTj* does indeed follow, but the argument stated in words clearly appears fallacious. This might lead to the suspicion (shared by the historian) that premise (1) might mean something different from what we expected. Thinking it over, we might agree that, in fact, the premise cannot be true, for we easily imagine that if Jesus had not suffered on the cross, then *Tj* would be false, contradicting (17). But then the theologian could argue that if *Tj* were false, then it would make the antecedent in (1) false and, consequently, the conditional in premise (1) would automatically be true! And from here we fall into the abyss of Contradiction.

To unravel the mess, let us return to the meaning of the premise and conclusion of the original theological argument. In the premise we affirm that *in any imaginable case in which the crucifixion occurred*, Jesus was crucified by someone. The conclusion, however, seems to mean that Jesus was crucified in *every* imaginable case. Of course, the range of possibilities considered has changed midway through the argument. Or, in other words, the meaning of “necessary” has changed. If we stick to the first meaning, then the argument becomes trivial. When necessary means true in every possible world considered, and we only consider worlds in which Jesus was crucified, then the crucifixion indeed becomes necessary.

So we have found another way out of the problem: we accept the premises and conclusion, but argue that the premise implicitly restricts the cases considered, making the conclusion apply only to a specific range of possibilities. In technical terms, we would say that in order for (1) to be true, the accessibility relations must be set up in such a way that *Tj* is true in every possible world accessible from the actual world⁵. Otherwise we end up with a contradiction. In this sense, stating (1) implicitly constrains the range of possibilities.

⁵ Hughes and Cresswell, 1996, 75–80, include a more formal discussion to explain how the range of possibilities considered (in their terms, “what is conceivable”) determines the conditions of accessibility.

Possibilities and probabilities

At this stage, we have transformed the intriguing subject of a logical fallacy that is nevertheless syntactically correct into the disappointingly obvious observation that, in modal reasoning, you are not allowed to change the range of possibilities within which the premises were true. We will therefore shift our attention to an application of this insight.

To set up the next example of modal fallacy, we will evoke the 2009 Olympic Games, when Usain Bolt (Jamaica) was expected to break the World Record in the 100-metre dash. It was assumed that Tyson Gay (U.S.A.) and Bolt's compatriot Asafa Powell, were much less likely to win the race, but that Gay was faster than Powell. Before it started, the fictional philosopher Herbert Strawmann argued that "there must be something wrong with the rule of *Modus Ponens* because, if the U.S.A. does not win, then if Bolt does not win, then Powell wins". "However," he continued, "we know that bolt wins, so we can affirm the antecedent and (*Modus Ponens*) reach the conclusion that if Bolt does not win then Powell wins. But this is the wrong conclusion because we should have concluded that if Bolt does not win then Gay wins." More formally, with b standing for 'Bolt wins', g for 'Gay wins' and p for 'Powell wins'⁶,

- | | |
|--|-----------------------------------|
| I. $\neg g \rightarrow (\neg b \rightarrow p)$ | (Premise) |
| II. $b \rightarrow \neg g$ | (Premise) |
| III. b | (Premise) |
| IV. $\neg g$ | (II, III, Modus Ponens) |
| V. $\neg b \rightarrow p$ | (I, IV, Modus Ponens, Conclusion) |

Again, we have an impeccable deduction, but the desired conclusion was $\neg b \rightarrow g$. Of course, p and g are incompatible (if Powell wins, then Gay does not, and vice versa), but this does *not* make the statements $\neg b \rightarrow p$ and $\neg b \rightarrow g$ incompatible. In fact, since b is true (premise III), the antecedent of these conditional statements is false, making them both true. So *Modus Ponens* still works and Strawmann was wrong.

⁶ The structure of this argument was taken from Vega Reñón, 2007, 53–54.

If we want to understand the point that Strawmann was making, though, we need to consider a range of alternative scenarios. We have to imagine cases in which the victory goes to Jamaica and cases in which Bolt does not win. But then we run into the modal fallacy again. Indeed, we can restate Strawmann's premise as $\neg g \rightarrow L(\neg b \rightarrow p)$, but this restricts the conclusion to cases in which g is false, so we can in fact conclude that $L(\neg b \rightarrow p)$ because g has become impossible. Conversely, if we insist on including the possibility of g , that is $\neg L\neg g$, among the premises, then we are led to a contradiction and anything follows.

When rethinking the argument in the language of probabilities, the constraints on the range of possibilities appear explicitly. Let us assign probabilities to the mutually exclusive (and, for simplicity, exhaustive) events b , g and p such that $P(b) > P(g) > P(p)$. It follows that the conditional probabilities when Jamaica claims victory satisfy $P(b \mid \neg g) > P(p \mid \neg g)$ and $P(b \mid \neg g) + P(p \mid \neg g) = 1$. Premise (I) amounts to stating that $P(p \mid \neg g \wedge \neg b) = 1$, which of course does not contradict Strawmann's desired conclusion that $P(g \mid \neg b) > P(p \mid \neg b)$, with $P(g \mid \neg b) + P(p \mid \neg b) = 1$. All these relations hold simultaneously thanks to the conditions (or should we say constraints) on the range of cases considered.

Modal logic accepts an interpretation in terms of probabilistic semantics⁷, but here we will take the opposite route and regard probabilities from the perspective of possible outcomes, that is, from the point of view of modal logic. We define a *possible outcome* (or simply *possibility*) as a set of propositions consistent with a series of (often implicit) logical constraints⁸. The probability of a proposition is simply defined from an operational point of view as the number of possibilities in which that proposition is true divided by the total number of possibilities⁹. In the words of Laplace, the probability of an event is "the ratio of the number of favourable cases, to that of all the cases possible"¹⁰. Considered this way, *by definition* all outcomes have the same probability (namely, one divided by the number of possibilities).

⁷ Hughes and Cresswell, 1996, 227–228.

⁸ Contrast this definition with that of *possible world* as a *maximal* consistent set of propositions.

⁹ We would need a more rigorous mathematical definition in problems with an infinite number of possible outcomes, but that would lead us too far astray for the purposes of this article.

¹⁰ Laplace, 2012, Chapter III, First principle.

One might object that when you throw a six sided die, you could regard the outcome as one of two possibilities: either you get a five or you do not. In that case, according to the definition above, the probability of getting a five on the roll of a die should equal one half, which is clearly wrong. On the contrary, I would argue, one half is in fact the correct answer in *that* model of the possibilities involved, but I would submit that you get a much better model if you contemplate six possible outcomes, one for each side of the die, in the sense that it better predicts the outcomes of actual experiments with dice. Then you get the standard answer of one sixth. In the latter case, it makes no sense to argue that we have not taken into account the probability of the die balancing on one of its vertices or of a cosmic ray destroying it completely. True, we have not, but when we equate the probability to one sixth, we simply mean that a five is one of the six outcomes considered.

Will quantum logic supersede classical logic?

We will conclude by applying this conception of probability and our diagnosis of the modal fallacy to an interesting problem in the philosophy of science. Here is a Bell-like inequality which constitutes a theorem in probability theory:

$$P(A \wedge \neg B) + P(B \wedge \neg C) \geq P(A \wedge \neg C)$$

You can easily understand the theorem by drawing a Venn diagram with three sets, A , B , and C and observing that $A \cap \bar{C}$ is always contained in the union of $A \cap \bar{B}$ and $B \cap \bar{C}$.

Quantum mechanics professors commonly claim that quantum systems may violate this inequality. The argument works like this¹¹. Some two-particle systems in quantum mechanics¹² have the following property: if a measurement determines that the spin of one particle points in a given direction, then the same type of measurement on the other particle will find it pointing in the opposite direction. We can only measure along determined directions, and the measurement always yields a spin either in the direction of the measurement or in the opposite direction. Suppose

¹¹ For a technical discussion of these issues, see Griffiths, 2011.

¹² Two entangled electrons in the singlet state, but the details need not concern us here.

that A , B and C represent the statements that the spin of particle one points upwards, at a 45 degree angle or right, respectively. We wish to determine the probabilities that appear in the inequality above. Now by measuring along the direction A , we can easily determine $P(A)$, but how to find a probability like $P(A \wedge \neg B)$? We are not allowed to simultaneously measure particle one along the directions A and B , but we can use a clever trick. Here is where the second particle comes into play. By determining the number of times that particle one points in the A direction and particle B points in the B direction, we can calculate $P(A \wedge \neg B)$ because a spin in the B direction for particle two corresponds to a $\neg B$ for particle one. We can set up similar experiments to determine the values of $P(B \wedge \neg C)$ and $P(A \wedge \neg C)$ and verify that, surprisingly,

$$P(A \wedge \neg C) > P(A \wedge \neg B) + P(B \wedge \neg C),$$

contradicting the inequality above. Does this mean that we should abandon our old rules of logic and adopt a new quantum logic?¹³ Let us not be too hasty.

Recall what it means to measure a spin along B for particle two:

Premise: *If we had measured the spin of particle one along the B direction, then (necessarily) we would have observed $\neg B$.*

Ah, modal fallacy, we meet again! The statement above holds only assuming that we measured particle one along the B direction. But we did not, we chose the A direction instead, and thus implicitly changed the set of possibilities available. Though the particle spins point in opposite directions when measured along the same line, what can we infer from this if they were measured along different lines? Van Fraassen illustrated the problem with a classic example¹⁴: if Bizet and Verdi had been compatriots, would Bizet have been Italian or would Verdi have been French?

¹³ A serious discussion of this topic would land us in deep waters, well beyond the scope of this essay. See Griffiths, 2011, and references therein. Here we simply show that the argument against classical logic as stated above is, at best, incomplete.

¹⁴ Van Fraassen, 1991, 122–125.

Conclusion

In summary, when reasoning about an array of possibilities, one way to commit a fallacy involves implicitly changing the range of possibilities considered. This modal fallacy can introduce subtle errors, especially when considering probabilistic reasoning, as the set of possible outcomes is very often not stated explicitly. Hopefully, the preceding pages will have persuaded readers to pay close attention to the set of cases regarded as possible in discussions involving modal or probabilistic reasoning. Philosophers tend to regard the ability to easily identify a new fallacy as a welcome addition to their analytical toolbox.

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Public announcements, belief expansion and abduction

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Abstract

Abduction has been recognized as the most important problem of modern epistemology. In order to detect when an abductive problem has arisen, three parameters should be considered, namely, a background theory, a fact to be explained (in terms of such theory) and the inferential parameter, that is to say, the underlying logic used in the corresponding scientific practice. In our proposal any logical system could represent the underlying logic, though we rehearse with the system KD45 as a basic one. Then, according to abductive triggers, epistemic operations and belief closure of a theory are defined as the set that contains believed formulas of such theory for an agent a , and, when a formula does not belong to the theory, then the negation of the formula that says that the former is believed by the agent is a member of this closure. New operators could be defined: after the expansion (or revision, or contraction) with a formula, another one holds. On the other hand, by means of several easy examples we look how applying a dynamic perspective to the logical analysis of abduction in order to open roads or continue the initiated ones in the field of epistemology.

1 | Introduction

As it is known, Peirce defined the concept of abduction, as a form of inference, though he first named the corresponding process as to formulate a hypothesis. Inferences could be classified as deductive (or analytic) and

inductive, which may be subdivided in induction or formulation of hypothesis (finally, the last was called abduction). This classification gives reason to Whewell's point of view¹ since he considers that abduction is a form of induction, because of which this is a perspective that could be taken as a precedent of the Peircean conception.

Peirce's perspective does not avoid a very important question: since both inferences, induction and abduction, have synthetic characteristics, is abduction a form of induction after all? A clear answer to this question is given by Hintikka² who has studied the Peircean notion of abduction and qualified it as the "central problem of contemporary epistemology". This could be discussed, but the Finnish philosopher, by adopting Kapitan's thesis, point out the specific character of abduction and, as a consequence, that it is due to an inferential process that is not identifiable with deduction nor induction, as we shall see below.

The issue could be investigated by resorting to the history of sciences. Is there any example of scientific practices that can be seen as an exponent of investigation based on abduction? A positive answer may be given if, for example, the methodology of Kepler is considered³. In fact Kepler's work is seen as the best example of the ideal of scientific method, so that, in the last resort, if we consider the inference processes that take place in scientific practices, it is not very difficult to defend that abduction is a kind of inference different from induction and deduction, though all of them are present –whatever their intensity may be– in scientific practices.

The paper is structured in the following way. After this introduction, next section is devoted to settle what a scientific practices is and the associated with that the notion of underlying logic and the important notion of abduction. Section 3 is a short summary of belief revision in relation with abduction. In section 4 a basic logical system is proposed as underlying logic with the purpose of tackling the epistemological problems that are suggested through previous sections. Finally some concluding remarks are presented.

¹ Whewell, W., 1840.

² Hintikka, 1998.

³ It is shown in Ana Paula Silva, 2010.

2 | Scientific practices: the underlying logic and abduction.

In the process of constructing scientific theories, certain system of reasoning is adopted by a scientific community. Such system of reasoning, to simplify, shall be called the *underlying logic* of such practice. Of course, as any system of reasoning, the underlying logic can be codified in the so called a logic. A usual notion of (full) logic is given by means of a (formalized) language L , the corresponding semantics and the inferential device, in symbols

$$\text{LOG}(L) = (L, S, \vdash),$$

where L is the language, S its semantic, meanwhile $\vdash$ represents the device, usually a deductive calculus, so $G \vdash F$ expresses that the sentence⁴ F is deduced from the set of sentences G , and $G \nvdash F$, should be read as F is not deduced from G . To abbreviate, we can say that $\vdash$ is the genuine representative of the underlying logic.

In scientific practices, a developed theory T can be presented as a set of sentences of the corresponding language that are obtained by means of the underlying logic from a set of postulates P , provided certain relevance conditions are accomplished (specially, consistence of postulates and theorems different from tautologies). In symbols:

$$T = \{F \in L : P \vdash F, \& P \nvdash \perp \& \nvdash F\}.$$

Researchers could be on the look out of a fact expressed as a sentence F , because 1) $T \vdash F$, that is to say it is predictable, or 2) it may be suggested by phenomena, other facts or a base of knowledge. But F could be surprising because of it is not finally the case, against what has been suggested, or finally $T \nvdash F$. So that, when F is surprising, it is not verified that $T \vdash F$. This is an example that leave us to the realm of abduction.

Given a theory – or base of knowledge – Q , the corresponding underlying logic $\vdash$ and a fact –represented by de formula F –, in symbols $(Q, \vdash, F)$, it is an abductive problem if and only if (iff from now on):

⁴ As we shall move in the field of propositional logic, “sentence” and “formula” will be used interchangeably.

1. $Q \vdash \neg F$,
2. $Q, F \vdash \neg \perp$,

We omit the study of the case of structural abduction, which looks for a solution for a given abductive problem in a change of underlying logic. A sentence A is a solution for the abductive problem $(Q, \vdash, F)$ iff

1. $Q, A \vdash F$,
2. $A \vdash \neg F$,
3. $Q, A \vdash \neg \perp$,

Abduction has been named “the fundamental problem of contemporary epistemology”⁵. That notion, first studied by Peirce⁶, was considered a form of inference or, which is more exact, it is a process that leads to an explicative hypothesis. A deductive proof can also be seen as a process whose result is a sequence of propositions that begins with a set of premises and ends in a last proposition that is called conclusion. However, an abductive process, as indicated above, has a background theory and a conjecture as its starting point, that is to say, an initial set of propositions – formulas of the corresponding (formalized) language – but the last proposition, the “conclusion” of the process so to say, is now the explicative hypothesis. In order to fix characteristics of abductive processes, the Finnish philosopher adopts the following thesis:

1. Inferential thesis: abduction is, or includes, an inferential process or certain processes.
2. Thesis of purpose: the purpose of scientific abduction is 1) to generate new hypothesis, and 2) to select hypotheses for further examination.
3. Comprehension thesis: scientific abduction includes *all* operations whereby theories are engendered.
4. Autonomy thesis: abduction is, or embodies, reasoning that is distinct from, and irreducible to, either deduction or induction.

This idea of abductive inference as the source of all new hypothesis is against the hypothetic-deductive model of scientific research established from modern scientific revolution. Then, the most general problem, according to Hintikka, to which such model and the concept of abduction

⁵ Hintikka, 1999, 91-113.

⁶ Peirce, 1991, 186-199.

should put face to is how characterizing an enlarging reasoning as abduction, which is not a simple inference. An essential difference is that in (sound) deductive inferences the step from premises to conclusion is truth-preserving but in abduction the step from premises (a background theory plus the fact to explain) to hypothesis is not so but at most probable. That is to say, taken an abductive problem as the set of premises and the solution as conclusion of the corresponding inferential process, the last is not logical consequence of the former.

If inferences are considered as games, then the main concern in logical analysis is to determine conditions to win such games. This is in consonance with the game theoretical semantics and de perspective of how the “true” logic of scientific knowledge is⁷. Since a game could be given by means of rules, how are these rules would be the key to understand how is an inference. In general, to justify an inference, from a game theoretical point of view, we should distinguish two kind of rules (or principles), namely,

1. Definitory rules. These rules are similar to the ones that define a game like chess – deduction or scientific inquiry may be considered as a game –, they tell possible moves in a given situation through the game.
2. Strategic rules. These rules tell which moves are good in order to win the game.

On the other hand, Hintikka⁸ brings out a systematic characterization of inquiry that is consistent with the game theoretical semantics perspective and her idea about the true logic of scientific discovery. In short, it is taken as an activity that can be studied as an interrogative process, that is to say, as a process of questions and answers. In such interrogative approach, the difference between ampliative and nonampliative reasoning becomes a distinction between given interrogative (ampliative) and deductive (nonampliative) steps of argument. In interrogative inquiry the thing is to anticipate the epistemic situation brought about by the answer. All that remarks could be taken into account as an important set of accurate advice for tackling logical approaches to abduction. We shall retake that, though first we shall briefly review belief revision.

⁷ The semantics of game theory is stduied in Hintikka and Sandu, 1999 and the logic of scientific discovery in Hintikka, 1999b.

⁸ Hintikka, 2007.

3 | Belief revision

From the AGM model of belief revision we can adapt a form of treating abduction⁹. In AGM three kinds of changes are considered, namely *expansion*, *contraction* and *revision*. There a belief set K is a set of sentences of the corresponding language that is closed under (logical) consequence, in symbols $K = \text{Cn}(K)$. Given a belief set K , $K+F$ represents the belief set obtained after the expansion with the sentence F . This expansion operation guarantees that $K+F$ is the smallest belief set. By adapting that postulates to define an abductive expansion, we are taking consistent sets of sentences instead of the belief set taken in AGM, that is to say, instead of sets of sentences that were closed under consequence. So, let K be a consistent set of sentences of the (formalized) language L used in a scientific practice. Then, $K+F$ is the smallest set that verifies

1. $K+F$ is a consistent set of sentences,
2. $F \in K+F$,
3. $K \subseteq K+F$
4. If $F \in K$, then $K+F=K$,
5. If $K' \subseteq K$, then $K'+F \subseteq K+F$.

The contraction of a consistent set of sentences K with F is defined as the minimal consistent set of formulas that verifies the following postulates (in the case of the AGM contraction operation there are more postulates, but we only have selected those that are not controversial):

1. $K-F$ is consistent,
2. $K-F \subseteq K$,
3. If $F \notin K$, then $K-F=K$,
4. If $\neg F \in K$, then $F \notin K-F$,
5. If $F \in K$, then $K \subseteq (K-F)+F$,
6. If $\neg F \leftrightarrow H$, then $K-F = K-H$.

⁹ In a similar way to the proposed in Aliseda, 2006. In van Ditmarsch et al. 2008, a summary of AGM is presented to be studied from a dynamic perspective.

From both previous operations, considering consistent sets of sentences, by means of Levi's identity, a new one can be defined, the abductive revision, represented by means of *:

$$K * F = (K - \neg F) + F.$$

So revision is a process of contraction and expansion.

From that logical tool abduction could be studied whatever the underlying logic may be. However, if abduction were the fundamental problem of contemporary epistemology, it would be worth more to take a look at examples with concrete underlying logic. About that, in order to consider belief revision and abduction two ways of tackling problems of this area: 1) a theory is a set believed propositions, 2) a set of knowledged propositions. In next section a concrete system is adopted to represent the underlying logic and some examples are briefly discussed.

4 | A basic underlying logic: KD45

A KD45 system, as we have said, could be an initial underlying logic to certain epistemological studies. First, let us see a basic system with belief operator. The language is defined according to the BNF rule

$$F ::= p \mid \neg F \mid F \rightarrow F \mid BaF$$

where p is a propositional variable, a is an agent, $F \rightarrow F$ represents the (material) implication –other connectives are defined from that and the negation–, and BaF expresses that “the agent a believes F ”. The system KD45 consists of all propositional tautologies and the following axioms and rules¹⁰:

1. $Ba(F \rightarrow H) \rightarrow (BaF \rightarrow BaH)$
2. $\neg Ba \perp$ (or, equivalently, $BaF \rightarrow \neg Ba \neg F$)
3. $BaF \rightarrow BaBaF$
4. $\neg BaF \rightarrow Ba \neg BaF$

¹⁰ Van Ditmarsch et al., 2008, 39.

5 Rules:

- a) if F and $F \rightarrow H$, then H ,
- b) if F , then BaF .

Now, given an abductive problem $(Q, \vdash, H)^{11}$, F is an abductive solution if H is deduced from the abductive expansion $Q+F$. On the other hand, from KD45 a different form of closure could be considered, namely the closure under belief: given a consistent set of sentences K , it is closed under belief iff for all F and an agent a , $F \in K$ iff $BaF \in K$.

If we work with a background theory that we want to be closed under belief, then some nuances should be made. Let us see an example. Let K be the belief set $\{Ba(F \rightarrow H)\}$ that is closed under belief, and $(K, \vdash, H)$ an abductive problem, with H as a novelty, which is equivalent to say that $K \vdash \neg H$ and $K \vdash \neg \neg H$. Of course now $\vdash$ represents the underlying logic, that is to say, the KD45 system. $\vdash \neg BaH$ and $\vdash \neg \neg BaH$ by hypothesis. Since $K, F \vdash \neg \perp$ (consistency condition, provided F is not $\perp$), the expansion $K+B_aF$ contains abductive solution: in fact $B_aF \in K+B_aF$ by definition, and according to applications of axiom 1 and *modus ponens* (rule 5, a) $B_aH \in K+B_aF$. On the other hand, it is verified that $B_aB_aF \in K+B_aF$ and $B_aB_aH \in K+B_aF$ (it is the result of applying axiom 3).

What about taking a background theory that is closed under belief? If we take K as closed under belief, with $Ba(F \rightarrow H) \in K$, the same abductive problem and $H \notin K$, then $B_aH \notin K$ so that $\neg B_aH \in K$ and, since K is closed under belief, $B_a \neg B_aH \in K$. Of course, $K+B_aH$ is not consistent because of $\neg B_aH \in K+B_aH$ and $B_aH \in K+B_aH$.

Returning to general cases of belief revision, how the corresponding steps should be given to solve an abductive problem depends on the characteristics of such problem. When the surprising fact is a novelty, the way could be found through an (abductive) expansion. However, when the surprising fact is an anomaly, then a process of revision should be initiated, namely with sentences that imply the anomaly.

Let us now consider a further extension of the logical language, namely with dynamic modal operators for expansion, contraction and revision. The following expressions are also formulas *cita11*: $[+F]H$, $[-F]H$,

¹¹ Q represents the background theory or base of knowledge, in this case $\vdash$ represents this system, and H (and F) a formula of the language.

$[*F]H$, which represent “after expansion with F , H holds”, “after contraction with F , H holds” and “after revision with F , H holds”, respectively. We bypass details of the semantics of such modalities and focus on the consequence relation and matters involving abduction. If K is closed under consequence, then given an abductive problem $(K, \vdash, H)$, if $[+F]H$ in K , then the explanation is already contained by K , since adding F to K allows to conclude that after expansion with F , H is entailed by the expansion, i.e., H in $K+F$.

Consider the previous example wherein K is, additionally, closed under belief and contains $Ba(F \rightarrow H)$. We may assume that K also contains $[+F]BaF$, as this is a validity of the logic. Therefore, $K+F$ now contains H : given that $[+F]BaF$ is in K , $K+F$ contains BaF , and from BaF and $Ba(F \rightarrow H)$ and deductive closure follows BaH , the desired formula to be abductively explained. Note that, unlike in the previous example, closure under belief does not cause consistency problems, as the added formula BaF is not in K but only in $K+F$. However, with dynamic modalities for expansion and revision, we may still ask ourselves to what extent facts are still surprising, as the surprise is now, so to speak, inbuilt in the logical language.

We are taken again considerations inspired in game perspective. Whatever the underlying logic may be, a way of develop deductive steps in an abductive process is the tableaux method. In short, given an abductive problem $(Q, \vdash, H)$, with H as a novelty, if we construct a tableau with formulas of Q and $\neg H$ as its root –the initial set of the tableau– since $Q \vdash \neg H$, the tableau is open, but every open branch shows which formulas are necessary to close the branch. Then, a set of possible solutions can be found¹². However, to obtain a true logic of discovery, the tableaux method should be seen not only by a dynamic perspective, but also as an interrogative process¹³. Then the construction of the tableau should have to take into account that remark.

Let us see an example. Let the following theory be: $K = \{Ba(p \rightarrow q), Ba(q \rightarrow r)\}$, with p , q and r as atomic proposition, and a as an agent. The surprising fact is represented by r . Could Bar be explained from K ? Then the root of the tableau consists of $Ba(p \rightarrow q)$, $Ba(q \rightarrow r)$ and $\neg Bar$. There a branch

¹² In Nepomuceno-Fernández et al. this approach is more developed. Applications of tableaux method can be seen in Aliseda, 2006, who introduced the use of this method to study abduction.

¹³ In Hintikka, 2007 it is explained. In Hintikka 2014, 184-186, introduced a critic: the tableaux method is appropriated provided it reflected an interrogative process, as it is presented in Genot, 2011.

has at least these formulas: $Ba(p \rightarrow q)$, $Ba(q \rightarrow r)$, $\neg Bar$, $\neg Bap$, $\neg Baq$. If we construct the tableau with two columns to represent questions, we could see that Bar , Bap , Baq are suitable answers that, on the other hand, allow to close the standard tableau. But in the course of the interrogative process we reject Bar (otherwise, it would be a trivial explanation), and Bap and Baq would be good explanation. What is the best one? To choose that, several criteria can be adopted as, for example, the most increase of atoms (then, the best one would be p) or the simplest process of achieve a satisfactory explanation (then, the best one would be q). Whatever the case may be, the example shows that the standard tableau method needs to be completed with the mentioned perspective.

5 | Concluding Remarks

We addressed some matters concerning abduction in sets that are closed under deduction and also in sets that are closed under belief. As known, sets that are closed under belief cannot be expanded in a strict sense, as new information causes ignorance to disappear: such formulas should not be retained. Thus, more complex mechanisms are required to address abduction for sets that are closed under belief. One way out of this predicament is the enrichment of the modal logical language with dynamic modalities for belief expansion and belief revision. However, this may be at the price that the surprising aspect of abductive explanations disappears, as the abductive consequences of additional assumptions may now be explicitly stated in the initial belief set.

Regardless of this point of view, by considering the scientific discovery as an abductive process, the interrogative games can be implemented by means of the method of semantic tableaux, provided the game perspective is incorporated or, equivalently, the tableau is constructed with formulas and questions. That is to say, steps in scientific inquiry are now represented by means of formulas and questions, around which presuppositions, answers, etc. should be considered.

The work could continue by a KD45 system with a knowledge operator instead of the belief one. That is to say, with a positive and negative introspection with a knowledge operator. Besides, by taking a

public announcements logic as an extension of the former and considering epistemic operations as it is pointed out in application of AGM to abduction.

Finally, an important issue is how choosing, in an abductive process and from a logical point of view, the most suitable formula as the explanation when there are a wide variety of abductive solutions.

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Citizen science - Tinkering and heuristic strategies of scientific research

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Abstract

Citizen science is a contemporary movement of public participation in scientific research that agglutinates different proposals, all of them dependent somehow on the internet, the *World Wide Web* and *Web 2.0* tools. Through citizen science websites as *Foldit* or *CosmoQuest*, the user can collaborate in finding the best folding for a given protein or in building a more accurate map of the moon. This process emphasizes how Charles Sanders Peirce concepts of *abduction* and *diagrammatic reasoning* can relate to the heuristic strategies of contemporary scientific research. A user-friendly approach appeals to the human interaction and depends on human reasoning. The user is then *tinkering*, finds doubt and has to choose the best fit using the visual clues. Moreover, other Web 2.0 tools as forums and social media bring forward how citizen science character of public sphere is taken, with educational and community building social dimensions being central to this process.

Keywords: Citizen science; Tinkering; Abduction; Diagrammatic reasoning; Public Sphere; Heuristics

Participation in Science

In the nineteenth century's Great Britain William Whewell created the professional role of the scientist. Still Whewell probably had in mind enterprises that involved many human resources, as the non-professional

scientists, or “subordinate laborers”, as he referred to them¹. The openness to social participation in knowledge production continues today in modern sciences. Citizen science stands for this Public Participation in Science Research, boosted by the North-American Government, through the National Science Foundation (NSF) since 1992². In the turn of the twenty-first century in Europe and Oceania participation became a tool to be used to engage citizens, just like in the United States of America³. The strategies developed, these now popular methods of engagement, make use of internet and *World Wide Web* tools. *Crowdsourcing*, using the contribution of the virtual community of online users became an asset for scientific research. There’s more and more projects’ design that reflect this awareness of all the volunteers that could be reached and engaged. Still, most projects don’t need to outsource human resources on-line. Not all the research done is in dire need of opening up to volunteer work. One of the characteristics of these internet-based citizen science projects is the involvement of uncanny quantities of data. This phenomenon, referenced as ‘data deluge’⁴, implies the need for the multiplication of human resources. The nature of the scientific questions raised is another feature that can explain the need (or not) for further social engagement. William Whewell in 1835 conducted the ‘great tide experiment’, a big science experiment aimed to unfold the natural secrets about tides. Floods had the greatest social interest, being the cause of great devastation and mortality. In the course of this experiment, tides were measured every fifteen minutes for a fortnight at over 650 tidal stations that included Great Britain, France and the United States⁵. Since those days much changed in these Societies and States. Still, the “great tide experiment” of Whewell is in the same line of action than today’s citizen science. This concept of citizen science became popular in the turn of the century, mainly through the contribution of one researcher, Richard Bonney, of the North American *Cornwell’s Lab of Ornithology*. In the frame developed by Bonney, citizen science was defined as any project or activity in which the public was active collecting or analyzing data.

¹ Rozwadowski, 2004, 32;

² Bonney, 2009, 10, 15, Wulf, 1999, 133-134;

³ Osborne and Dillon, 2008, Braun and Schultz, 2010, Stocklmayer and Perera, 2011;

⁴ Interagency Working Group on Digital Data, 2009, 6;

⁵ Reidy, 2006;

Furthermore, the level of involvement of the citizen was defined in three categories of projects: collaborative, cooperative or communitarian⁶.

In the model developed by Bonney three categories measure the *intensity* of the participation in amateur scientific research. The volunteer that chooses to take part in a given citizen science project usually enters into it downstream, when the project is already defined. Mediated through technology, the person is challenged to fulfill a given task, to *collaborate*. The *WWW* and *Web2.0* tools allow still, other levels of engagement. Forums and other communication and social media tools allow the opportunity for deeper interaction between project designers and participants, to develop *cooperation*. Still, when the character of the participation established is defined as *communitarian* scientific research, it usually involves local geographical needs. The interests of scientists and citizens come together in a physical space and a community is formed that shapes the project. The projects we find online for virtual cooperation depend upon the Web 2.0 to build-up their own communities. Besides their user interface, they host forums, blogs and groups to enhance the informal learning process. Moreover, Toolkits and guidelines are usually shared⁷.

To be able to participate

The times we live now are quite different from Whewell's and still we have states that continue to dictate the 'public interest'. Through the veil of liberal mass media and opinion leaders, there's still the need for a "naïve faith in the idea of a rationalization of domination". Jürgen Habermas, the German sociologist and philosopher, expressed these words in *Social Structures of the Public Sphere*⁸. There he identifies a basic blueprint of the public sphere: how in Great Britain, France and Germany between the eighteenth and nineteenth centuries social spaces were created for the public use of reason. At their beginnings, all of them were identified with literature and the discussion of it. They maintained each their own particular cultures, of course.⁹ London, from the middle seventeenth century developed a tea, chocolate and coffee culture that came along with the

⁶ CAISE's Inquiry Group Report, 2009, 18;

⁷ CAISE's Inquiry Group Report, 2009;

⁸ Habermas, 1989, 238;

⁹ Habermas, 1989, 32-34;

emergence of a certain “parity of the educated”. At the first decade of the eighteenth century there were already 3’000 *coffee houses*, where men could discuss literary issues, at first, but from a certain point also economics and politics. The representation of moneyed and landed British nobles at the coffee house permitted that the discussions had practical consequences, quite differently from what happened in France. The French *salons* began to be active in the regency period of Louis XVI at a period that the nobles lost their social function. Therefore the discussions weren’t so entangled with political activism in these spaces. Women shaped the salons where nobility, the *grande bourgeoisie* and intellectuals took part. In seventeenth century Germany, the literary and table societies, the *tischgesellschaften* and the *sprachgesellschaften*, were fewer and even more removed from practical politics than the *salons*.

The *tischgesellschaften*, the *salons* and the *coffee houses* had, though, some institutional criteria in common¹⁰. The participants in such spaces had a social intercourse that disregarded status and were problematizing areas that weren’t before questioned. In such spaces, “everyone had to *be able* to participate”¹¹. The contemporary citizen science movement created such spaces virtually on the internet, amplifying the *public use of reason* in scientific culture.

The WWW medium is used to communicate and memorize information giving many new possibilities of opening up scientific methods to participation. Being able to cooperate and be involved in this process is something different from passively dealing with aspects of scientific knowledge, reading about ‘facts’ and ‘discoveries’ in a book or having an explanation given in class about natural phenomena. In other words, if we consider the educational role these platforms have, we are not facing a traditional form of formal education. The traditional diffusionist model of communication is being targeted, but, still, its roots are deep in our culture.

The Austrian logical positivism from the beginning of the twentieth century marked scientific methods that has logic as its tool and deals only with justification aspects of scientific discovery. It’s in this frame that the deductive-nomological model of Carl Hempel and Paul Oppenheim appears and, posteriorly, the method of conjectures and refutations of Karl

¹⁰ Habermas, 1989, 36-38;

¹¹ Habermas 1989, pp. 38;

Popper. The reverse of the medal, the one of the context of discovery has a more obscure history, as Imre Lakatos notices,

There is no infallibilist logic of scientific discovery leading in fallibly to results, but there is a fallibilistic logic of discovery which is the logic of scientific progress. But Popper, who has laid the basis for this logic of discovery was not interested in the meta-question of what is the nature of this investigation, so he did not realize that it is neither psychology nor logic, but an independent field, the logic of discovery, heuristics.¹²

These forms of online participatory science, through its blogs, forums and social media opens spaces where participants can exchange viewpoints and can problematize their experience. It is believed that such spaces can offer a window to the heuristics of scientific research. Citizens are invited to take part of the scientific venture with a multitude of defined objects used with intent, to be used and “discovered”, from the depths of Space to the Microworld. What defines this experience? What reasoning strategies are employed to find truth?

Playing a protein puzzle

FoldIt is a protein puzzle that can be installed on personal computers and portable devices. It is one of oldest and more iconic projects of citizen science. It started in 2008 and has three articles published through the *Nature Publishing Group*, taken as a very good recognition of professional value. *FoldIt* was formalized through the cooperation of the *Department of Biochemistry* and the *Center of Game Science* of the *University of Washington*. In 2013 *FoldIt* had more than 450 thousand registered users¹³. All of these people have, at least, tried to play this research game, a puzzle.

The attempts that *FoldIt* users do to solve puzzles are part of a protein modeling strategy. In this challenge, from the basic building blocks of proteins, the amino-acids, biochemists have to figure out: What is the structure of each of these proteins? Understanding it, other biocompounds can be designed aimed to fit the discovered structure and have a clinical or

¹² Lakatos 1976, 167 in Aliseda, 2004, 343;

¹³ FoldIt 2013;

industrial application. Computer models are manipulated to find out the perfect packing, the minimum *Free Energy* that such structure can have. Although experimental physical methods are more commonly used in the laboratory, computer modeling is an alternative, one that, with *FoldIt*, uses crowdsourcing to work. Solving puzzles corresponds to a guess at the final structures. The thermodynamics of protein folding is translated to a user interface, a game, where users can freely manipulate proteins from the start.

Any new member of *FoldIt* will have to go through the game tutorials (as represented in Figure 1). The puzzles start by being more simple and easy. The aim is always to make the energy bar to get full. The objects being manipulated are loops of amino-acids, arrangements that these structures take in space, commonly called secondary structure motifs¹⁴. These include the more popular α -*helixes* (spacially identified as spirals) and *beta-sheets* (that are represented as flat plaques). These concepts are part of the educational and professional process of Biochemistry met through traditional textbooks and educative media. The game *FoldIt* further presents other operative concepts, as the red balls that can be found in Figure 1, markers that get bigger with proximity between chains. The dynamical character of the manipulation of the protein made that concept useful. During that tutorial, the molecular size as a concept had a character of *firstness*, as a "thing-in-itself". The red balls warning the user of danger while he manipulates the protein chains had a character of *secondness*, while the apprehension of that concept, the creation of a new *habit of mind* was *thirdness*. The author of these concepts of firstness, secondness and thirdness was the polymath Charles Sanders Peirce, for whom the order of the World presented itself in triads.

¹⁴ These concepts are part of the typical introduction to the discipline of Biochemistry and can be found in traditional academic media, as in Berg, 2002;

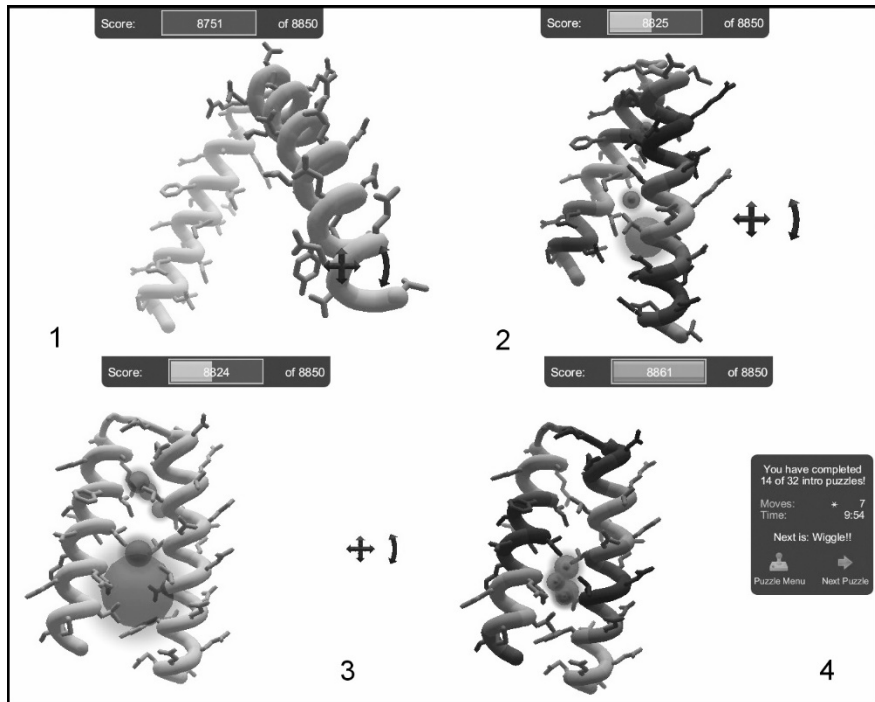


Figure 1 | **The manipulation of the citizen science game *FoldIt*.** Sequentially, from step one to four, the new user can manipulate the protein through a tutorial. The 'red ball' identified in the middle was the concept to apprehend in this step, as it was a marker for the proximity between protein chains. The user can 'grab' the protein chains and bring them closer together, producing a more stable protein conformation, as represented from step 1 to step 2, with the energy bar getting more points to the player. If the user brings the α -helices too close, although, the red balls in the middle of the chains get bigger, warning the player of the less stable conformation, as it happens from step 2 to 3. With a small move, bringing the protein arms a bit farther, as represented from step 3 to 4, a compromise is found and the tutorial protein puzzle is won, © University of Washington Center for Game Science, University of Washington Institute for Protein Design, Northeastern University, Vanderbilt University Meiler Lab, and University of California, Davis.

Diagrammatic reasoning

Three men met and decided to create *FoldIt*. Some of them had before released *Rosetta@home*, a screensaver that runs in personal computers. The step forward to *FoldIt* was to involve directly the user and take advantage of human's superior spatial reasoning. The reasoning process, how we think, was a problem that Charles Sanders Peirce reflected upon in the turn of the 20th century. Now, in the beginning of the twenty-first, as more on-line applications are developed, images are increasingly more important to learn. Through *FoldIt* scientific concepts met before through written text are translated into graphic and dynamic representations: *symbols*, using Peirce's terminology. The object defined by *FoldIt* is the protein molecular structure that the user can manipulate, which, for Peirce, would be a *form of relation*¹⁵. It is interesting to further consider the Peircean architecture of understanding of this manipulative action.

The diagrammatic reasoning is defined through *Existential Graphs* in Charles Peirce's latest writings. Peirce's Existential Graph, as the also called *Existential System of 1897*, "represents one recognized universe, real or fictive"¹⁶, later defined, in 1906, as a "system for diagrammatizing intellectual cognition"¹⁷ or as a "moving picture of the action of the mind in thought"¹⁸. The complex architecture of representations had a clear aim: "a method of representing diagrammatically any possible set of premises, this diagram to be such that we can observe the transformation of these premises into the conclusion by a series of steps each of the utmost simplicity"¹⁹. Existential Graphs can act as a key that opens the door of the interpreting mind, using Ahti-Veikko Pietarinen's metaphor²⁰. Still such a broad conception of a Logic of Signs wasn't integrated into Logic research, being even considered by Pietarinen as anti-foundational²¹. Semeiotics, on the other hand, was marked with Charles Peirce's concepts. Liking Charles Sanders Peirce's philosophy is like liking a "house that is

¹⁵ Peirce, 1958, Collected Papers (CP), 6.530;

¹⁶ CP 4.421;

¹⁷ Peirce, 1906 in Pietarinen, 2011;

¹⁸ Peirce, 1905 in Pietarinen, 2011;

¹⁹ CP 4.429;

²⁰ Pietarinen, 2006, 22;

²¹ Pietarinen, 2010, 44;

continuously rebuilt by the inside"²². His path is sinuous but returns to central concepts, as the one of abduction or hypothesis. The abduction or formation of hypothesis is one of the basic inferences defined by Peirce:

Hypothesis is where we find some very curious circumstance, which would be explained by the supposition that it was a case of a certain general rule, and thereupon adopt that supposition²³.

With deduction and induction, Peirce found the roots of the three inferences in Aristotelic syllogisms²⁴. These came to be later reframed as the operational tools for the transformation of signs, with the use of graphs.

Tinkering

Diagrammatic reasoning can explore other senses²⁵, but is highly understood as visual. The user interface of *FoldIt*, as many others, uses visual clues to work. Our manipulation of parts of the protein model, transforming it, uses a mesh of the reasoning processes indicated by Peirce. Our hypothesis at hand is tested by manipulation and deduction. When finding an unusual object we rehearse a usage, looking for the best fit. We build our models of understanding of the problem and try our hypothesis. We look for a solution for the puzzle and we might even just be aware of it after the solution presents itself, after manipulation or facing of the problem from a different angle. Quietly or abruptly, we are *tinkering*.

In 1903, during one of his conferences at Harvard, Charles Peirce made an important statement about the nature of abduction: "abductive inference shades into *perceptual judgment* without any sharp line of demarcation between them"²⁶. The perceptual judgment is a product of the *percept*, from the sensorial activity, but, for Peirce, this judgment is itself an extreme case of abduction. The sudden abductive suggestion is an act of insight, "although of extremely fallible insight", he adds²⁷. While tinkering, we try different possibilities, either aware of the try out, or just playing

²² Murphey, 1992 in Peirce, 1992;

²³ Peirce, 1878, CP 2.624;

²⁴ Peirce, 1878, CP 2.619;

²⁵ Pietarinen, 2010b;

²⁶ CP 5.181;

²⁷ CP 5.181;

through without being aware of this process. We are prone to error in this abductive path. But it is the error that makes the learning process meaningful, through tryouts that lead us to solutions. The creation of new hypothesis associated to the cognitive accomplishments of this method serve as a base for more inductive generalizations²⁸.

Citizen science interfaces, like the 3D model manipulation of *FoldIt*, offers a good window to the experience of this biochemistry challenge. But experience isn't restricted to the manipulation of objects. *Web 2.0* forums and social media are part of the participants' experience.

Creating habits of mind

CosmoQuest is another citizen science game, one coming from a domain of knowledge also associated with the '*data deluge*'. There are plenty of applications for the participation of the classroom and the general public in research in astronomy and astrophysics. In opposition to the before appreciated *FoldIt*, *CosmoQuest* is a small platform. It had roughly 300 thousand registered members and 700 active users²⁹. It was created in 2011, by the *Center for Science, Technology, Engineering and Mathematics* of Southern Illinois University Edwardsville in cooperation with *Astrophere New Media* and other partners. The first project of *CosmoQuest* was *Moon Mappers*, a simple application to map the Moon. Participants can immediately start spotting and cataloguing Craters by size on images taken from remote telescopes. Doing so, they can collaborate in generating catalogues of Craters for age dating. Of course there have been computer programs' running algorithms for decades. But the accuracy of such methods isn't great. Involving human cognition for spotting craters, the scientists want to understand the effects of Sun angle on Crater identification and measurement.

CosmoQuest promotes many events, physical and online for the community of members. Not all such citizen science games allow and stimulate community build-up. While travelling through the forum, one of the users, drhaugh, started out an interesting thread of discussion (presented in Figure 2). While drhaugh was strolling through the park and

²⁸ Magnani, 2004;

²⁹ CosmoQuest, 2013;

spotting craters through *Moon Mappers* he found an image that puzzled him. The identification of rare objects on the Moon's surface (other than just craters) is one of jackpots you might get lucky enough to get. Most times they're connected with ancient exploratory missions.

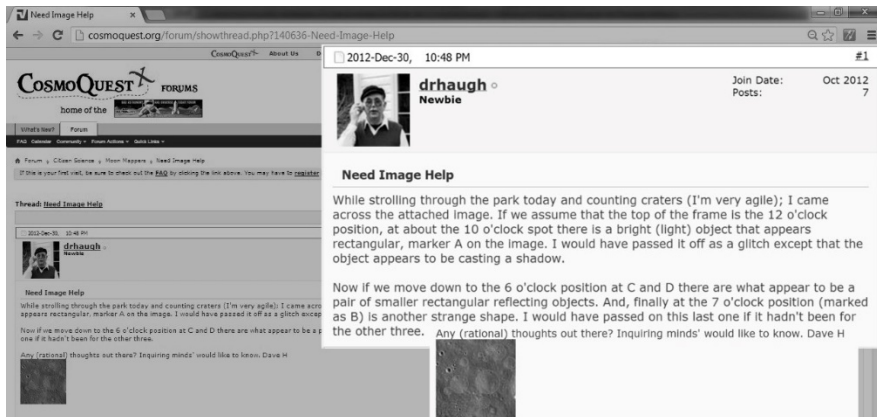


Figure 2 | **drhaugh's doubt on Moon Mappers' forum.** drhaugh found a strange object while he was recognizing and cataloguing Craters on the Moon through the *Moon Mappers* application in his portable device. The continuation of the discussion with the replies of other members could be followed by all users in one of the forum's threads. When he started the discussion, drhaugh came to express belief that the rectangular shaped objects he found could be parts of an old lunar rover, regarded as a possible and rare event to happen through this process of image recognition. Through the exchange that soon followed, a team scientist debunked drhaugh's belief. His finding was just a boulder. drhaugh expressed then his frustration over the experience³⁰. CC BY-NC 4.0 CosmoQuest.

Some hours after the post of the new member drhaugh in *Moon Mappers'* online forum, an older user replied. He hinted that the strange bright object might be part of the old Russian rover *Lunokhod 1*. The following research on image databases seemingly confirmed that, drhaugh was assured of his finding, he believed it was a rover in that image of his. Still, next day, after the New Years Eve of 2013, a much more experienced Team Scientist commented on the discussion thread. It couldn't be the old Russian rover in that image, it was probably just a big rock, a boulder.

³⁰ CosmoQuest, 2013;

The answer of drhaugh, still on June 1st revealed his frustration. "Good thing we're starting a New year or I'd be really depressed"³¹.

drhaugh's experience frames well the process that inquirers go through. Charles Sanders Peirce polarized doubt and belief as guidelines for reasoning in *The fixation of belief* from 1877³². After the answer of the team scientist to drhaugh's doubt, he still decided to open up the discussion to further members. What if that bright light isn't a rock, what else could it be? But still, he didn't consider even to stubbornly say it's a rover, the expert knew better. With this process maybe drhaugh and other forum users could learn a bit more about the identification of boulders or, using Peirce's words, to form a new *habit of mind*:

That which determines us, from given premises, to draw one inference rather than another, is some *habit of mind*, whether it be constitutional or acquired. The habit is good or otherwise, according as it produces true conclusions from true premises or not; and an inference is regarded as valid or not, without reference to the truth or falsity of its conclusion specially, but according as the habit which determines it is such as to produce true conclusions in general or not³³.

Forums in citizen science projects serve as a communication tool between participants and scientists. Still, most of the protein puzzle *FoldIt* members have a formal education in Biological Sciences, but not all. This community differs from the scholar's, it's open. Cheese, one of the top solo players that made an highscore in 2008, was at the time just 13 years old. During these years many *FoldIt* top-gamers, even the ones playing in groups, were non-scientists³⁴. With citizen science, the scientist can be mixed with the public or, as in the given examples, located upstream from the observation. But is this open group able to integrate the academic community? Mixing to the fold all these untrained minds, is this wise?

Charles Peirce's contribution to the social epistemology of science is relevant to the subject. His commonly taken to be consensual theory of truth says that:

³¹ CosmoQuest, 2013;

³² CP 5.358;

³³ Peirce, 1877, CP 5.367;

³⁴ FoldIt 2013;

Different minds may set out with the most antagonistic views, but the progress of investigation carries them by a force outside of themselves to one and the same conclusion. (...) The opinion which is fated to be ultimately agreed to by all who investigate is what we mean by truth, and the object represented is the real³⁵.

This passage is often read as meaning that the truth is whatever the community of inquirers converges on in the long run. Other interpretation is that truth (and "the real") depends on the agreement of the community of inquirers or that it is the effect of the real that it will in the end produce agreement among inquirers. Whatever the correct reading of this particular statement, Peirce made it clear that, in his view, truth is both attainable and beyond the reach of any individual:

We individually cannot hope to attain the ultimate philosophy which we pursue; we can only seek it, therefore, for the community of philosophers. Hence, if disciplined and candid minds carefully examine a theory and refuse to accept it, this ought to create doubts in the mind of the author of the theory himself³⁶.

Discussion

Web 2.0 tools, as different interfaces and forums, allow the build-up of communities that are enjoying since the turn of the twenty-first century the possibility of contributing to the construction of new scientific knowledge. The interaction through 3D-model interfaces, as the one of *FoldIt*, allow *tinkering* processes that are closer in nature to the moving pictures of the mind in action, as defined by Charles Sanders Peirce. As part of these reasoning processes, doubt and critical interaction are means to attain scientific knowledge, also acknowledged by Peirce. If continuing thinking about the relevance of citizen science as a social movement, it considers scientific research as a public sphere. Scientific research is represented in a form more similar to coffee places and salons. Argumentation, dialogue and social interaction also have also their space in the virtual world of internet. Just like the noble Londoner took part of a

³⁵ Peirce, 1878, 133, CP 5.407 Cross-Ref 407;

³⁶ Peirce, 1868, 40, CP 5.265 Cross-Ref 2;

literary discussion at the coffee place, so did drhaugh's discussed with other players in the *Moon Mappers* forum. Having the human interaction mediated through WWW and Web 2.0 allows a diverse communicational media. Formal academic methods of learning and integration in a community of peers traditionally disregard tinkering processes and don't give enough credit to the social gatherings and dynamics for the creation of new knowledge. Citizen science seems to make the scientific practice and knowledge more public and magnifies the 'learning by doing' approach of this activity that can be considered as a window for the heuristics of scientific research.

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Minimalism, Dispositions and the Contingent Liar

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The Liar paradox jeopardizes any approach to the notion of truth. In particular, it creates problems to Paul Horwich's theory of truth, Minimalism. In this paper, we are going to introduce both the minimalist theory of truth and the Liar paradox and we are going to see Horwich's stance in front of the latter, which turns out to be analogous to his treatment of the phenomenon of vagueness and its related paradox, the sorites paradox. Horwich's strategy to overcome the conundrum posed by the Liar consists in offering a restricted version of Minimalism. In order to do that, he has tried to characterize the notion of paradoxical axioms of the theory in terms of our dispositions to accept them. In this paper we will argue that, in the face of the so called 'contingent Liar paradox', Horwich's argument with the use of dispositions is not successful in determining which axioms must be excluded from Minimalism.

In the first section, we will present Minimalism and the role that its axioms have in Horwich's view about truth. Next, in the second section, we will look into the problems that the Liar paradox suppose for Minimalism and how Horwich intends to solve them. Finally, the third section will discuss if Horwich's view is tenable in front of a variation of the Liar paradox that uses propositions that are not intrinsically paradoxical.

1 | The Basic Regularities of Truth and Vague Expressions

Paul Horwich has presented and defended, in a number of places,¹ his theory of truth, called 'Minimalism'. One of its main theses is that the instances of the T-schema,

(T-schema) $\langle p \rangle$ is true if, and only if p ²

are epistemologically, explanatorily and conceptually fundamental with respect to the notion of truth.³ In the first place, they are conceptually fundamental because the basic and fundamental regularity of use that determines the meaning of 'true' (which is the concept of truth, for meanings are concepts, according to Horwich) is our disposition to accept all instances of the T-schema. In the second place, they are explanatorily fundamental because the instances of the T-schema are all we need to explain all our uses of 'true'. And, finally, they are epistemologically fundamental due to the fact that the instances of the T-schema are "immediately known",⁴ they cannot be deduced from anything more basic. Given these considerations, it is not surprising that the minimalist theory of truth contains as axioms all instances of the T-schema applied to propositions, and nothing else.⁵

The role that the T-schema plays in Horwich's theory of truth mirrors the role that certain partial functions specifying the subjective probability of the application of vague predicates to objects play in Horwich's theory of vagueness. Specifically, Horwich defends an epistemic account of vagueness according to which vague predicates have sharp boundaries unknown to us. This view allows Horwich to preserve the Law of Excluded Middle (LEM), which is seen as a basic law of thought, and, consequently, he claims, the Principle of Bivalence (BIV).⁶ The general idea Horwich defends with respect to vagueness is that vague predicates are governed

¹ See, for instance, Horwich 1998b, 2001, 2010.

² The symbols ' $\langle$ ' and ' $\rangle$ ' surrounding a given expression e produce an expression referring to the propositional constituent expressed by e . Thus, when e is a sentence, ' $\langle e \rangle$ ' means *the proposition that e* .

³ Horwich, 1998b, 145.

⁴ Horwich, 2010, 36.

⁵ That characterization is not completely accurate; as Horwich admits, the theory should also have an axiom claiming that only propositions are bearers of truth (Horwich, 1998b, 23).

⁶ See, for example, Horwich, 1997, 2005.

by partial functions which specify “the subjective probability of its applying as a function of the underlying parameter n (i.e. ‘number of grains’ for ‘heap’, ‘number of dollars’ for ‘rich’, . . .)”.⁷ As in the case of the T-schema, such regularities are the fundamental facts about our use of vague predicates; they are epistemologically, explanatorily and conceptually fundamental with respect to vague predicates; they explain all our uses of vague predicates.⁸ Then, given the fact that these functions remain silent with respect to the application of the predicates to certain objects explains why we also must remain silent in front of such applications. These considerations also explain why we cannot know whether borderline cases are in the extensions of vague predicates; the only way we can be justified in applying a given vague predicate to an object is via the fundamental facts underlying the vague predicate and that, as we have seen, is not possible. Hence, believing an ascription of a vague predicate to a borderline case will never be able to constitute knowledge.

Horwich defense of LEM and BIV is, in the face of the sorites paradox,⁹ highly problematic for this paradox seems to show that classical logic (and, hence, LEM and BIV) imply the following claim:

(SB) vague predicates have sharp boundaries, that is, they divide the world into two sharp groups of things; the ones that have the property expressed by the predicate and the ones that haven’t.

which, given our understanding of vague terms, is highly counterintuitive. As we mentioned, Horwich accepts (SB) and claims that it is the impossibility to know where the sharp boundaries are what explains why it seems to us that there are no such boundaries.

⁷ Horwich, 1997, 933.

⁸ Horwich, 1997, 934.

⁹ The sorites paradox is one of the seven puzzles made famous by Eubulides of Miletus (fourth century BC), contemporary of Aristotle (see Laertius, 1972, 237). In the contemporary discussion in philosophy of logic, this paradox is presented in the form of an argument:

1. A man with no hairs on his head is bald.
2. If a man with n hairs on his head is bald then a man with $n+1$ hairs on his head is bald.
3. Therefore, a man with a million hairs on his head is bald.

According to Horwich, then, the argument is not sound, for the second premise is false. For details on the sorites paradox see, for example, Oms and Zardini, forthcoming.

2 | Horwich and The Liar

Another of the puzzles that Eubulides of Miletus held was the Liar paradox. Suppose we have a proposition, say λ , that asserts its own untruth (I will call such a proposition ‘the Liar’). Suppose, now, that λ is true. Then, if λ is true what λ says is the case (is not that what truth is about?); but what λ says is precisely that λ is not true, which contradicts our initial supposition that λ is true. Thus, we confidently conclude, by *reductio*, that λ is not true. Consequently, if λ is not true, then what λ says (that is, that λ is not true) is the case and, thus, λ is true after all. This means that λ is both true and not true and, hence, we end stuck in a contradiction.

As we are going to see, Horwich’s solution to the Liar mirrors his treatment of the sorites. Horwich’s response to the problem posed by the Liar was, in the first place, very succinct. He claimed that what the Liar implies is that not all the instances of the T-schema are axioms of his theory of truth.¹⁰ Consequently, he claimed, the minimalist theory of truth must consist of a restricted collection of instances of the T-schema; only those that do not engender Liar-like paradoxes. Which of the instances of the T-schema should be removed, though, was left undetermined. Later, Armour-Garb,¹¹ Beall,¹² Restall,¹³ and, though succinctly, Horwich himself,¹⁴ elaborated this proposal into what was called, by the former, ‘Semantic Epistemicism’.

Specifically, Horwich’s solution to the Liar consists of the two following claims:

1. The Liar is true or The Liar is false.
2. It is conceptually impossible to know whether The Liar is true and it is conceptually impossible to know whether The Liar is false.

¹⁰ Horwich, 1998b, 42.

¹¹ Armour-Garb, 2004.

¹² Armour-Garb and Beall, 2005.

¹³ Restall, 2005.

¹⁴ Horwich, 2010.

Let us see which are the justifications Horwich gives for both of them.

With respect to 1, Horwich claims that it is just an instance of BIV.¹⁵ To see this, we need to define, first, falsity. Horwich proposes to define it in terms of absence of truth:

$\langle p \rangle$ is false if, and only if it is not the case that $\langle p \rangle$ is true.

With this definition of falsity at hand, LEM gives us claim 1 above, for we can derive BIV applied to any proposition and, hence, in particular to the Liar. To see this, take any proposition p . Then, we have that p or $\text{not-}p$ (that is just an instance of LEM) and, hence, in particular when p is of the form q is true, we have that *either q is true or it is not the case that q is true*; that is, according to our definition of falsity, *either q is true or q is false*. Since this is the case for any proposition q , it is the case with the Liar. So 1 is vindicated. This argument, though, is not satisfactory.

The problem with that line of thought is that it begs the question regarding the definition of falsity in terms of absence of truth. The idea is that Horwich is supposing that the only way in which a proposition can be not-true is by being false; but that is, precisely, BIV! Semantics that posit truth-value gaps will not accept the definition of falsity in terms of non-truth, for propositions that are neither true nor false are, in particular, not true, but they are not false.

A non question-begging way of defining falsity in this context is in terms of truth of the negation:

$\langle p \rangle$ is false if, and only if $\langle \text{not } p \rangle$ is true.

Then, we can reason as follows. For any proposition p , as we have already seen, LEM guarantees that we have that p or $\text{not-}p$. Then, given the T-schema for $\text{not-}p$, this implies that p is true or $\text{not-}p$ is true, which, given our new definition of truth, implies that p is true or p is false. The problem now is that we needed p 's instance of the T-schema. But, as a consequence of the Liar paradox, some of the instances of the T-schema are not in the theory (at least the Liar instance) and, hence, the argument above cannot go through in general. Hence, it is not clear at all how Horwich can advocate in favor of claim 1.

¹⁵ Horwich, 1998b, 71.

What about claim 2? According to Horwich, the reasons why the truth value of the Liar is indeterminate are the same as in the case of vagueness.¹⁶ Specifically, they are rooted in the already mentioned fact that the instances of the T-schema are explanatorily fundamental, which means that they must be in the basis of any explanation of any fact concerning our use of the truth predicate and, moreover, that they fix the meaning of 'truth', that is, the concept of truth. As in the vagueness case, Horwich can define a notion of determinateness based on his account of meaning. He uses, first, the claim that meanings are concepts and, second, that meaning properties are constituted by use properties, which, in turn, stem from some given fundamental acceptance properties,¹⁷ which, in the case of the truth predicate, are our dispositions to accept the instances of the T-schema. Some of these dispositions, though, are overridden by the fact that we realize that some of the instances of the T-schema lead to inconsistency. These instances, then, will not be in the minimalist theory of truth.

The idea, once we have restricted the truth theory, is that some ascription of the truth predicate to a given proposition p is indeterminate when we do not know whether p is true or not and the unknowability of the ascription has its roots in the facts (that is, the fundamental acceptance properties; our dispositions to accept the instances of the T-schema, in the case of truth) that make our words mean what they mean, and that happens because the instances of the T-schema needed to know whether p is true or not are not present in our truth theory; Horwich claims that, then, it is *conceptually impossible* to know whether p is true. And that is precisely what happens with the paradoxical instances of the T-schema.

In sum, since we do not accept the paradoxical instances of the T-schema and, hence, the minimalist theory of truth does not contain them as axioms, it is conceptually impossible to know any fact concerning the truth value of the paradoxical sentences involving truth. As we said, the reason Horwich's gives in his book *Truth* for such a restriction of the minimalist theory of truth is the fact that our dispositions¹⁸ to accept the paradoxical instances of the T-schema are "overridden by the discovery that

¹⁶ Horwich, 2010, 91.

¹⁷ Horwich, 1998a, 44.

¹⁸ Horwich uses indistinctly 'disposition' and 'inclination' (see, for uses of the former, Horwich 1998b, 126, 128, 135, and for uses of the latter, Horwich 1998b, 24, 37, 136) so I will treat them as synonymous.

they lead to contradiction”¹⁹. As we are going to see, this position is at odds with a variant of the Liar paradox, to be introduced in the next section, usually called ‘the contingent Liar’.

3 | The Contingent Liar

There is another alleged version of the Liar paradox attributed to the Cretan philosopher Epimenides (sixth century BC), which is interestingly different from Eubulides’: according to St. Paul’s Epistle to Titus, Epimenides said that all Cretans were liars.²⁰

The first thing we need to note is that Epimenides’ assertion is not necessarily paradoxical; as a matter of fact, it will very likely be plainly false (for example, if there is some Cretan who is not a liar). The paradoxicality of Epimenides’ claim only arises in certain situations like the following one. Suppose, first, that a liar is a person who has never (even unintentionally) said anything true. Consider, second, a situation where all Cretan utterances are false (except perhaps Epimenides’ one). Then, in this case, if the sentence ‘all Cretans are liars’ is true, this very same sentence is false, for it has to be a lie (since it has been uttered by a Cretan). But if it is false, then it is a lie and, consequently (given that the rest of Cretan utterances are false) it is true.

The distinction between Eubulides’ and Epimenides’ version of the Liar brings up one important feature of this paradox: some sentences (like Eubulides’ one) are intrinsically paradoxical in the sense that they can generate a paradox independently of the way the world is; they are paradoxical in all possible worlds. On the other hand, some sentences (like Epimenides’ one) generate paradoxes that do not depend only on semantic stipulations, but also on empirical facts in the world.

Kripke, in his *Outline of a Theory of Truth*,²¹ remarks precisely this distinction when he shows the following way of achieving self-reference. Suppose *P* and *Q* are two predicates of a given language that apply to sentences of that very language. Then the sentence

$$\forall x(Px \rightarrow \neg Qx)$$

¹⁹ Horwich 1998b, 136.

²⁰ Titus, 1:12.

²¹ Kripke, 1975.

can be paradoxical if some empirical facts determine that this very same sentence is the only one that satisfies P and, moreover, Q is the truth predicate. For instance, suppose that the predicate P is the predicate 'the sentence written on the blackboard of room 202'. Now, depending on the way things are, the sentence $\forall x(Px \rightarrow \neg Qx)$ can be paradoxical; it is only needed that this very same sentence, and only it, is written on the blackboard of room 202 and that the predicate Q is interpreted as the truth predicate. That means that utterances in which we use the truth predicate might be paradoxical without our being aware of it. Following the habitual usage, I will call paradoxes like this 'contingent Liar paradoxes' and the sentences that generate them 'contingent Liar sentences' or, more succinctly, 'contingent Liars'.

We are now in a position to see why the contingent Liar is a problem for Horwich's proposal. As we just saw, according to Horwich, we are not disposed to accept some of the instances of the T-schema because we realize that they lead to inconsistency.

According to the sometimes called Simple Conditional Analysis,²² a subject S has the disposition to accept a given instance of the T-schema when confronted with it if, and only if, S would accept it if it were the case that S was confronted with it. So, to say that we do not have the disposition to accept the paradoxical instances of the T-schema is to say that we would not accept such instances in case we were confronted with them. That may be so in the case of The Liar; we may claim that when a subject S is in front of the instance of the T-schema applied to The Liar, S can conclude *a priori* that such instance is not acceptable, because it leads to a contradiction.

But as we saw in the last section, some paradoxical sentences are not intrinsically paradoxical; that means that they are paradoxical depending on some features of the world; consider, again, the sentence

(CL) the sentence written on the blackboard of room 202 is not true.

If the world is such that a token of this sentence is written on the blackboard of room 202, then the sentence is paradoxical but if, for example, the sentence written on the blackboard of room 202 is '2+2=4', then (CL) is just false. Now, when a subject S is confronted with the instance

²² Defended, for example, in Quine, 1986.

of the T-schema applied to (CL), she cannot say *a priori* if this instance is acceptable or not.

Horwich could try to overcome this difficulty by idealizing the situation; he could idealize the subject *S* and suppose that she has access to all the relevant facts (in our example, she would have access to room 202). This, though, will not do due to the notion of indeterminacy that Horwich has in mind; it is not just that we are not able to know the semantic value of the Liar sentence (or the ascriptions of vague predicates to borderline cases), but it is conceptually impossible to know it. Let us elaborate this point.

As Hartry Field has noticed,²³ Horwich's proposal implies that "the concept of an omniscient being is conceptually incoherent"²⁴. For an omniscient being would know the location of the boundary between the objects that satisfy a given vague predicate and the objects that do not but, since it is conceptually impossible to know such boundary, we must conclude that the notion of an omniscient being cannot be conceptually coherent. Indeed, Horwich claims that an omniscient being (and any being with a different language of ours) with a language *L* can only judge whether a given term α (of our language) is true of a given object *k* via a term β of *L* with the same meaning as α . That means, having in mind Horwich's theory of meaning, that α and β must have the same "conceptual role"²⁵. But, then, α and β will be governed by the same fundamental facts about use and, therefore, if α is a vague predicate, then the speaker of *L* will not be able to find the location of its sharp boundaries, for he will be neither capable of finding the location of the sharp boundaries of β .

The same story applies to the truth predicate; since the instances of the T-schema are the fundamental facts about the use of the truth predicate, any being capable of knowing the truth value of the Liar would have to be using a truth predicate governed by a theory containing the Liar instance of the T-schema, but then this being would not be using our truth predicate, and what this being would have knowledge of would not be the truth value of the Liar sentence, but something different.

Returning now to the problem that the contingent Liar poses to Minimalism, recall that Horwich could try to idealize the situation and

²³ Field, 2010.

²⁴ Field, 2010, 2.

²⁵ Horwich, 2005, 96.

suppose that when we evaluate whether a given instance of the T-schema is in our theory of truth we know all the relevant facts. We can create now a contingent Liar whose paradoxicality depends on an indeterminate sentence in Horwich's sense. In order to do that take the proposition expressed by the following sentence:

(L) A and L is not true

where A is any sentence whose truth value is conceptually unknowable to us; A can be the Liar sentence itself, or any paradoxical sentence, or an ascription of a vague predicate to a borderline case. Next, we need to note that L is paradoxical (that is, its instance of the T-schema can be used to obtain a contradiction via liar-like arguments) just in case A is true. To see this, suppose A is true (is A is false, then L is just false, for it is a conjunction with a false conjunct). Under this supposition, suppose L is true. Then, since it is a conjunction, it must have both conjuncts true, in particular, the second one must be true. It follows, then, that L is not true. From this we conclude, by *reductio*, that L is not true. Then, though, L is a conjunction with both conjuncts true and, hence, L is true after all. Therefore, L is true and it is not true. Contradiction.

So L is paradoxical (that is, its instance of the T-schema can be used to obtain a contradiction via liar-like arguments) just in case A is true. Moreover, it is conceptually impossible to know the truth value of A . Consequently, it is conceptually impossible to know whether the L instance of the T-schema should be in the minimalist theory of truth. Hence, even idealizing the situation and supposing that the subject S who is taking the decision whether to incorporate or not a given instance of the T-schema into her theory of truth has all the possible information available in front of her, even in a case like that there will be undecided cases like L . Minimalism remains, thus, indetermined.²⁶

²⁶ Horwich has offered other ways to try to determine which instances of the T-schema are in the minimalist theory. Specifically, he has proposed a constructive specification similar to the one offered by Kripke, 1975. Still, such a construction lacks a reasonable philosophical motivation and the resulting theory is too weak (see Oms, 2017).

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